

SMITHIAN FOLD THEORY - CHEMISTRY BRANCH PAPER 001

From Fold to Chemistry

An Exact, Parameter-Free and Machine-Closed Reconstruction of Chemical Science
from Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Third clean-room reconstruction - Chemistry inventory complete
86 required Chemistry claims and 7 explicit upstream claims

24 July 2026

DOI: 10.5281/zenodo.21531455

Paper: CC BY 4.0 - Code: Apache-2.0

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Abstract

This paper reports the Chemistry branch of the third clean-room reconstruction of Smithian Fold Theory. It documents all 86 obligations in the frozen inventory and the seven newly required upstream physical prerequisite/validation claims. Every claim passed one fail-closed admission engine with a complete generated grammar, exactly one survivor, trace to the premise-free foundational theorem, zero axioms, zero free or fitted parameters, exact permitted arithmetic, four adverse controls and an implementation-distinct reconstruction. Empirical claims add capability-closed prediction, target custody, post-seal opening, complete authoritative rows and tampered controls. The required Chemistry support contains 22016 generated candidates and 86 survivors; the terminal-prediction prerequisite appendix contributes 4224 candidates and seven survivors. The formal walk forces orbit widths 2, 6, 10, 14 and 18; nuclear closures 2, 8, 20, 28, 50, 82, 126 and 184; Smithium at $Z=126$, $N=184$, $A=310$ with active 8s2 5g6 organization; g-block opening at 121; and a structural endpoint at 137. NIST/CODATA, IAEA and IUPAC checks occur only after derivation seals. Element 121, Smithium and endpoint 137 remain explicit unobserved predictions.

1. Publication, authorship and license boundary

PUBLISHED OPEN-ACCESS BRANCH PAPER. DOI: [10.5281/zenodo.21531455](https://doi.org/10.5281/zenodo.21531455).

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The paper is prepared for CC BY 4.0 distribution; code is Apache-2.0. Copyright preserves authorship while the licenses preserve open inspection, reuse and criticism. The Ernos Labs designation denotes conformance to the published empirical and community standards; it is not transferred merely by copying the work.

2. Exact scope

The inventory contains 86 obligations in ten ordered subbranches and all are admitted. Closure means each registered Chemistry question has one machine-admitted law at its exact generated and empirical boundary. Materials bulk properties, biological function, clinical intervention, Earth-system fate and applications remain later branches. Observed absence beyond element 118 is not promoted into proof. The 121, 126 and 137 results are standing predictions.

3. Constitutional arithmetic

Structural absence is empty One, never numerical zero. Proof magnitudes are positive generated counts and exact positive rational parts or ratios. Opposition, orientation, gain/loss direction and charge sense are held labels, never negative proof quantities. Irrational, imaginary and floating proof values, completed infinity, imported equations, fitted coefficients, learned parameters and target-selected forms are prohibited. Host counters may enumerate artifacts but have no scientific value authority.

4. Single-engine admission

Every claim enters `SFTAdmissionEngine`. Registration verifies root identity, zero axioms, zero free parameters, admitted dependencies and exact source identity. The engine then requires complete candidate identity/cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four controls and an implementation-distinct reconstruction. Empirical claims additionally require an evaluator-verified derivation seal, capability isolation, target absence before seal, matching custody release, all-row retention and a falsification condition. Failure halts and is retained; it cannot be edited into admission.

5. External evidence and blind prediction

IUPAC terminology and periodic-table records test categorical consequences. NIST and IAEA records test sealed numerical or ordered-sequence consequences. External data never select the grammar or survivor. The capability-closed predictor cannot read the filesystem, network, clock, environment, subprocess, dynamic import or foreign function. A distinct target custodian releases only after a matching prediction seal, every registered row is compared, and a deliberately changed row must fail.

6. Preserved failure and lawful correction

The original prerequisite programs retained two capacity survivors, two nuclear-schedule survivors and no endpoint survivor. Their rejection receipts remain immutable. V3 then derived new discriminators: generator three, stable three-space, boundary rank two, exact orbit orientation, exact colour coupling, complete three-direction shell composition, the finite inverse-alpha promotion ladder and a greatest-whole order certificate. The later admissions do not rewrite the failed runs; they change the admitted dependency surface through separately proved claims.

7. V1/V2 reconciliation without proof import

V1/V2 identify mandatory questions, including magic numbers, Smithium, the g-block and endpoint. They supply no answer-bearing input to V3. `census/lineage_reconciliation.json` binds those sources by hash and now records the elements/nuclear/island-of-stability group as closed at the current V3 standard. Other lineage groups remain open for their own branches and the final TOE paper.

8. How to read the claim ledger

Every section below projects the complete machine evidence: statement, dependencies, grammar, boundary, candidate cardinality, sole survivor, axis eliminations, controls, independent reconstruction, external rows, exclusions and immutable receipt identities. The JSON claim packages remain authoritative and are included in the release archive.

9. Measurement Identity

Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability.

1. Chemical entity and retained identity

Claim identity: SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

Question and theorem. A chemical entity is the least singular Fold carrier that remains separately distinguishable after constitution, isotopic identity and the observation context's required precision are retained.

singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

Generated grammar. Generate the literal product of the registered entity carrier, constitution, isotope, distinction, precision, record, provenance and extension choices.

Boundary: Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `singular-carrier__constitution-retained__isotope-retained__separately-distinguishable__context-bounded-precision__held-observation-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	singular-carrier	ensemble-without-member	An ensemble alone does not identify one chemical entity.
constitution	constitution-retained	constitution-erased	Erasing constitution merges chemically distinct carriers.
isotope	isotope-retained	isotope-erased	Erasing isotopic identity merges distinguishable entities.
distinction	separately-distinguishable	indistinguishable-answer-label	An answer label without a retained distinction cannot identify an entity.
precision	context-bounded-precision	unbounded-or-fixed-precision	A fixed universal precision either loses a required distinction or adds an unforced one.
record	held-observation-record	result-without-observation-record	Without an observation record the identity boundary cannot be audited.
provenance	source-bound-proof-trace	unbound-provenance	Unbound identity cannot be traced to the Fold dependency spine.
extension	no-extra-rule	free-extra-identity-rule	An extra identity rule can arbitrarily split or merge entities.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5f110d5c9e6a40120b2d1a5cb5a6dfa2738fc19b611acdb41f21092208f2a84e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:564377b083b5fd48f74bcd6ffe5e88fbe00d970d9fb10253723fd39ff85ba41f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5f6268c44c846a4e91dea61554c4b01f5c85b882c9e894009c0d04e325d2a0a2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:eec730468fa661f596b4ac9b005665b9817aa9b7a63bb0878ca3cf0ec33b255d.

Independent reconstruction. Implementation hash

sha256:1e162cfb721018f6da4cb2c45eccff8324a1cba92e801169174696005e395938; independent certificate sha256:ab5e4ae45a37d0b2ac8f02f883990992b7c6492fcd59e809620be8a416d6fd86; external-validation hash sha256:1b952ccd2e4c3740a1bb5774c459db17faff80d5e6a4cb7daa93e4794f2a1e62. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03986-2026

Comparison records:

- `chemical-entity-iupac-m03986`: predicted
singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; source-derived
singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the current source-derived IUPAC entity record lacks singular distinguishability, constitution/isotope identity or contextual precision, or if any altered row is accepted.

Measurement receipt: sha256:6a9b6ee1de808a674690a9a99fd5f30323d47069e59eefe8585b5947bdda7f.

Isolation certificate: sha256:96a541e22fd08099bd11c4e7d615141525231b3347b80e2bd9c78caf14ccfa28.

Custody certificate: sha256:5784d9539afd879ed716b07d55763fa6bbdb94bec65bfe0a47728f00beca35bd.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03986/json> (sha256:6b0f6e183ef109d6f3483bcd1d70ca73a6d57e496f9ca32da23964bca652de8e)

Registered target identities:

- chemical-entity-iupac-m03986 from IUPAC-GOLD-BOOK-M03986-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

Immutable identities. Source manifest

sha256:9aac86602d9aeafe5045ac6aa7c5d8f91e1e90faca9d244bb831aa65f3003d63; derivation seal

sha256:9de339302a77f6504cf71743ec67ac5e3ec4abfadcc74da783418fd88e2a9f4d; engine receipt

sha256:130c21788e898b4b6f0cb012bc72b4cb29d5aeda5b86e67257c6008c13318f66 at

receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-ENTITY-001-130c21788e898b4b.json;

empirical hash sha256:141bf9f942bd9ffff44a8fe96e7a49f31931130413064ecaeba0e805491e64de;

measurement receipt sha256:6a9b6ee1de808a674690a9a99fd5f30323d47069e59eefe8585b5947bdda7f.

2. Chemical species and observation-equivalence class

Claim identity: SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

Question and theorem. A chemical species is the least Fold quotient whose support is an ensemble of chemically identical entities sharing accessible state support at one declared experimental timescale.

ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001

Generated grammar. Generate the literal product of the registered species carrier, member identity, state support, timescale, solid support, record, provenance and extension choices.

Boundary: Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary__molecular-or-solid-unit-support__held-member-and-class-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	ensemble-carrier	singular-only	One member alone is not a chemical species class.
member_identity	chemical-identity-equivalence	mixed-chemical-identities	Mixed chemical identities do not form one species.
support	shared-energy-support	energy-support-erased	Erasing accessible-state support merges experimentally distinct species.
timescale	experiment-timescale-boundary	timeless-equivalence	Interconversion can change whether two forms are experimentally distinct.
solid_support	molecular-or-solid-unit-support	isolated-molecule-only	A species law limited to isolable molecules loses solid structural units.
record	held-member-and-class-record	class-without-member-trace	A quotient without member trace cannot be reconstructed.
provenance	source-bound-proof-trace	unbound-provenance	An unbound class can silently change its equivalence rule.
extension	no-extra-rule	free-extra-equivalence	A discretionary equivalence can force any desired species partition.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a7193b15135c4dbbd92429905bc94c36075dd22dcadc3a76f4198f85f360f0a4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e41ee4909562f76a1c4e66cce01a19dc0f86a6e9eb482555c81b63598cf9d228`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ecff63e0a1fe45c9d7427360c1ff52460290dafcb6e8b33ace20a61491aa215e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4363a31941fcec233e36ff41c3cc5d6f7fdef7a860258c722f8ef96cae519289`.

Independent reconstruction. Implementation hash

`sha256:0d4dfc16c46530663cc1eb405e4ca6b2ed0599e69792c046629a88fb4f98ff35`; independent certificate

sha256:cee56a524f22fcf251ad18689682ad1b6469ea97710e2683277a0350c05c517e; external-validation hash sha256:5b7fb1676de0d49c2e985ad3dc0f7c8361560ea6ce3bfa8586079c67bbf368b5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT01038-2026

Comparison records:

- chemical-species-iupac-ct01038: predicted
ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; source-derived
ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC species record does not retain ensemble identity, common accessible support and experimental timescale, or if an altered row is accepted.

Measurement receipt: sha256:cef190fad510ddb2239cffed977546900c68888b9357a3e6825289630d02dbed.

Isolation certificate: sha256:6caa3ed29c98414d4e888a779c419cba0c579faa5725435c5bbb57bc1f89bc8d.

Custody certificate: sha256:ed48c33c70767039f305f77ee378a8980046322871a4819c5dbd0c6e766b1620.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/CT01038/json>
(sha256:538dd1087bc057eec8b68fdb713987cc8e1f0ff7548b4d03192d67b3e328eafe)

Registered target identities:

- chemical-species-iupac-ct01038 from IUPAC-GOLD-BOOK-CT01038-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

Immutable identities. Source manifest

sha256:1e419ad3d572f88eb13a42e738886b7e6dd56e382dfdfa5c52121021a9c087b0; derivation seal

sha256:5ae3a8fef3137c61ff35cab5e3bb870ee623f3799971bf3c757fcb143efaf4c4; engine receipt

sha256:5cde2482e85a9e7469dac34f4379a164979abac66a484aed5ebcb7b12639bd5b at receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-SPECIES-001-5cde2482e85a9e74.json; empirical hash

sha256:be270d0c83872e53b8772bf007dbdc8b09790e24b3e44bfd3aceb575a740d43a; measurement receipt

sha256:cef190fad510ddb2239cffed977546900c68888b9357a3e6825289630d02dbed.

3. Chemical substance and composition identity

Claim identity: SFT-CHEM-MEAS-SUBSTANCE-001

Question and theorem. A chemical substance is the least source-bounded Fold carrier that retains constant composition, its constituent entity support and the complete registered property characterization.

constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

Generated grammar. Generate the literal product of the registered substance composition, constituents, properties, source, phase, record, provenance and extension choices.

Boundary: Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity__phase-record-retained__complete-property-record__source-bound-proof-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	constant-composition	variable-unrecorded-composition	Unrecorded variable composition cannot identify one substance.
constituents	constituent-entity-support	anonymous-matter	Matter without retained constituent entities loses chemical identity.
properties	property-characterization	composition-only-answer	Composition alone can leave observation-equivalent candidates unresolved.
source	source-bounded-substance-identity	source-free-substance-label	A source-free label cannot distinguish sample and convention boundaries.
phase	phase-record-retained	phase-silently-collapsed	Silently collapsing phase can erase an observed substance distinction.
record	complete-property-record	property-selection-only	Selecting one favorable property is not a complete characterization record.
provenance	source-bound-proof-trace	unbound-provenance	Unbound composition and property claims cannot be audited.
extension	no-extra-rule	free-extra-substance-rule	A discretionary rule can split or merge substances after observation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: ad2432d5ad2ab35e20e37628d4c7a34f5468b37f79470eebbe483f318bea4b2a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: d3b75c1e2afca58bd1ace3b6c79c0db08b4790c8b5614a06f227e517436b2806.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b2afcb7ccd08f7de7686f06d0ee02426d69c0b76843fe38af1c13d393267e211.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: aa8c859776acb4b9507dadcd36885aa443478164dce3f33ad26e21c9f415a45.

Independent reconstruction. Implementation hash

sha256: c2a593300db78987c89c58b98d64c4f82f87cf79cbd340127f5f33ff93303b38; independent certificate sha256: bb690d7cc17309542fe5180ead6830f124ab9fa1db52c15739778d23046a5210; external-validation hash sha256: d069634773fcd9bfd04bf555fc350469bd45b504e44847c22f87bd6b61e21d5d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01039-2026

Comparison records:

- chemical-substance-iupac-c01039: predicted
constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity;
source-derived
constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC substance record lacks constant composition, constituent entities or property characterization, or if an altered row is accepted.

Measurement receipt: sha256: c66662a1a574bf8b3358a0a4308abad3c8d232b612cb6d78ed2d4b1e944db4a6.

Isolation certificate: sha256: 21e6ccdcbe853b5176f0fdf5dbeeae2531bad16581a6135ab163ca1fc0659177.

Custody certificate: sha256: 4fa96f00cf7136b60fec8f4b6b73c71998dcdbf1e904c29bfb6fa42007f1e699.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01039/json>
(sha256:97af068278c31568a03402f8dcb3e5de2327833ab9ac8da78c1322e730997498)

Registered target identities:

- chemical-substance-iupac-c01039 from IUPAC-GOLD-BOOK-C01039-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor

- external target content is absent from this specification and opens only through post-seal custody
- Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

Immutable identities. Source manifest

sha256:a5d94367f08d367b2c628c0a7e9b4cbab4bfcace5aa5042433ef1e6255b6b87a; derivation seal sha256:e0134f8366c295cf62cd3a8cd00af0a215c3f9b4f0f2812017ea6b81006b128a; engine receipt sha256:264b52127a204f58e2bcce66e422f2f8b3c4c41d7bfd0f9d64bf6a2413454220 at receipts/engine/model_admitted/SFT-CHEM-MEAS-SUBSTANCE-001-264b52127a204f58.json; empirical hash sha256:0b6623170beda97be63c42b4b4cbf6ee8d23e11811d8c13a8e61ce69c6acb73a; measurement receipt sha256:c66662a1a574bf8b3358a0a4308abad3c8d232b612cb6d78ed2d4b1e944db4a6.

4. Amount-of-substance count and reference carrier

Claim identity: SFT-CHEM-MEAS-AMOUNT-001

Question and theorem. Amount of substance is the exact Fold measure carried by a positive-finite generated count of one specified elementary-entity kind, with the entity reference and count trace retained.

amount-carrier__count-measure__specified-entity-reference__generated-entity-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-SUBSTANCE-001

Generated grammar. Generate the literal product of the registered amount carrier, entity reference, count, measure, support, record, unit and extension choices.

Boundary: Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *amount-carrier__specified-entity-reference__generated-cardinality__count-measure__generated-entity-support__member-and-reference-trace__separate-unit-comparison__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	amount-carrier	answer-only-amount	An answer label does not retain the entities being counted.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
entity_reference	specified-entity-reference	unspecified-entities	A count without a specified elementary-entity kind changes meaning under substitution.
count	generated-cardinality	mass-proxy-count	A mass proxy can vary with isotope and cannot establish entity cardinality.
measure	count-measure	unrelated-count-and-quantity	An unrelated quantity does not measure how many referenced entities occur.
support	generated-entity-support	single-presupposed-entity-kind	Presupposing one conventional entity kind excludes other generated elementary carriers.
record	member-and-reference-trace	total-without-members	A total without its entity reference cannot be independently reconstructed.
unit	separate-unit-comparison	unit-scale-as-proof	A conventional scale cannot select the amount law.
extension	no-extra-rule	free-amount-factor	An extra multiplier is a free answer-selecting parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:530a1ab514b800a0675a6f969439574a59117f367012e3ef8055b6cec3ebcc12.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:caafe4fbfd3413d2a4eeb2d4e1e893b744e6af78312e4a16209bccb295e91fef.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b4ba2f18586ba7dd2c311b1e9d1a1be3fbd7f3980668604bd20becae6b4cd02.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:939ec2105ede1067163abcfe2685d946baa15236b910a9e6f925a5d31ef0d46f.

Independent reconstruction. Implementation hash

sha256:7876fb1d2c9a700c5019ad446b918591401f0b872b74241cd017f0208c2f1a22; independent certificate sha256:64d8d926722782a15875c9eaf0a522aae7879683644af93d38b2082865374622; external-validation hash sha256:d2f7b45fd1f7add2b9d3206a58cca2999f9519aaecc41f2d3dafc8eddc94834a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00297-2026

Comparison records:

- amount-of-substance-iupac-a00297: predicted
amount-carrier__count-measure__specified-entity-reference__generated-entity-support; source-derived
amount-carrier__count-measure__specified-entity-reference__generated-entity-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC amount record lacks number measure, specified entities or general elementary-entity support, or if an altered row is accepted.

Measurement receipt: sha256:47cffd8c0f8e1ee747770cf58fb42a626bb62cba92c8b3431fdf548fd299db26.

Isolation certificate: sha256:2a8dd4298c04a59a3f32910aefd475e794e2f6d440c16dee3ac4770d0aac7d31.

Custody certificate: sha256:110d0ab7d20d5d6fb319b23299c4a88d5ca2f1eab0247796c4442bbe274af9e2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00297/json>
(sha256:ddf59230caa12b510125795af6f624797e904a1724fe970b3c144af47167077e)

Registered target identities:

- amount-of-substance-iupac-a00297 from IUPAC-GOLD-BOOK-A00297-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

Immutable identities. Source manifest

sha256:2583444b420c81a2c7b1d7ca3040e4d3801feafef2830b73ef1392be6cceece2; derivation seal sha256:e123e8f72a4f87a8188e392ea5f1e1526d778296998b7ba1cbec1429ffc0adc6; engine receipt sha256:c0dd23aedf3372bf9012540d9f80474ab7e34fa07b38664c9b0d3633f0cafa03 at receipts/engine/model_admitted/SFT-CHEM-MEAS-AMOUNT-001-c0dd23aedf3372bf.json; empirical hash sha256:d4e5ab18313a931276ffb357cb2851acd039d3ccd496b054715835c6474e3b01; measurement receipt sha256:47cffd8c0f8e1ee747770cf58fb42a626bb62cba92c8b3431fdf548fd299db26.

5. Chemical formula as exact composition encoding

Claim identity: SFT-CHEM-MEAS-FORMULA-001

Question and theorem. A molecular formula is the least decodable Fold word that retains a discrete molecule's constituent entity labels, exact positive multiplicities and any structure coordinate required by the declared identity boundary.

discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001

- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001

Generated grammar. Generate the literal product of the registered formula domain, carrier, constituents, multiplicity, mass, structure, decoding and extension choices.

Boundary: Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `discrete-molecule-domain__formula-encoding__held-entity-symbols__exact-positive-multiplicity__composition-mass-correspondence__structure-correspondence__composition-decodable__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
domain	discrete-molecule-domain	unbounded-object-label	A formula without a declared chemical carrier domain can encode unrelated objects.
carrier	formula-encoding	name-only-string	A name alone does not retain exact constituent multiplicity.
constituents	held-entity-symbols	anonymous-symbols	Anonymous symbols erase chemical-entity identity.
multiplicity	exact-positive-multiplicity	unrecorded-repetition	Unrecorded repetition cannot reconstruct composition or relative mass.
mass	composition-mass-correspondence	mass-unrelated-to-composition	A mass label not reconstructed from the constituent record is not a formula consequence.
structure	structure-correspondence	order-erased	Equal counts can represent distinct molecular structures.
decode	composition-decodable	nondecodable-inscription	An inscription that cannot return its constituent record is not an exact encoding.
extension	no-extra-rule	free-formula-exception	A discretionary exception can manufacture a desired formula.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3845fb5f1a0403ba69cc287efe987ff69f8c3e9f1acf7a480f9aea44f03fdadd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2a15902eb8c0b79689cca8acb12b7c410d6e39222d9f24e4135db5e49740ba4f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6d635c7023bcc25b4967611c84b32596233cfd7d5cf35afa38b87999e78d31a0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aa7e7ab21abb28659b496319db0ef2d0e3706b69d707305a9813d1f5ef243140`.

Independent reconstruction. Implementation hash

`sha256:16580eadc4660d4c0de70006350f24437e5dbafe427f71d0925f4cb62a62eb45`; independent certificate `sha256:cae7f79d06ad779f29123b54de65efd692475f4b1b10617a6ff41ce3049089a9`; external-validation hash `sha256:f5e253f8e69d5774f6dc6be627dbd002f626eabefc0d09c95519379d10b4246a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03987-2026

Comparison records:

- molecular-formula-iupac-m03987: predicted
discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence; source-derived
discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC molecular-formula record lacks its discrete-molecule domain, formula carrier, relative-mass or structure correspondence, or if an altered row is accepted.

Measurement receipt: sha256:c002bc140bf7f658b9bd74f3cdccdb5940d9de7b0d41e8a7788042e3c9f5d49d.

Isolation certificate: sha256:ec7b847ec7b441599c22fd7a69884695d8977e86eb2b7f7c1b53a531d74e402d.

Custody certificate: sha256:28d7f9b6ad63d9a800288da13612553c37c13bb97e838098fd9ae08952a13086.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03987/json>
(sha256:b33959596338e0e2797247c379e8214ecd6a58abd3e929a4dfad36f42b081704)

Registered target identities:

- molecular-formula-iupac-m03987 from IUPAC-GOLD-BOOK-M03987-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

Immutable identities. Source manifest

sha256:a27c10a35cee4c3af142d2a480cb927c84c9cb972fe22356725e274ed4e439ae; derivation seal

sha256:921e694badb6f35e3da05f404a690f9dc9c5762d5a3ad062c76027127895889; engine receipt

sha256:7cf0d7fa8f26712a49a5dab92f8019e46473a27e61cb1d94dc0ef8d98c861319 at

receipts/engine/model_admitted/SFT-CHEM-MEAS-FORMULA-001-7cf0d7fa8f26712a.json; empirical

hash sha256:366a61563aaefc5fb17c55e4e15ecd7faa08268006f6e77b74c2d1410f3fff56; measurement

receipt sha256:c002bc140bf7f658b9bd74f3cdccdb5940d9de7b0d41e8a7788042e3c9f5d49d.

6. Reversible chemical nomenclature and identity boundary

Claim identity: SFT-CHEM-MEAS-NOMENCLATURE-001

Question and theorem. A systematic chemical name is a complete ordered Fold word assembled from registered name parts and optional count prefixes; exact reversibility is admitted only on the generated injective name-to-identity boundary.

complete-name-carrier__generated-name-parts__count-prefix-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-FORMULA-001

Generated grammar. Generate the literal product of the registered name carrier, parts, prefix, order, formula binding, decoding, record and extension choices.

Boundary: Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-name-carrier__generated-name-parts__count-prefix-support__held-name-order__formula-bound-name__injective-boundary-decoding__complete-name-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-name-carrier	partial-name-fragment	A fragment alone does not carry one complete registered chemical name.
parts	generated-name-parts	unregistered-free-text	Free text can import arbitrary answer-bearing distinctions.
prefix	count-prefix-support	implicit-count	An implicit multiplicity cannot be recovered from the name.
order	held-name-order	unordered-syllable-bag	Unordered parts can merge names with distinct composition.
binding	formula-bound-name	name-detached-from-formula	A detached name cannot preserve chemical identity.
decode	injective-boundary-decoding	unconditional-reversibility	Not every conventional name is uniquely decodable, so reversibility cannot be assumed.
record	complete-name-trace	surface-name-only	A surface name alone hides its assembly and decoding boundary.
extension	no-extra-rule	free-naming-exception	A discretionary naming exception can select any desired identity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:363d36bc6f6549e4e0530b91da9cf081f8d8618f41d78e08b72d9307ca63e793`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5eed2098f5a9a626e7aa35bb5ca2da5d3c9ae4c9a5bf310a635b7bcfc898655e`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 213e9b19aa90d67471f010fc8b8919b0c63247f7c71195d457576cf1b0da6e6e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 19185c053f8a90242fa96295f5bcd2329eb4dee3c69ece49d5d51993607c075a.

Independent reconstruction. Implementation hash

sha256: 35e60d416791137fa320e94b41a12eed0e18a1099c1b862537158c24d45f4fb9; independent certificate sha256: 33534ecd55d79bd46e31fa8e85c98a78d1a0d48f961fa421774c1c17b1936a14; external-validation hash sha256: 8735d81e6460be0dc718bd57b42f95898b7419725482134f4100650b95f5dc67. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06236-2026

Comparison records:

- systematic-name-iupac-s06236: predicted complete-name-carrier__generated-name-parts__count-prefix-support; source-derived complete-name-carrier__generated-name-parts__count-prefix-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC systematic-name record lacks complete name composition, selected syllable parts or optional numerical prefixes, or if reversibility is claimed outside the registered injective boundary.

Measurement receipt: sha256: d712d539a814c6a82ea1bdb5448a4f07f9bd1c481211d208234b6d83f68c07e4.

Isolation certificate: sha256: 20ebbcbdf200ebc000e4026c0676d8fff1cfd1bfe8ccde37bda93fccfc5b884af.

Custody certificate: sha256: c5a80bffa574ea176c61f65735cbb6b9d06cbc34d56447c7423d725ed1d29d5.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06236/json>
(sha256:a3163fe9e9694283c7280d87679fb49286dd44631d908c19fcc164ce8f2984ac)

Registered target identities:

- systematic-name-iupac-s06236 from IUPAC-GOLD-BOOK-S06236-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

Immutable identities. Source manifest

sha256: 80c5d76427e08fd47b0cf27d682c64de46a3874b5b1e7e16dab7ce6a3c5da173; derivation seal sha256: 02607a2faf000f2fb512cdc4dc3e555a32eb5342469e3ac7cafcbad1dc54d4c30; engine receipt sha256: 33b780889dc8278c1dc64c63d16eb372d3222c6955f9b6c3f9077929411a4e8c at receipts/engine/model_admitted/SFT-CHEM-MEAS-NOMENCLATURE-001-33b780889dc8278c.json; empirical hash sha256: 7ab983fba64884599030fec4cbdda2fd8069b531d4e84be88f32238500a8be98;

measurement receipt sha256:d712d539a814c6a82ea1bdb5448a4f07f9bd1c481211d208234b6d83f68c07e4.

7. Chemical measurement uncertainty and retained alternatives

Claim identity: SFT-CHEM-MEAS-UNCERTAINTY-001

Question and theorem. Chemical target uncertainty is a source-bound Fold record of unresolved measurement alternatives with a preregistered upper limit fixed by intended use and attached to the exact result identity.

`uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary`

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-AMOUNT-001

Generated grammar. Generate the literal product of the registered uncertainty carrier, alternatives, bound, use, result, source, record and extension choices.

Boundary: Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `uncertainty-carrier__retained-alternative-class__upper-bound-support__use-bounded-resolution__result-record-boundary__source-bound-uncertainty__complete-uncertainty-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	uncertainty-carrier	single-answer-without-resolution	A lone answer erases the unresolved observation class.
alternatives	retained-alternative-class	discarded-alternatives	Discarding compatible alternatives falsely turns resolution into exact identity.
bound	upper-bound-support	unbounded-vagueness	Unbounded uncertainty cannot falsify a measured correspondence.
use	use-bounded-resolution	context-free-resolution	Required resolution depends on the declared purpose of the result.
result	result-record-boundary	bound-detached-from-result	A bound detached from its result cannot be applied or audited.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	source-bound-uncertainty	source-free-bound	A source-free bound can be changed after comparison.
record	complete-uncertainty-record	favourable-row-only	Selected rows cannot test uncertainty transport.
extension	no-extra-rule	free-uncertainty-margin	An adjustable margin can force agreement with any target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : f33978481bee18a785e521a82b83da05e66ad23051bf34af7e6ba8e6222893ac.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 2f46e0952e59ff9947a89d6942dd04a2175a7182e033d5c137c0d7d097b50848.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 259fc17f95d9dab542b48fca5a0d0362fabb280b7315b7130540653701ef78d6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : de5d0150aeeccffac4ac229cb1552a22b70e6afa6c0f27ecb72cdd228a34927.

Independent reconstruction. Implementation hash

sha256 : 8b0ba2478d537cb6f9383f894be69dab1485aecb826ef31b9205e556bdc6fbaf; independent certificate sha256 : e9f1432b80fd5ed55c607e74ee387292fbb941e02ed52f76f923b53cca2f70dd; external-validation hash sha256 : 21608d02fd9a335c3693b3f297002da11da3af14faf5c0376041274c73179df5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08133-2026

Comparison records:

- target-uncertainty-iupac-08133: predicted
uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; source-derived
uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC target-uncertainty record lacks an upper limit, intended-use boundary or measurement-result attachment, or if an altered row is accepted.

Measurement receipt: sha256 : 146e60e995906ada26b57276843c2fe43560331cda2643c439a385911896cda3.

Isolation certificate: sha256 : 549188b9c54e2df8fb1c36f07e85fdfcba7c45e10061c391c5ed4e056282630a.

Custody certificate: sha256 : 5592e7a03248c5b1b239fead57d09a5ae99899b692cf9f79db6ebaf29555aa28.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08133/json>
(sha256:014b6b034695f2b25c07cd4e5474b856aeeccf8907e4022595cc93df2f6674dc1)

Registered target identities:

- target-uncertainty-iupac-08133 from IUPAC-GOLD-BOOK-08133-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

Immutable identities. Source manifest

sha256:28c377ca52afd1ce1bf78e732ad37fa441f4dc6ad80d4c31eac50f86621b42b3; derivation seal sha256:158d9174013868a2a0218a64a4806c9891e067cecd1d40fc913c76878ff743c6; engine receipt sha256:32f136b934cb1114201bb7126bb4c83116018780973264d902def43937d23cbe at receipts/engine/model_admitted/SFT-CHEM-MEAS-UNCERTAINTY-001-32f136b934cb1114.json; empirical hash sha256:0637803ce25d0b17f78078fafe60eb873203497641208a42c08a71aabebdd6f1; measurement receipt sha256:146e60e995906ada26b57276843c2fe43560331cda2643c439a385911896cda3.

8. Chemical reference traceability and complete record

Claim identity: SFT-CHEM-MEAS-TRACEABILITY-001

Question and theorem. Chemical traceability is the complete Fold predecessor chain binding a property value to an identified reference material while retaining uncertainty, material identity and its certificate record at every link.

reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-CHEM-MEAS-UNCERTAINTY-001

Generated grammar. Generate the literal product of the registered property, reference, uncertainty, trace, certificate, identity, record and extension choices.

Boundary: Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `reference-material-property__identified-reference-realization__uncertainty-retained__traceability-retained__certificate-bound-record__material-identity-held__complete-chain-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
property	reference-material-property	value-without-property	A value without a property identity cannot be traced.
reference	identified-reference-realization	anonymous-reference	An anonymous comparison object cannot anchor a trace chain.
uncertainty	uncertainty-retained	uncertainty-erased	Erasing uncertainty overstates what each comparison preserves.
trace	traceability-retained	broken-predecessor-chain	A missing predecessor prevents reconstruction to the reference.
certificate	certificate-bound-record	uncertified-result-label	A result label alone does not bind property, reference and trace.
identity	material-identity-held	substitutable-sample	Silent sample substitution breaks traceability while preserving a surface value.
record	complete-chain-record	terminal-value-only	A terminal value cannot reproduce the chain.
extension	no-extra-rule	free-trace-link	An unregistered link can hide a discontinuity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 97e310c97ca4153e8085aa29d2b588832b6425e4220e59493ed9a11b6d0a9c48.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 0a72df6d765e4516c4b2e09d9092b1bcef4c9665423eb32a35aa1eec79899488.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b1151e9de01fe193c6b7934ff2bca5000776b28dc01480a985bb08dc9de5e007.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 6743deb0003e20c5af2dac5d032ad5d67023c98dd7446991ec75a06c01abee11.

Independent reconstruction. Implementation hash

sha256: 87aa49285efc58f739ef9b98990ae232ca5b846a3d052775904c353924df74bd; independent certificate sha256: 1de8c820992757a3ec3499dda3159ad7f6e874af66e63b62d67c21a873a17a8c; external-validation hash sha256: 5349788bb8f683ad56d0ec122803afd7568f82ded454f8dbc49a2605502ba541. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08013-2026

Comparison records:

- `certified-property-iupac-08013`: predicted
reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; source-derived
reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC certified-property record lacks reference material, associated uncertainty, traceability or certificate binding, or if an altered row is accepted.

Measurement receipt: sha256:bbaf754c42c20b65ad35ff2c5245f8058ec184990f96e3ae024e462c39f8f1dc.

Isolation certificate: sha256:be1c1fdcc2d815663fdaac62ebb81e2ccc704729c7b5406fafae95777dfaec0e.

Custody certificate: sha256:d1fcabab015c47623821e98f2026ef93d20f4cd20d464b5939139e5a30b5fce6.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08013/json> (sha256:34656c42ec7d741c00a0006b462487115d51dcd3d918d2fa64982ce100e24868)

Registered target identities:

- certified-property-iupac-08013 from IUPAC-GOLD-BOOK-08013-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

Immutable identities. Source manifest

sha256:ad5f1df58a9beab76e6e3034a81a8e2e8fd65d318f6467bddc09d6f0ef21918b; derivation seal

sha256:23fcc09debac7c710bfa5bbbed6d301d1229db7fc994fc4202eb4755cc2ff8a86; engine receipt

sha256:7482443dfef4205a7bcd2a89f0ac89014486ab7f8d960a0e0821f917247108fe at

receipts/engine/model_admitted/SFT-CHEM-MEAS-TRACEABILITY-001-7482443dfef4205a.json;

empirical hash sha256:1eff947400af8d1bc6293865d4476377c46cc6df48de302129f7e37dbca253c6;

measurement receipt sha256:bbaf754c42c20b65ad35ff2c5245f8058ec184990f96e3ae024e462c39f8f1dc.

10. Elements Periodicity

Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries.

9. Chemical element identity

Claim identity: SFT-CHEM-ELEM-ELEMENT-001

Question and theorem. A chemical element is the complete Fold class of generated atoms sharing one held nuclear proton count; isotope mass refinements preserve that identity, while pure-substance use remains a separate correspondence.

atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001

Generated grammar. Generate the literal product of the registered element carrier, identity, location, membership, isotope, substance, record and extension choices.

Boundary: Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atom-species-carrier__common-proton-count__nuclear-identity__all-generated-atoms-with-identity__mass-refinement-permitted__substance-carrier-correspondence__complete-nuclear-membership-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atom-species-carrier	substance-name-only	A substance name does not define the atom class.
identity	common-proton-count	mass-selected-identity	Mass can vary between isotopes of one element.
location	nuclear-identity	protons-without-nuclear-carrier	An unlocated count does not preserve atomic constitution.
membership	all-generated-atoms-with-identity	sampled-atom-list	A sample cannot close the complete generated class.
isotope	mass-refinement-permitted	one-mass-number-only	Restricting an element to one mass number erases its isotopes.
substance	substance-carrier-correspondence	species-substance-conflation	An atom species and a pure substance are related but not identical carriers.
record	complete-nuclear-membership-trace	element-label-only	A label alone cannot reproduce class membership.
extension	no-extra-rule	free-element-exception	A discretionary exception can split or merge element classes.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:76dd7261cbd3b67042302e0d1db175f36c10d018ae297517613179d05638a9d5`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:13b7765dcc0f65b3252315344aad5c8a70c3fc576d445f435ec090dcd81e8e61.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:09f6d83fe72cd5c693cec7b72ad03d414521081d8b0934b0289a7580b0df232f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f00796ad6f37a439619cea9a44469afcd69aa1d406af0949ab0b2172048d418a.

Independent reconstruction. Implementation hash

sha256:099700048127608e9d9b7d95a855a9c6cf5f79d2e8a37ebc5198fadbba1ca6b5; independent certificate sha256:be152c894873bde8bc0c4fa13caed3619dcde05f4a97b6d1351f0fcbf5bc3027; external-validation hash sha256:a33fb7f0513f61005f561158de442bccb71f5d3a62ea43fe1bd3bf86ee76b87a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01022-2026

Comparison records:

- chemical-element-iupac-c01022: predicted
atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; source-derived
atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC element record lacks an atom-species carrier, common proton count, nuclear identity or pure-substance correspondence, or if an altered row is accepted.

Measurement receipt: sha256:f79af5351c60739735dc774cc20043ed452b76ee03ea6088cc064d01fac53829.

Isolation certificate: sha256:26eb6acd9dbf27a6b226638bb7d6b485f8a05e4cb6c6337002c95c4338461d43.

Custody certificate: sha256:f030e4eea55cdd5a6703a715589e085ad181ff0a9ea72af42c408e7df840fc90.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01022/json>
(sha256:6d75c5b2d5299402b7f35243e248cb60d4972553d3bb622490236e7b889d9575)

Registered target identities:

- chemical-element-iupac-c01022 from IUPAC-GOLD-BOOK-C01022-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

Immutable identities. Source manifest

sha256:b9d75c92672c039d7f4a5deb483bcdfe8ee1a6f61a2c8a0fe13853de6f027e83; derivation seal

sha256:1aa52affee6653414842ec851ef6a5e55473108f1638ef8939184c330e1bac21; engine receipt sha256:ea7aa6a543e4710c2901ad39a6d67445bebda012e735322cff802f05720434d7 at receipts/engine/model_admitted/SFT-CHEM-ELEM-ELEMENT-001-ea7aa6a543e4710c.json; empirical hash sha256:6977b7243cf2ec9f6c028016233c7ba0fbb3bba0bbb789a00450822f9a1c30af; measurement receipt sha256:f79af5351c60739735dc774cc20043ed452b76ee03ea6088cc064d01fac53829.

10. Atomic-number ordering and element distinction

Claim identity: SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Question and theorem. Atomic number is the exact generated positive-finite count of protons in an atom's nucleus and therefore the unique order coordinate of its chemical-element class.

exact-proton-count__nuclear-count-carrier

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-ELEM-ELEMENT-001

Generated grammar. Generate the literal product of the registered atomic-number carrier, location, order, element map, isotope, charge, record and extension choices.

Boundary: Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `exact-proton-count__nuclear-count-carrier__positive-count-order__one-count-one-element-class__isotope-invariant__charge-state-invariant__count-nucleus-element-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	exact-proton-count	symbol-position-only	A table position cannot define the nuclear quantity.
location	nuclear-count-carrier	whole-atom-particle-count	Counting every atomic particle changes across isotopes and charge states.
order	positive-count-order	unordered-label	An unordered label cannot supply periodic succession.
element_map	one-count-one-element-class	many-elements-one-count	Two element classes with one proton count violate their identity condition.
isotope	isotope-invariant	mass-dependent-number	Mass-number variation must not alter element order.
charge	charge-state-invariant	electron-count-dependent	Ion formation changes electrons without changing the element.
record	count-nucleus-element-trace	number-without-nucleus	A bare number loses the physical carrier and class mapping.
extension	no-extra-rule	free-order-offset	An offset can relabel the periodic order arbitrarily.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 4ddd54cd7947aa0e3b2aa3d03493a8c34c888edcfc7dea36e9282471cc6c308c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ed80251591ba927846d8a1247ed87eb592801a08102e5dbde5cac482563e7ef7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : be2dd24a591be28884b941432fdaefc7b12120c27fd0008b55795cad15fab594.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : db5a33c3d4c0529fb072f64aaf74b52b89281189de1ad94dc76b985ad0466b81.

Independent reconstruction. Implementation hash

sha256 : a562cb4295bee04704e6d6f298f9400a671a00306a4c3e99f78b3192e9dec8c7; independent certificate
sha256 : 50a96aa5a9fc736f9b208bcbbaa5ad98be89a75d160feb645174f208c0c28123d; external-validation
hash sha256 : 6efd610e28967cfbb5efa2627358944c4d44a76332c5f1b1e98230c81ccf9abe. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00499-2026

Comparison records:

- atomic-number-iupac-a00499: predicted exact-proton-count__nuclear-count-carrier; source-derived
exact-proton-count__nuclear-count-carrier; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC atomic-number record is not the proton count in the atomic nucleus, or if an altered row is accepted.

Measurement receipt: sha256 : f59895f2b153d17da12ab9b09f875868e0451fc789fb53d6a979dc8bf3ad9eaa.
Isolation certificate: sha256 : b9634abd89c1680c95f58fe81bd47e7fb90589acb8fd9f0abed916127fa01fd.
Custody certificate: sha256 : dff8feac17003dea3a7af44fa7a350a6e945797a2d2d6a2dd92179d4953c3605.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00499/json>
(sha256:6ba53740329bcdde38d8a8734282fb9414a1c7e62fad09743d0890890ff3e127)

Registered target identities:

- atomic-number-iupac-a00499 from IUPAC-GOLD-BOOK-A00499-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

Immutable identities. Source manifest

sha256:94777a429f8e0c22d9c3e5647936c8340570eca54e9f73a2c1d4e45cb3ea77f4; derivation seal sha256:7cd1aa9a886c6f4665acd5de9967d9c13b46eb3079c9f2e78e7e16e251904b22; engine receipt sha256:d0fccb17f5ec07aefad94d4b1118dc69e8e90be2ed1b2c9f742215b0fc9e50c6 at receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-NUMBER-001-d0fccb17f5ec07ae.json; empirical hash sha256:4a059aea0ff9e110133f3beafc0986860757123ac9a632aece87e9dea0a6b421; measurement receipt sha256:f59895f2b153d17da12ab9b09f875868e0451fc789fb53d6a979dc8bf3ad9eaa.

11. Isotope identity within an element

Claim identity: SFT-CHEM-ELEM-ISOTOPE-001

Question and theorem. Isotopes are the generated Fold family of nuclide carriers sharing one atomic number while retaining different exact mass-number coordinates.

nuclide-carriers__element-identity-retained__mass-number-distinction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001

- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Generated grammar. Generate the literal product of the registered isotope carrier, element, mass, complementary nuclear count, chemistry, observation, record and extension choices.

Boundary: Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `nuclide-carriers__element-identity-retained__mass-number-distinction__complementary-nuclear-count-held__common-element-chemistry-boundary__resolution-bounded-distinction__complete-nuclide-family-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	nuclide-carriers	atom-label-without-nucleus	An atom label alone does not establish nuclide identity.
element	element-identity-retained	mixed-atomic-numbers	Different proton counts are different elements, not isotopes.
mass	mass-number-distinction	mass-number-erased	Erasing mass number merges distinct nuclides.
neutron	complementary-nuclear-count-held	unrecorded-nuclear-difference	The source of mass-number distinction cannot remain anonymous.
chemistry	common-element-chemistry-boundary	different-element-chemistry	Changing element identity does not define an isotope relation.
observation	resolution-bounded-distinction	always-indistinguishable	Isotopic distinctions are resolvable in registered measurements.
record	complete-nuclide-family-trace	isotope-name-only	A name cannot reconstruct proton and mass coordinates.
extension	no-extra-rule	free-isotope-equivalence	An arbitrary equivalence can merge different elements or erase mass distinctions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8b484361d78650d9a8282675607175a29a3f07500429fe30cdbc8be83fb806e8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d12deb5f406b68526427218796918b712da246387d65d777d60f2500d0d4f4a0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3b803889d00f08f745238ff5c3ba28066c73e87c0d9722104dc3c1b7ef514893`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aca608666ff8ba3aa7e88511159717815c73d806611043bb4da1be7d86e7cc4b`.

Independent reconstruction. Implementation hash

`sha256:d5f0ab0003ff6099b3b93cd20d190c1ff1c2b231255d493a8e3275e5c580c8d8`; independent certificate

sha256:79130b42b41148a1bc0c2d74d50493a322b3eba6375c8a84a6f992c33a84db56; external-validation hash sha256:b9c97d6ebf639546d9cd99cc7a53152fec627e8b5c36f65a93654d4e6684133c. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03331-2026

Comparison records:

- isotopes-iupac-i03331: predicted nuclide-carriers__element-identity-retained__mass-number-distinction; source-derived nuclide-carriers__element-identity-retained__mass-number-distinction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC isotope record does not retain nuclides, common atomic number and distinct mass numbers, or if an altered row is accepted.

Measurement receipt: sha256:92be2b66a7789623a970c015af9c8534dd7e798570078bee0e560045632cd1b5.

Isolation certificate: sha256:2dfcbbd8c82ae8a225a018b4947bcd82c6cbdc21ec302d88b7fa8a580f2fb222.

Custody certificate: sha256:b9649c33739d67c8af62b4c078d4a74c7aa85d22a1d618b223988a5ed0d70f09.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03331/json> (sha256:7b5ca7fe293902804ca76194a6ac35a57a35bfa489245c93979eb81370a6635c)

Registered target identities:

- isotopes-iupac-i03331 from IUPAC-GOLD-BOOK-I03331-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

Immutable identities. Source manifest

sha256:3313ef2e5af4b187fd2ef43be40fb53c248fbfd67283100d91b1cf219f569470; derivation seal

sha256:d6dd20174811be1754c06fe8d2ad4de01aec548b5656a840206850b8e3cb051b; engine receipt

sha256:814344a440e051068f0e54e91276fddd45b2d388ea1fe438f630e3aae0a4b28f at

receipts/engine/model_admitted/SFT-CHEM-ELEM-ISOTOPE-001-814344a440e05106.json; empirical

hash sha256:8a4eede1026a036dc1160429f93a3ad2c26572173e864dc6c9c83299ab551304; measurement

receipt sha256:92be2b66a7789623a970c015af9c8534dd7e798570078bee0e560045632cd1b5.

12. Atomic-weight record and isotopic composition boundary

Claim identity: SFT-CHEM-ELEM-ATOMIC-WEIGHT-001

Question and theorem. Relative atomic mass is the exact source-bounded Fold ratio of a finite atom population's retained average mass to a separately registered unified atomic-mass reference, with isotope support and measurement uncertainty preserved.

exact-ratio-carrier__average-atomic-mass__reference-unit-comparison

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-QUANTITY-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-ELEM-ISOTOPE-001

Generated grammar. Generate the literal product of the registered atomic-weight carrier, population, isotope support, mass pairs, reference, source, uncertainty and extension choices.

Boundary: Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `exact-ratio-carrier__average-atomic-mass__isotope-support-retained__isotope-mass-pairs__reference-unit-comparison__source-bounded-population__measurement-interval-retained__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	exact-ratio-carrier	decimal-answer-only	A decimal answer hides population, isotope and reference provenance.
population	average-atomic-mass	single-atom-mass	Atomic weight concerns a population average rather than an arbitrary single atom.
isotopes	isotope-support-retained	isotope-mixture-erased	Erasing isotope support loses the source of population variation.
mass	isotope-mass-pairs	unbound-mass-labels	Unbound masses cannot be paired with their isotope parts.
reference	reference-unit-comparison	unit-as-derived-law	A conventional unit scale cannot select the structural ratio.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	source-bounded-population	universal-fixed-composition	Natural isotope composition can vary by source.
uncertainty	measurement-interval-retained	exact-decimal-pretense	A reported decimal must not erase its measurement interval.
extension	no-extra-rule	free-weight-adjustment	An adjustable correction can force a measured target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:90d954fc2cd84571376d947dca60f0e84ee9fb1325ede38bfd707aa41ef2ea0f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ff64c465529ab6a5c91e2ea3023965c9e499826385218803b3983644b4974ce.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c341944a48f79104fa347f17ecb18ceeeddad0388d4e720e6ad69745b6e70ee8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fc1f589db5a76a8aadd4fd38cd7cce85db80eced58a1aa027d11df0c552742eb.

Independent reconstruction. Implementation hash

sha256:039223c857c830d8414ee0110d8b2216a5f2c4f22714d0c2242d71e34dbb29e2; independent certificate
sha256:a0c1484aa9c368048506eda8a5164f4678cc44bcde54be62e38a94bbcb126bb9; external-validation
hash sha256:4969b91031a41ce1744454b4f52e0287a9002aed4ddf700ec0a25694f1a7884a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-R05258-2026

Comparison records:

- relative-atomic-mass-iupac-r05258: predicted exact-ratio-carrier__average-atomic-mass__reference-unit-comparison;
source-derived exact-ratio-carrier__average-atomic-mass__reference-unit-comparison; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC relative-atomic-mass record lacks a ratio, average atomic mass or unified atomic-mass reference, or if an altered row is accepted.

Measurement receipt: sha256:21313a418ae7996bdf96d2506cc95bf578404e55d8ba7b5fd7e33972d4a37041.

Isolation certificate: sha256:7ed3d2c02eb1da57572bb9cd3be50d176e74daf990e644f474862c07416d43fd.

Custody certificate: sha256:4ddd880e74630e1a4f4488a4d90669b6fefff6fc9230481d81b8b22be693659e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05258/json>
(sha256:9a08b1dfe8699f2be546b1a0fcb9b359d49c78419099d853603c6c750a04f17a)

Registered target identities:

- relative-atomic-mass-iupac-r05258 from IUPAC-GOLD-BOOK-R05258-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

Immutable identities. Source manifest

sha256:5a744926c46b43835db76bd0e8deabed8ca4561e5c72675601f4ace437897719; derivation seal
 sha256:114a1de5d3715ee6d73bc7400aa8cb90e6e928a6bb500dabf706da100c64d279; engine receipt
 sha256:611aeb63a9ed5877d30862065a427b279c52aa8cd1f3b94bc81920643951fe4a at
 receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-WEIGHT-001-611aeb63a9ed5877.json;
 empirical hash sha256:3b2b46429f7461bc13d9eff3a61aaac08e2eed4fc365830d70e8a5e8e7e601db;
 measurement receipt sha256:21313a418ae7996bdf96d2506cc95bf578404e55d8ba7b5fd7e33972d4a37041.

13. Periodic ordering by retained nuclear identity

Claim identity: SFT-CHEM-ELEM-PERIODIC-ORDER-001

Question and theorem. Chemical elements possess one invariant exact total order: the generated positive proton-count order, with the complete element class retained at each coordinate.

complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Generated grammar. Generate the literal product of the registered periodic-order carrier, coordinate, order, identity, succession, observation, record and extension choices.

Boundary: Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained__one-proton-successor__post-seal-known-table-correspondence__number-element-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-element-class-support	names-without-element-records	Names alone do not preserve the generated element classes.
coordinate	atomic-number-coordinate	layout-selected-coordinate	A drawn location can be changed without changing nuclear identity.
order	positive-count-total-order	unordered-element-set	An unordered set cannot express successor structure.
identity	element-identity-retained	order-erases-isotopes	Ordering must not discard refinements of an element record.
succession	one-proton-successor	free-table-rearrangement	A discretionary rearrangement can select any pattern.
observation	post-seal-known-table-correspondence	known-table-as-proof-premise	The observed table cannot create the order law.
record	number-element-source-trace	atomic-number-only-list	A number list loses symbols, identities and source boundary.
extension	no-extra-rule	free-order-exception	An exception can silently permute successors.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: fb79be00e6697a390e2401a3f35d4a36e790dd0e25f17ce2d82cdf109459fdc8.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2468f6a821c5f46777ffec9979273168f4fb4a44c9a3b07a459d3885cd591fe4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: fcd4a72f1f0a76bcc6ab20b90e22587f439fd13e32a9a32b98ab2c2a256b2cf2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 8f2e3cc1418f0d11a932467a232d2806e7e2ad150c76df6cbf655f22d58a3a91.

Independent reconstruction. Implementation hash

sha256: 1ab5b9c59e4bf934778ced8adb3532d3129d525e09be9ba7215d8b53249aee1d; independent certificate
sha256: e3e8d00b305566a1ce3a029a1b2fe23d581ff06b4c904848b79cadf71121781e; external-validation
hash sha256: f4f6cbb70c7fd3237d454f6a746fed076402174d152c4fea167c0db37d1888f3. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-order-iupac-2022: predicted complete-known-element-support__atomic-number-coordinate-order; source-derived complete-known-element-support__atomic-number-coordinate-order; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official source lacks the complete registered element support or atomic-number coordinate, or if a changed row is accepted.

Measurement receipt: sha256:d6bf48b62ba038232ce2da4f0d9f6a776405f68b04b8301f248c1548ef4dc775.

Isolation certificate: sha256:e98f2de32003bb5ea5bf83bdc912a7f748ac5a5ab36346c62098915ea1e67efa.

Custody certificate: sha256:a8c2761729e8122a72c1aea1ee0e98da0f0a8d830a7aa76e0f89507c316fa22c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
 (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-order-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

Immutable identities. Source manifest

sha256:9b2389237f9ad69dd83892791369de88baee13f1c90a5209de80dac6286d0332; derivation seal

sha256:6885e8766a0e478de2a057c2c68db53263289d2a942ba9e293b0539c7796c37c; engine receipt

sha256:2c3e161729975fa71a7baffe345e886c873fdec44e54e05547ba1ce105a10b4c at

receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-ORDER-001-2c3e161729975fa7.json;

empirical hash sha256:43e7886c804f8cd2df62f76d209094f5861f7ed55856611bc38f3b33a8da9dc3;

measurement receipt sha256:d6bf48b62ba038232ce2da4f0d9f6a776405f68b04b8301f248c1548ef4dc775.

14. Periodic recurrence of outer chemical organization

Claim identity: SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001

Question and theorem. Periodicity is the return, along atomic-number succession, of an earlier complete outer chemical observation class generated by finite bound-state recurrence and exclusion-preserving occupation.

boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__return-of-observation-class

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001

Generated grammar. Generate the literal product of the registered recurrence carrier, progression, equivalence, closure, occupation, periodicity, record and extension choices.

Boundary: Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__finite-bound-state-recurrence__exclusion-preserving-occupation__return-of-observation-class__support-transition-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	boundary-closed-outer-support	property-name-series	A list of property names contains no physical recurrence carrier.
progression	atomic-number-successor-progression	mass-selected-progression	Isotope mass does not determine element succession.
equivalence	complete-chemical-trace-equivalence	visual-column-equivalence	A drawing cannot establish common chemical behavior.
closure	finite-bound-state-recurrence	unbounded-orbital-continuum	An ungenerated continuum violates the finite exact domain.
occupation	exclusion-preserving-occupation	duplicate-state-without-trace	Unrecorded duplicate occupation violates exclusion.
periodicity	return-of-observation-class	assumed-fixed-period	A fixed imported period can manufacture recurrence.
record	support-transition-recurrence-trace	recurrence-label-only	A repeated label does not reproduce the equivalence test.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extension	no-extra-rule	free-shell-rule	A discretionary shell equation would import the desired organization.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 79f9f2d5c12e6b82db58e3e375f1eacc44ecb05fab2fa50516b603680f28412c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 28b430eac9c449e5d0a8d0fb4bba092ed7abb057120feb4adf7acffbdcc2322d4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: bbc1a7eb2265d08c410bfba1e8e59977b6d22242f4ecc31d573126547e3564e9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: beaad842b2bffc5a9a6ff2d5afadcabaa2e10ef7901d916f73c931394e1d43f4.

Independent reconstruction. Implementation hash

sha256: f41c5f8015683765486787511a50e71e5470ce686a2d01b82e98842cc0aa0c64; independent certificate
sha256: 497e7cb1c9504707525c356e731610f93b86140f319380641e6854ed1e9e1d85; external-validation
hash sha256: f1f1dc9ad812144cd280bd0e7ec8f882baf24b26f8d7d38e0fa49d7e51b48229. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-recurrence-iupac-2022: predicted recurrent-element-organization__ordered-periodic-table-support; source-derived recurrent-element-organization__ordered-periodic-table-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official periodic source lacks recurrent ordered element organization at the registered boundary, or if a changed row is accepted.

Measurement receipt: sha256: 5e525ae34da77af4a8ca56a8c5d781501603c089b924bdf38b98d7cb0622fce.
Isolation certificate: sha256: b4ed887f73fbdd06b09c8c63fec1e2ac3c9451e2669f433cede0ded45da8430d.
Custody certificate: sha256: 3ba444f8def3a520da6c2db126b1bcad5e56af2490e26a447820074181e0a859.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256: ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-recurrence-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

Immutable identities. Source manifest

sha256:a495a7d16184cb6a720db529dcc86261f9e6149e3686d05c78e502d3a3c7b46e; derivation seal
 sha256:c1593f4d94b272c104efa540ff269c897523d2841b7630baf0880f96ea778779; engine receipt
 sha256:c14383f6326b6c3cb950e80d48e1864d60ac7a8e794272abd80664e35338ca83 at receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001-c14383f6326b6c3c.json; empirical hash
 sha256:46ace4f08e9f48c8e0f447369c6a46466de8403197d7accc83bf7aec4639c08e; measurement receipt
 sha256:5e525ae34da77af4a8ca56a8c5d781501603c089b924bdf38b98d7cb0622fce.

15. Group and period compositional coordinates

Claim identity: SFT-CHEM-ELEM-GROUP-PERIOD-001

Question and theorem. A Fold group is an outer-chemical-trace equivalence class and a Fold period is its counted closure-to-reopening depth; their pair is independent of how a table is drawn.

periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001

Generated grammar. Generate the literal product of the registered coordinate carrier, group, period, pairing, ordering, equivalence, record and extension choices.

Boundary: Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate__atomic-number-order-retained__complete-trace-quotient__element-order-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	periodic-element-carrier	printed-cell-coordinate	A page coordinate is representational rather than chemical.
group	outer-trace-equivalence-class	column-number-as-premise	A conventional column number cannot select equivalence.
period	closure-reopening-depth	row-number-as-premise	A conventional row number cannot select recurrence depth.
pairing	joint-group-period-coordinate	group-only-or-period-only	Either coordinate alone loses a required distinction.
ordering	atomic-number-order-retained	layout-dependent-order	Alternate drawings must not alter chemical coordinates.
equivalence	complete-trace-quotient	namesake-family-only	A family name cannot prove matching transition support.
record	element-order-recurrence-trace	coordinate-without-witness	A coordinate without a recurrence witness is not auditable.
extension	no-extra-rule	free-layout-exception	A layout exception can move an element without a chemical distinction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c3f412be7e6caf1ae399956b649c37ddf393710eddf9fdb12016eea631e0b019`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:98da74bd924801244f5e8caa8728810ec0af692816c78ff6a8d630faccfb69f0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:60a6d8de4a10afab9cd43595398e1dead9b886863e69e3047bce476d9cfe5b9d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:137fa1bbe50de52fb9ced8a59a7e7c9aaba62081905c588226eb4b55378df7bd`.

Independent reconstruction. Implementation hash

`sha256:70bed5c2690f2e67d9c06dae6a5d91346efc2ffdd3a831a7be754f51740875b3`; independent certificate `sha256:f4762e4f73f2b81be48db4aee3269731dd14ed8cf2cd2ee5f6e57bb13db1c7f2`; external-validation hash `sha256:5825dc7247ed53812a05f89b525015475d9d597df304ef669aa1af067f466a4e`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- group-period-iupac-2022: predicted group-column-classes__period-row-progression__atomic-number-order-retained; source-derived group-column-classes__period-row-progression__atomic-number-order-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official table does not retain column classes, row progression and atomic-number records together, or if a changed row is accepted.

Measurement receipt: sha256:308a30c9c2357b6e6e2a1f5b0378a1dcb16d5f3230abf0ccb108b961ce6a8d77.

Isolation certificate: sha256:ee7c666ade956cf327e9ba9cca9ac4e4ffe4ee9eb946355a7afb830040fc789b.

Custody certificate: sha256:8317dd343d712159cc95d6ce534d6dafc844faf4122cf3c88bd3066033bd4ae5.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- group-period-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

Immutable identities. Source manifest

sha256:c87091d13914ec3c6740b16efe6029eb769b82f253673ab88a8aa14409ee1e00; derivation seal

sha256:181bc9d5fdf0cbb0dd587f7a8706a9cc4cfd0ea94d104f00073dd582099f5641; engine receipt

sha256:9e7a0fd48fc23d0b8ec3bb2656b1ba78fe5d760aea358e51c897a1ae9bd630da at

receipts/engine/model_admitted/SFT-CHEM-ELEM-GROUP-PERIOD-001-9e7a0fd48fc23d0b.json;

empirical hash sha256:67dd185229d14dc8365a73b060046f333914e105df3bc20bd4361037811afd89;

measurement receipt sha256:308a30c9c2357b6e6e2a1f5b0378a1dcb16d5f3230abf0ccb108b961ce6a8d77.

16. Valence availability and chemical combination boundary

Claim identity: SFT-CHEM-ELEM-VALENCE-001

Question and theorem. Valence is the exact greatest realized univalent joining or substitution count in the complete generated combination support of a retained atom or fragment and chemical context.

atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001

Generated grammar. Generate the literal product of the registered valence carrier, support, unit, selection, substitution, context, record and extension choices.

Boundary: Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count__combination-or-substitution__chemical-context-retained__combination-maximum-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atom-or-fragment-carrier	element-name-only	An element name contains no combination witness.
support	complete-generated-combination-support	assumed-valence-number	An assumed number imports the answer.
unit	univalent-partner-count	mixed-partner-count	Partners of unrecorded multiplicity cannot define one comparison.
selection	maximum-realized-count	typical-count	A typical value can omit a lawful larger combination.
substitution	combination-or-substitution	combination-only	Valence also constrains lawful substitution for the carrier.
context	chemical-context-retained	universal-context-free-number	Chemical environment may alter available combinations.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	combination-maximum-trace	number-without-traces	A bare number cannot reproduce its maximum witness.
extension	no-extra-rule	free-valence-correction	An adjustment can force a conventional value.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5806447da7a755ea31ac1d6fd82ead9fe731b639e24caddb330920603419c201.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c1fe82f9e1c25112465debbf7a0addc85fad42abe7789fe1bd67cb615eaf18d2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:091cb85be6fa237c253e1c4bc53ae5f474f1aba2031682b507ccab02ce6a5752.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:549fd50e5b28eee9a33c5d54cf5a6e7e30bd1ea06f63e8cbcb36d869dd1c2b78.

Independent reconstruction. Implementation hash

sha256:65ca8bb5246cd20eb6c823f8fffd231729d9340a5330aca2a3c2b19751167f6d0; independent certificate sha256:40019dd01e7a18c83373b3ae05cfe65130dac33c549325a093ecdb169e7d389d; external-validation hash sha256:f9beaa50054128034e329f57e42c26db09c5cc90d10f50e1123d3039fa80310f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-V06588-2026

Comparison records:

- valence-iupac-v06588: predicted maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; source-derived maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC record lacks a maximum univalent combination count, atom/fragment carrier or substitution boundary, or if a changed row is accepted.

Measurement receipt: sha256:1f1d5e7911ec5bd684094de0543366afaa5146e4dfe1dde279d608bf1913a310.

Isolation certificate: sha256:f0e86d8677ea4dea42bcda19d6aebe2ddddd42cea345f8919cbbba017f42eec2.

Custody certificate: sha256:0dc036295c288a80524fc61eaaaf11c6ab8be89ced27645f1423ae9b13a0601c2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/V06588/json> (sha256:99ae406a9ac89a8f0a89bf3c775ff9ba7c5d9d0c96f89f83a57a0b35c76c4077)

Registered target identities:

- valence-iupac-v06588 from IUPAC-GOLD-BOOK-V06588-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

Immutable identities. Source manifest

sha256:51369027d90ed6e989ad967b42c4796457065baaee933c83d46692ec406ce46c; derivation seal sha256:18a74251e5eca02417c3febb4357618634c5aa241a8e33bee56d281bd287f0b1; engine receipt sha256:2110f0a13f9081622880753f9a8c1eb5948486d92b8376d780b8e428eb7abd61 at receipts/engine/model_admitted/SFT-CHEM-ELEM-VALENCE-001-2110f0a13f908162.json; empirical hash sha256:305f7d43aa0e56ee708a9301489fa21ca684838a4fd117b07a946550fec5be90; measurement receipt sha256:1f1d5e7911ec5bd684094de0543366afaa5146e4dfe1dde279d608bf1913a310.

17. Ion formation with retained elemental identity

Claim identity: SFT-CHEM-ELEM-ION-001

Question and theorem. An ion is an atomic or molecular Fold particle whose closed electric-label transfer leaves a net held charge orientation while retaining its nuclear and chemical identity.

atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-VALENCE-001

Generated grammar. Generate the literal product of the registered ion carrier, charge, formation, conservation, element, orientation, record and extension choices.

Boundary: Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer__atomic-number-retained__cation-anion-held-fibres__carrier-charge-transfer-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atomic-or-molecular-particle	charge-label-without-particle	A charge label detached from matter is not a chemical ion.
charge	net-held-electric-charge	signed-proof-number	A signed proof magnitude violates the positive exact domain.
formation	label-transfer-with-identity-retained	identity-destroying-change	Destroying the carrier does not create a charge state of it.
conservation	closed-charge-transfer	unpaired-charge-creation	Unpaired creation violates closed label flow.
element	atomic-number-retained	proton-count-change	Changing proton count changes element identity.
orientation	cation-anion-held-fibres	positive-negative-proof-scalars	Signed scalars are not SFT proof objects.
record	carrier-charge-transfer-trace	ion-name-only	A name cannot reconstruct carrier, charge and transfer.
extension	no-extra-rule	free-charge-exception	An exception can create or erase charge without transfer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2ae6650a4e3754c24fb9a4f8f09536c51e7b81ac2b96455c63b56e597cb94fe0.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3b2456e07702e7ce9cbe5e02573a4d787dcd52859d789ab8bbc3c418bc1633ae.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: df200a2b7556795d20a2d92057fdcb90f979904158d594e9d314cd2223d281f1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 75999b23650345a5d5e4effbad9ba3bbf7ec85d09e0d64c8e675cc381cb72a3c.

Independent reconstruction. Implementation hash

sha256: 87f51143641461df3f93e8bc4ab0dc75357216b47b04bf5df604d469ee8f223f; independent certificate sha256: 7f6be160af0765a688078df30a929390192b67361cb72598c39e46183ca9d4863; external-validation hash sha256: 38523bcd1b0ff10c6c6cfe24dc3a64d839a02d31534d387d63e01c1bc1e15be2. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03158-2026

Comparison records:

- ion-iupac-i03158: predicted atomic-or-molecular-particle__net-electric-charge; source-derived atomic-or-molecular-particle__net-electric-charge; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC ion record lacks an atomic-or-molecular particle and net electric charge, or if a changed row is accepted.

Measurement receipt: sha256:124ef271dda53df2fd9cf81cc15dc01fdc9c2339d437f79cded6f0d634a8dec3.

Isolation certificate: sha256:85d33896465594805f1d8c7ce9f57f543a261bf737ed8bc8462562b533374859.

Custody certificate: sha256:78024287855816755f0c6c958ba47ffc77b434abd705cc8217983ba785eb190f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03158/json> (sha256:6b54a0f6d9cf83f30251501b58b75764a39396e56d819b420b1556476f9f13f6)

Registered target identities:

- ion-iupac-i03158 from IUPAC-GOLD-BOOK-I03158-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

Immutable identities. Source manifest

sha256:664be51c66193e7f6227d15f9fa7dce3ae3d0c44235d204918cc3e9dac4696a2; derivation seal

sha256:6560d04c7ffdaec794c13af0400a37d9632dadfe10d59b2144e84036b3057e89; engine receipt

sha256:9015e9496cd2acac1ded1af87e474a69cf1d8cad0eb15f6f928464b761148c8f at

receipts/engine/model_admitted/SFT-CHEM-ELEM-ION-001-9015e9496cd2acac.json; empirical hash

sha256:169141dd9bda0ebe1813a601cb3a190c7e2d13dbb50df93f9277731449b3e410; measurement receipt

sha256:124ef271dda53df2fd9cf81cc15dc01fdc9c2339d437f79cded6f0d634a8dec3.

18. Known periodic-table observation boundary

Claim identity: SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001

Question and theorem. The known periodic boundary is the complete source-dated IUPAC element census through its greatest listed atomic-number coordinate; it is an observation boundary and not a theorem forbidding successors.

complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__unobserved-successor-open

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-GROUP-PERIOD-001
- SFT-CHEM-ELEM-ION-001

Generated grammar. Generate the literal product of the registered known-boundary carrier, extent, identity, time, completeness, extension, record and rule choices.

Boundary: Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__release-date-retained__all-listed-rows-retained__unobserved-successor-open__source-hash-census-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-source-bounded-census	largest-number-only	A largest number alone omits the observed element census.
extent	greatest-listed-coordinate	timeless-terminal-claim	A current release cannot prove that no successor can exist.
identity	element-record-retained	count-without-element	A number without its element record is not an observation boundary.
time	release-date-retained	undated-table	An undated table cannot define a reproducible empirical boundary.
completeness	all-listed-rows-retained	selected-terminal-row	Selecting only the last row cannot establish the registered census.
extension	unobserved-successor-open	observed-means-impossible-beyond	Absence from one release is not impossibility.
record	source-hash-census-trace	boundary-without-source	An unsourced endpoint cannot be reproduced.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
rule	no-extra-rule	free-terminal-rule	A terminal rule chosen from the table would overfit observation.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:07957a12479b7fc21cb0f19bd56d222e157ebb86d5ecc6de767af082f09b4529.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9529beed9e6c984672c342003db5011e89be4e68d0e1591ab8de1c93c429482e.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ef04b83aec104f822c81ac402cd320bdea3fd18aa95096286e409badaa6e4a4f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2eec45d3cb99a30f59046b3098e2f4659fb5d313949bb9dd9f1ac6d56ffd2cf6.

Independent reconstruction. Implementation hash

sha256:c99ea6d2271b92fe7b4dddb45245248dc45abe31f824d86bde302e418fe92c65; independent certificate sha256:f7c1c445a2435d2d0997ca0b49d94cc60bf174d5f973464f39c4f256593279be; external-validation hash sha256:9b2b3f3e4fe85ffed184cc29ae0ccf5fd1a7d1e9cdc5f4b9f1660e991fca420d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-boundary-iupac-2022: predicted source-bounded-known-element-census__greatest-listed-atomic-number-118; source-derived source-bounded-known-element-census__greatest-listed-atomic-number-118; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the registered official release does not contain the complete stated current census through atomic number 118, or if a changed row is accepted.

Measurement receipt: sha256:68205bb3cf64338c6cb4df3bbcbdc3a2fd7200c5996771517de4670997755f6.

Isolation certificate: sha256:626e0b70b2d68de0b1f30887d0bd047017c22600d889c933da04f3c3f88366c7.

Custody certificate: sha256:11f824e4af628af363afff788480c72da16a213b2e09456e08eff4879bdf0822.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-boundary-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

Immutable identities. Source manifest

sha256:c19bd53cdeb7132b39723a417b54955507500f1b2b905fb3bfe87c0b32bdbc51; derivation seal
 sha256:686eeab8f18d74ee44e4b60cb62514ce3fcad82000980177b5ea5671b4e4d74d; engine receipt
 sha256:8fbea0a9f06035f41313a38182243b9e7c9e5510957ce95c3069d30213e73fc8 at receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001-8fbea0a9f06035f4.json; empirical hash
 sha256:8fa21da784af33f3540672619aec09eed73b9e7ca318766f0209204bd473db39; measurement receipt
 sha256:68205bb3cf64338c6cb4df3bbcbdc2aa2fd7200c5996771517de4670997755f6.

11. Composition Stoichiometry

Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization.

19. Exact chemical composition

Claim identity: SFT-CHEM-STOICH-COMPOSITION-001

Question and theorem. Chemical composition is the complete finite Fold carrier of distinct constituent identities paired with exact positive rational parts that exhaust the One.

complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-One

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001

Generated grammar. Generate the literal product of the registered composition carrier, components, quantity, closure, absence, identity, record and extension choices.

Boundary: Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-one__absent-component-omitted__constituent-identity-retained__identity-part-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-chemical-carrier	bulk-label-only	A bulk label erases constituent support.
components	distinct-constituent-identities	anonymous-parts	Anonymous parts cannot preserve chemical identity.
quantity	exact-positive-rational-parts	floating-percentages	Floating percentages cannot certify exact closure.
closure	parts-exhaust-the-One	partial-selected-composition	Selected components need not exhaust the carrier.
absence	absent-component-omitted	zero-valued-components	Numerical zero is not an SFT composition object.
identity	constituent-identity-retained	interchangeable-constituents	Equal amounts do not make substances identical.
record	identity-part-source-trace	composition-answer-only	An answer loses part and source provenance.
extension	no-extra-rule	free-normalization-correction	A correction could force closure after the fact.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a72500ce5db6e8ec585702f2e1a8dbfed043e7e52c2ee8304e5647b7e3abf5e7.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : c75c9b3e1be8fe6f88b5afb87d59e106cfc119080312c51c41d0e99922b5167a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : fcecb84871bf9e043bc807dccc5e3c148e05b26ac43b05af6806cfd0025fc304.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 6961ea1b574acf85e4ab8c4dc2c15a306a8853e50ddf3e7c30bb6e707858c87f.

Independent reconstruction. Implementation hash

sha256 : 17727dd7b5815338c9679ab3cc2682e0e423ae50b27cf1305f044b57369c4959; independent certificate sha256 : ad7e04bbe9d538d460664d08bba33e86cee6c30555ff31191f5fc4cbce190eb7; external-validation hash sha256 : a57b4f689cfd58532a3ba9d3348415a52c28454b049eaba3dc8b4c90bc9ae336. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03722-2026

Comparison records:

- composition-iupac-m03722: predicted constituent-carrier__constituent-part-over-total-mixture; source-derived constituent-carrier__constituent-part-over-total-mixture; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks constituent identity, a constituent-to-total relation or mixture support, or if a changed row is accepted.

Measurement receipt: sha256:d4a6cc0ba56b7f6b94481f21014a9be2deca14df0f4ba7cc26b95a7fef4374b2.

Isolation certificate: sha256:2a8f74b85fac0e37d95b18ad205da144f96cf966913800109647a4cf545da143.

Custody certificate: sha256:0a657320e42bc4eefc3aeedbb705dd197c259007ae0d2dbd23cde01147eeac8e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03722/json> (sha256:cc715cdd306779e5032685105dacbbc7d1ef79ea8241269496befb1dc37a7f2f)

Registered target identities:

- composition-iupac-m03722 from IUPAC-GOLD-BOOK-M03722-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

Immutable identities. Source manifest

sha256:8f2e8d5cc8143850e17f1f542360dd2e0bfbbaa0f272eef399b72d1bd3a2b57b; derivation seal

sha256:165b5200764d9d49f6881b554db180129167625b1e98c79e12037a7bc4dfd5a0; engine receipt

sha256:4749de58750e305820c39678a3a2f0f6d8332e721a17921651eafca98a8a1dbc at

receipts/engine/model_admitted/SFT-CHEM-STOICH-COMPOSITION-001-4749de58750e3058.json;

empirical hash sha256:58069201b824519064d26333ef7a65bc5843bf825a54910f51a007904f4f5750;

measurement receipt sha256:d4a6cc0ba56b7f6b94481f21014a9be2deca14df0f4ba7cc26b95a7fef4374b2.

20. Conservation of elemental identity through reaction

Claim identity: SFT-CHEM-STOICH-CONSERVATION-001

Question and theorem. A chemical reaction is closed only when complete held reactant/product words have exactly equal positive multiplicity totals for every retained elemental identity, including named boundary carriers.

complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__all-element-totals-paired

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COMPOSITION-001

Generated grammar. Generate the literal product of the registered conservation carrier, orientation, identity, count, closure, boundary, record and extension choices.

Boundary: Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__exact-positive-multiplicity__all-element-totals-paired__named-boundary-carriers__species-formula-total-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-reaction-word	product-list-without-input	Products alone cannot demonstrate conservation.
orientation	held-reactant-product-sides	signed-stoichiometric-scalars	Signed proof quantities violate the positive domain.
identity	elemental-identities-retained	mass-only-balance	Equal bulk mass can hide changed element labels.
count	exact-positive-multiplicity	approximate-decimal-balance	Approximation cannot prove equality of discrete carriers.
closure	all-element-totals-paired	selected-elements-only	Omitting an element can manufacture balance.
boundary	named-boundary-carriers	unrecorded-open-system	Unrecorded boundary transfer can mimic loss or creation.
record	species-formula-total-trace	balanced-label-only	A label cannot reconstruct the equality.
extension	no-extra-rule	free-balancing-exception	An exception could excuse an unpaired carrier.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5108a653085bf72b6824763bc3d1b16ed015738b65ff8dcd78f9367699bc2f27.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:079af8a29365c118d92b3732ad4284de9697833b1c3698b2e3da9b48679a85fc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:455afb7670db3dca84beea9f74cbfc874dd8b6a3bbaa348262b0aafd2c7282d2.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bf89eebc5d713f6278f738bfc83f579a0c871960d060961c54922ca77d71ce0.

Independent reconstruction. Implementation hash

sha256:77092730858fcc8d021aca7483bbc7c25c64ef8b6cd071c34b0f409bb5e6901a; independent certificate sha256:85f9be325874c16051fc4b181a3689b3516089ad25438316ae14404254699402; external-validation hash sha256:25a3c688ffe9847880d57fcbcb28559933ac4e2aaf0715a004e15b7277e365f75. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06026-2026

Comparison records:

- stoichiometry-iupac-s06026: predicted reactant-amount-support__product-amount-support__reaction-relation; source-derived reactant-amount-support__product-amount-support__reaction-relation; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The formal conservation theorem fails on any unequal element map; the categorical correspondence fails if IUPAC lacks related reactant and product amount supports or if a changed row is accepted.

Measurement receipt: sha256:cf0214f22de339efff956717d2cdc8acb0ff8e6fdef23090126f6d18be727b27.

Isolation certificate: sha256:8cd441eb74abde90e29877ce9f3ad59fb77f7f6251baccdef4791383bf21c93a.

Custody certificate: sha256:c7e3c8ea5cabfd382ce210a899b3f7e10136eecb58f4642c611adb86b991e755.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06026/json>
(sha256:0354d161fb1754d50dee2cfd3bf051c6a005dbde765a112398f8e717e9952bc)

Registered target identities:

- stoichiometry-iupac-s06026 from IUPAC-GOLD-BOOK-S06026-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody

- Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

Immutable identities. Source manifest

sha256:9b6bf9c5cd0488c8496f56be542d82c20afd343c7a85219e42394867ce67079d; derivation seal
 sha256:f0b0c46ec94d704cc104131476b8ee493a63b542f3f7792ce173996a9a4ff714; engine receipt
 sha256:a510b7f8cba3f007052fd1bab433ad8e5085cb0beab3eb85f247f8614be23d81 at
 receipts/engine/model_admitted/SFT-CHEM-STOICH-CONSERVATION-001-a510b7f8cba3f007.json;
 empirical hash sha256:e052d1b6079e4ea26822a82ccb4c1640bfd1c140570ff0c01b35f5f0ae660661;
 measurement receipt sha256:cf0214f22de339efff956717d2cdc8acb0ff8e6fdef23090126f6d18be727b27.

21. Positive stoichiometric coefficient and reaction balance

Claim identity: SFT-CHEM-STOICH-COEFFICIENT-001

Question and theorem. Stoichiometric coefficients are the unique primitive vector of positive generated species multiplicities balancing every element map, with reactant/product direction held structurally and nonparticipants omitted.

formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

Generated grammar. Generate the literal product of the registered coefficient carrier, orientation, balance, scale, absence, uniqueness, record and extension choices.

Boundary: Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector__*

`_nonparticipants-omitted__unique-up-to-common-multiple__formula-matrix-reduction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	formula-bound-species-support	coefficient-list-without-species	Counts detached from formulae have no chemical meaning.
orientation	held-side-positive-count	negative-reactant-count	Negative proof counts are inadmissible.
balance	element-balance-preserved	unbalanced-count-vector	An unbalanced vector violates element conservation.
scale	primitive-positive-vector	arbitrary-common-multiple	Common multiples are equivalent but not minimal.
absence	nonparticipants-omitted	zero-coefficient-species	A zero coefficient imports numerical zero.
uniqueness	unique-up-to-common-multiple	one-picked-balancing-vector	Picking one vector does not eliminate rescalings.
record	formula-matrix-reduction-trace	coefficient-answer-only	A count list cannot prove its balance.
extension	no-extra-rule	free-coefficient-adjustment	An adjustment can force a desired equation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c078d573e8a1e9e8a75c4bf44728a255e36bf2ce05c9bdb553302bc964024fb0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9a54b1e4fc558b7b721a27cfb8aef9fd9fd2b1b8a7f2a46eceed85bdba37fd4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:32577dd9fbd901b3ba894763049cc1ed4f6a95e6b0741fb56d9befed0823016f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e7f45e303be1880d3bd54dde084ced16329f3cbfbcd507ceb19c40ddd1200de9`.

Independent reconstruction. Implementation hash

`sha256:11bea52dca022dcd31511c76455602348cffc106f1c11d30a6f48454a49b6f41`; independent certificate `sha256:864a5d480e06f6a3c429374da8dc9e45ef4c20957a0a4713cbf3f9f00d91f0ca`; external-validation hash `sha256:a787dc6241c9679c6d5db09d5cd017b8d480ac33ea5285b60e8a88115e3a3017`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01034-2026

Comparison records:

- `reaction-equation-iupac-c01034`: predicted
`reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients`; source-derived
`reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients`; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC equation record lacks reactant/product sides, absolute stoichiometric counts or formula binding, or if a changed row is accepted.

Measurement receipt: sha256:55d7db99d8d29620cccf07bd1ee2bea8b0b5de14c7f53be0b825c719ef320afb.

Isolation certificate: sha256:f60f3161e2c07f23daf83f18c763c813fb51cf2acb05b743a0990e3825c7f6cf.

Custody certificate: sha256:d82d11e8dd27c83edf8e3e7252904837dcd770d5679c76350c6a48e47da2de70.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01034/json> (sha256:a582f71bc6c555d5513f365d18d5d46a6a5da61e0c1668d231ad860b9a32dea3)

Registered target identities:

- reaction-equation-iupac-c01034 from IUPAC-GOLD-BOOK-C01034-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

Immutable identities. Source manifest

sha256:e5eed4f4b0dae1eaf2e61c9a650168b57729440ac461ec4ab898e98740db3639; derivation seal

sha256:efe5701f6eedd86927d41aeb0b7a2549c9e8ea6690c674ef065d66b67beafd; engine receipt

sha256:fa3142d8b967e420b605b436d4fe897322b2976dc1ab730dfa79bb5d29622947 at

receipts/engine/model_admitted/SFT-CHEM-STOICH-COEFFICIENT-001-fa3142d8b967e420.json;

empirical hash sha256:bbc5100b4599ade423c2cd3b9282d2403595649ca8001f116e59bbe37136c5d3;

measurement receipt sha256:55d7db99d8d29620cccf07bd1ee2bea8b0b5de14c7f53be0b825c719ef320afb.

22. Limiting-component and complete-consumption boundary

Claim identity: SFT-CHEM-STOICH-LIMITING-001

Question and theorem. The limiting support is the complete tied least set of exact available-amount to primitive-required-count ratios; it bounds reaction progress before any unavailable carrier would be required.

available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COEFFICIENT-001

Generated grammar. Generate the literal product of the registered limiting carrier, requirement, comparison, selection, consumption, excess, record and extension choices.

Boundary: Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support__complete-consumption-boundary__retained-excess-support__amount-coefficient-ratio-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	available-reactant-support	named-reagent-only	A name without amount and coefficient cannot limit progress.
requirement	primitive-coefficient-requirement	unbalanced-requirement	Consumption requirements must come from the balanced reaction.
comparison	exact-available-required-ratios	floating-ratio-ranking	Floating comparison cannot certify ties or minima.
selection	complete-least-ratio-support	first-considered-reactant	Ordering of inspection cannot select the limiter.
consumption	complete-consumption-boundary	negative-remainder	Overrunning the least support would require unavailable matter.
excess	retained-excess-support	excess-erased	Nonlimiting remainder is part of the result.
record	amount-coefficient-ratio-trace	limiter-name-only	A name cannot reproduce the comparison.
extension	no-extra-rule	free-limiter-rule	A preference could choose a nonminimal reagent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:6ebd09dcebbbc38118e45114f8cc922497e0587e6cdfe8cd4eabb024443338070.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:74667f499635c03544e37dbda2e53204e74a332e76fe799df64c6be0f44a9754.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:656b446ccde1d50e6e371ff7dc0cad490f962eef10fabddfd8a43fcbcc4bfd4f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:c0ba9f1cd9197180791356d6e661445fe7460b90b6b9ce6446d2f8f18bcf07bc.

Independent reconstruction. Implementation hash

sha256:8495ce255510f5059960137d6e1e9b3ee7214201e8d830661353e4c7d1274ecf; independent certificate sha256:1406ee0f53cea26f83d16e7cd23a3d49c5a694e77edd6020d3544febd31e8c34; external-validation hash sha256:519a8a00f02ccc7a3f7fc3da597c67113efc403ccc7c61a865fec4559dfc746e. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-D01771-2026

Comparison records:

- limiting-reagent-iupac-d01771: predicted limiting-reagent-carrier__amount-boundary; source-derived limiting-reagent-carrier__amount-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a limiting reagent and its amount relation, an exact nonleast carrier is selected, or a changed row is accepted.

Measurement receipt: sha256:af4b2a71f1cf3ff3049af90956d74c521b1fb6bd1f967619e433e1299d44382b.

Isolation certificate: sha256:49ca936b63fb7bc678ad1191e82c2d8c6eaf963667978a997220527ede439d67.

Custody certificate: sha256:c45284506b0557a4d8b86cdca4c78e0f4df93c446c56b96ae4ff409c8b27349e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/D01771/json> (sha256:352e753acf025b61ce54a5b00b61d9dd5b6d112d0d35133b48f2eba212c3b807)

Registered target identities:

- limiting-reagent-iupac-d01771 from IUPAC-GOLD-BOOK-D01771-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

Immutable identities. Source manifest

sha256:9da9d441722db6967b140f8ff0849f7e757695f59bdc80ff8c23c853a26b08c3; derivation seal sha256:b784c5f3c18d8664ea883c1c9bbba2321a6db47675ddb505a48367c1fb478f4; engine receipt sha256:9fa6f3b55c889c675a54c5e137b08f9357ecbe860c59b9ced2b506925a73468b at receipts/engine/model_admitted/SFT-CHEM-STOICH-LIMITING-001-9fa6f3b55c889c67.json; empirical hash sha256:bfe6a293e23cb72a148862a36162aa6c4d713fc8ed68d4e26df6b547ac31d69a; measurement receipt sha256:af4b2a71f1cf3ff3049af90956d74c521b1fb6bd1f967619e433e1299d44382b.

23. Reaction yield as retained product share

Claim identity: SFT-CHEM-STOICH-YIELD-001

Question and theorem. Chemical yield is the exact positive Fold part of a retained element-or-compound product amount relative to its registered expected support after a specified reaction or separation.

element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__exact-positive-amount-fraction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-LIMITING-001

Generated grammar. Generate the literal product of the registered yield carrier, process, numerator, reference, quantity, absence, record and extension choices.

Boundary: Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__registered-expected-support__exact-positive-amount-fraction__empty-product-outcome-form__product-process-reference-trace__no-extra-rule*. Closure is *depth_independent*; minimality and named-shape

uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	element-or-compound-product	percentage-without-product	A percentage detached from a product is not chemical yield.
process	specified-reaction-or-separation	unspecified-process	Yield depends on the declared reaction or separation.
numerator	complete-observed-product-amount	selected-favorable-product	Selecting a favorable amount hides other outcomes.
reference	registered-expected-support	unbound-denominator	A fraction without its reference is uninterpretable.
quantity	exact-positive-amount-fraction	floating-percent	Floating output is not exact proof.
absence	empty-product-outcome-form	numerical-zero-yield	Zero is not an SFT proof value.
record	product-process-reference-trace	yield-answer-only	An answer loses product, process and reference provenance.
extension	no-extra-rule	free-yield-correction	A correction can inflate the observed result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : d25409ab09b66f54e66658f55ef35bd4c7b994814725c3cecd49c5a18ffc6ede.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 9a81c40f9ff9716ea4d44351b4d379929c6a2e124154eaa6bb55b4e775388ee4.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : f7c821f8d7494f3c08d13fd43f4971cebcd6ea3d5f45d85f62b112202583abcb.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 2819fb08332303920b39a0fc659c93cea8485c85ea07e708d3be83a8055bf9d7.

Independent reconstruction. Implementation hash

sha256 : e84db5b8638298ffd8290fb69d355e310f64a6201fd700ce8d188b614972d04c; independent certificate sha256 : 27ce7f5b9a159971c4c513624866a8244cf854b3cd65b9bb3f465fd4e67f8dfb; external-validation hash sha256 : dcface3941413fec1009e46c661ae31364b0b44b93042c4ed596278d4e57bbf1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01041-2026

Comparison records:

- chemical-yield-iupac-c01041: predicted amount-fraction__element-or-compound-product__specified-reaction-or-separation; source-derived amount-fraction__element-or-compound-product__specified-reaction-or-separation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks an amount fraction, element/compound carrier or specified reaction/separation, or if a changed row is accepted.

Measurement receipt: sha256 : f2825991402f7f43d0d997ed2187e32820013df3201d200377060d94424c4335.

Isolation certificate: sha256 : fdb2d43fefde6ac4a5a114c3f1f1ad979158e4f8aa3fbe254ad7a57359bbad6e.

Custody certificate: sha256:0fd75ee7e7f7ce4bc1badc51e44cbcd9a740bef3bc071676c1d6a737bca09bee.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01041/json> (sha256:47979dfe9e05be8a7c207779853ce25538a8753e05b648a8a43471ed040b732f)

Registered target identities:

- chemical-yield-iupac-c01041 from IUPAC-GOLD-BOOK-C01041-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

Immutable identities. Source manifest

sha256:a0116438e6f83db3d0d99e824726e84c88b578f7437cfe799bc4ddc360d92adc; derivation seal sha256:127d3ab3ebb275a09332cd72b1af9c50849b46a51bf96aad1fb561c2d39f8886; engine receipt sha256:87383bf6cedbbc08dae3f91f6932bbd4b151d8288c32ac5908cd6f62fff649d6 at receipts/engine/model_admitted/SFT-CHEM-STOICH-YIELD-001-87383bf6cedbbc08.json; empirical hash sha256:c9261af3ac67d4ec093db933e4af89fe002d4ac1693c29b5984b240eef512249; measurement receipt sha256:f2825991402f7f43d0d997ed2187e32820013df3201d200377060d94424c4335.

24. Mixture components and composition support

Claim identity: SFT-CHEM-STOICH-MIXTURE-001

Question and theorem. A mixture is a finite portion-of-matter Fold carrier with multiple retained chemical-substance identities and exact positive composition parts, without assuming formation of a new substance.

portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-YIELD-001

Generated grammar. Generate the literal product of the registered mixture carrier, multiplicity, identity, composition, bonding, phase, record and extension choices.

Boundary: Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts__constituent-identities-retained__phase-support-recorded-not-forced__component-part-phase-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	portion-of-matter-carrier	substance-name-only	One substance name does not define a mixture.
multiplicity	multiple-constituent-support	single-constituent	A single constituent is not a mixture.
identity	chemical-substance-identities	anonymous-components	Anonymous components erase chemical distinction.
composition	exact-positive-composition-parts	unmeasured-presence-list	Presence alone cannot reconstruct composition.
bonding	constituent-identities-retained	new-substance-conflation	Mixing does not by itself force a new chemical identity.
phase	phase-support-recorded-not-forced	single-phase-required	Mixtures may contain one or several phases.
record	component-part-phase-trace	mixture-label-only	A label cannot reconstruct components.
extension	no-extra-rule	free-mixture-equivalence	An arbitrary equivalence can merge distinct mixtures.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: acb80f57db5c57ab1df7b007f1543f85ba3748e02f7d0e3d95d125a1cfb31630`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: de36e78e252d653293e37ec113b0dc81ece173288e089f8d49fe23ac16133fdd`.

- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:56afba8866e13bcfdc397b0f3eee00ada26e353e74d5095ac4466e87ba6615f3.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:97238131131cae2f7c598609c639c60476b18cb2e6290608d7b8f37dfbd1b0de.

Independent reconstruction. Implementation hash

sha256:85d3ff3531cced11d325c2211b6ea9680e0d6626a89522b8cd3e8872a322d103; independent certificate
sha256:96bb6f5fffb2813d6bd37c857ebcefbaea52fb4dbc0be4cf181c61e211b5ed55f; external-validation
hash sha256:d52d2fb26eae22622359bc13b920674b4633d994cc2c8cb1b782b9f304c3b6ae. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03949-2026

Comparison records:

- mixture-iupac-m03949: predicted matter-portion__two-or-more-substances__constituent-identities; source-derived matter-portion__two-or-more-substances__constituent-identities; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a matter portion, multiple chemical substances or constituent identities, or if a changed row is accepted.

Measurement receipt: sha256:3b5d8220c456ede703a29c70a4a194fdb42cca38d763f4ef18bf947d8750d067.

Isolation certificate: sha256:a2712fae535f20dd5d7c7f128176cc00c95fa410975d4e5b33030389a30e47fc.

Custody certificate: sha256:d976c23c2014b49392fc31020aa7d4ed08f8d4995e7dceee239129204bb9bb50.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03949/json>
(sha256:71ba494096d995967227b9ef9fb4e6145b98c9cbcd9b46460bc63b94c06966f)

Registered target identities:

- mixture-iupac-m03949 from IUPAC-GOLD-BOOK-M03949-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

Immutable identities. Source manifest

sha256:bd9b1e23fa1e8fcda55ede4f1803e82363189aa77af123609c2ebf8c2c33695; derivation seal

sha256:b3b68cc273a3c95181481c58276e713fb631d80cadb79f154682f68e9b266d02; engine receipt

sha256:d77d8e9e7e5fafc9a4a986b30bb99a80af48adafccc45031c9190df75b29167 at

receipts/engine/model_admitted/SFT-CHEM-STOICH-MIXTURE-001-d77d8e9e7e5fafc9.json; empirical hash sha256:f490b60c9eb1cd26612a22e17d370715a1fb767dc32eda7613731c58b1a5fd1a; measurement receipt sha256:3b5d8220c456ede703a29c70a4a194fdb42cca38d763f4ef18bf947d8750d067.

25. Solution composition and solute-solvent distinction

Claim identity: SFT-CHEM-STOICH-SOLUTION-001

Question and theorem. A solution is a liquid or solid single-phase mixture containing multiple retained substances under an explicit contextual solvent/solute role partition.

liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-CHEMICAL-SPECIES-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-MIXTURE-001

Generated grammar. Generate the literal product of the registered solution carrier, phase, composition, roles, identity, context, record and extension choices.

Boundary: Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition__substance-identities-retained__declared-context-role__phase-component-role-trace__no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	liquid-or-solid-phase	gas-or-unclassified-matter	The registered solution boundary is a liquid or solid phase.
phase	single-phase-support	multiple-unresolved-phases	Unresolved phase multiplicity does not define one solution carrier.
composition	multiple-substance-support	single-substance-phase	A pure phase lacks solute/solvent distinction.
roles	solvent-solute-role-partition	unpartitioned-components	A solution record distinguishes solvent from solutes.
identity	substance-identities-retained	roles-replace-identity	A role label must not erase substance identity.
context	declared-context-role	intrinsic-permanent-role	Solvent/solute assignment can depend on the description context.
record	phase-component-role-trace	solution-name-only	A name cannot reproduce phase and roles.
extension	no-extra-rule	free-solution-exception	An exception can relabel a multiphase mixture as one solution.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4fe30a8a55d56b5ab42bf2b742148820ad68162cf9d5e70ee5a38004e0f457b9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:4ea951e01a0a70099147774aea8b7a15b2ec3dc32a151b75faf81596cee969d7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:24316754e898abeac2d19748aaf72de6a2f28f7cd99dd250b68a28cc3fa74def.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e1aad5725bea83acba570f25c4e0438cbff9c9620e5629755c76672e3f2b015a.

Independent reconstruction. Implementation hash

sha256:c0b00a98c9f8125cb8cf283d7fa1e8a33b328efeb815cb3e559cedc6740126bd; independent certificate sha256:105ee387d1fc186480013933b78a95f785fb02c759ac692c1cb36547a421997f; external-validation hash sha256:3e720b08c2d95c2a714a714fc6432fbc6ab362d7c67d1a62567725f2a7566dae. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S05746-2026

Comparison records:

- solution-iupac-s05746: predicted single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; source-derived single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC solution record lacks liquid/solid phase, multiple substances or solvent/solute distinction, or if a changed row is accepted.

Measurement receipt: sha256:715f072b9ecb06487c766125d3dddb7cfadad15cbf708a3a420b7effbdbcceacc.

Isolation certificate: sha256:29378a1c0c44592cbc595fe9461242a245f0d11fe67e586dad244badd67fda25.

Custody certificate: sha256:13f2783c070b3017363e017a98bba399b8db9cbfdd35ca942a4ffcd5a4210087.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05746/json> (sha256:39f8916fdced6993ceb67e374641b95375570866a24123508ce26f9ec0049f28)

Registered target identities:

- solution-iupac-s05746 from IUPAC-GOLD-BOOK-S05746-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

Immutable identities. Source manifest

sha256:db699f4ed4734f107bf620736f458a9f033160b02451a20467a4619f0f81c4d2; derivation seal sha256:6e5542e15258a967197185e1f07e767e97459a180c9fe7347d0e46ae97f96199; engine receipt sha256:9775d542364d5db8cb2ce3a2491cc224abfde1709129db3d42f0e64ed5cac371 at receipts/engine/model_admitted/SFT-CHEM-STOICH-SOLUTION-001-9775d542364d5db8.json; empirical hash sha256:02cfbf84dce854f2b64540c9bf6b036d8dc337ad39637bfc42e50dbb0179d3d8; measurement receipt sha256:715f072b9ecb06487c766125d3dddb7cfadad15cbf708a3a420b7effbdbcceacc.

12. Bonding Molecular

Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures.

26. Chemical bond as retained interaction closure

Claim identity: SFT-CHEM-BOND-CHEMICAL-BOND-001

Question and theorem. A chemical bond exists when complete source-bound interaction support between distinct atomic carrier occurrences or groups closes into stable joint recurrence distinguishable as an independent chemical carrier.

atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001

Generated grammar. Generate the literal product of the registered bond carrier, interaction, closure, independence, energy, reversal, record and extension choices.

Boundary: Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity__measured-stability-record-separated__opening-closes-to-separated-support__carrier-interaction-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atoms-or-atomic-groups	bond-name-without-atoms	A bond name detached from carriers has no chemical support.
interaction	source-bound-interaction-support	coincidental-proximity	Proximity alone need not produce a stable entity.
closure	stable-joint-recurrence	transient-contact-only	A contact without recurrence is not an independent bonded carrier.
independence	independent-chemical-entity	parts-only-observation	If only separated parts persist, no joined entity is formed.
energy	measured-stability-record-separated	imported-potential-function	A consensus potential cannot select closure.
reversal	opening-closes-to-separated-support	irreversible-by-definition	Bond opening may be lawful and must retain carriers.
record	carrier-interaction-recurrence-trace	bond-label-only	A label cannot reproduce interaction and stability.
extension	no-extra-rule	free-bond-exception	An exception can call any contact a bond.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : c79949b0bfe9498cf4d374c3041cad4bd3d2c76f13ee438fb68d1e4fed80a468.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 67d9bddfde9089152683c855abe345423e079bb9c69cbb4c2ec01eaea88c89ff.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256 : 829ed6daf9d8afea57c415f332a212cf2dd68a7834c99709e75ed5ca411522b8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256 : 8647853beaa9c2398420258b61194b99ef938fb4a29bf75625cae36762f09e7b.

Independent reconstruction. Implementation hash

sha256 : 13efe64c7d43bb48869c408cbab8c44f2ccf1d7ed6e0e875e332e021c448ec6b; independent certificate sha256 : 13b94ba61ffe44d8d734add0368d83768a5e8e2384b92b4517df7bc2422418a6; external-validation hash sha256 : fc484bd2b42299be96d1d68707cafab3022c9e9c7c2fb5524e92e97aadeed349. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT07009-2026

Comparison records:

- chemical-bond-iupac-ct07009: predicted atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; source-derived atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks atomic/group carriers, stable independent molecular entity or interaction-force support, or if a changed row is accepted.

Measurement receipt: sha256 : 5b96d8da59dcd2182001f4b15391758cf3342d766bd58d70bc73e794aecdd082e.

Isolation certificate: sha256 : 5460f966fa502eaab9a5a9b4fbc7149e6d85791abb800b101079296116777c1d.

Custody certificate: sha256 : 9d7857250559bc276af0441c1abd3f5d9a9ee8e424d50efa68144e0a21b979bd.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/CT07009/json> (sha256:691e8522684b2e4290c207ae988a4689d1ab255cdd5fca4ed66a1bd557fa4207)

Registered target identities:

- chemical-bond-iupac-ct07009 from IUPAC-GOLD-BOOK-CT07009-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody

- Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

Immutable identities. Source manifest

sha256:ad51f358991f2d60e1c5d5cfa71c4c37f8f7f7be1652cc5a5b3968d8058ba194d; derivation seal sha256:523dfb58508e28f1d3bbafdcf7d0b0b224e2a4522504ce4ab6a5b10646f3c786; engine receipt sha256:6f5a76351dd92e06a0982b6a611e3bf04d2d69dae0635e8dd6b4a3b4e92586eb at receipts/engine/model_admitted/SFT-CHEM-BOND-CHEMICAL-BOND-001-6f5a76351dd92e06.json; empirical hash sha256:3241f15a85770f40e37f28ca05b10426a7e3cd63b037a0a3f5bedc80a78c9e86; measurement receipt sha256:5b96d8da59dcd2182001f4b15391758cf3342d766bd58d70bc73e794aec082e.

27. Covalent shared-support bond

Claim identity: SFT-CHEM-BOND-COVALENT-001

Question and theorem. A covalent bond is stable joint recurrence in which one or more exclusion-preserving electron-labelled channels are jointly accessible between identified nuclear carriers without duplicating the conserved carrier.

identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001

Generated grammar. Generate the literal product of the registered covalent carrier, electron, sharing, density, attraction, occupation, record and extension choices.

Boundary: Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region__stable-contractive-response__exclusion-preserving-channels__nuclei-channel-sharing-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-nuclear-pair	bond-without-nuclei	Covalent support relates identified atomic centres.
electron	joint-electron-support	electron-assigned-to-one-centre	One-centre support alone is not shared bonding support.
sharing	one-support-shared-access	copied-electron-label	Copying one conserved carrier violates exclusion and conservation.
density	internuclear-accessible-region	empty-internuclear-path	No connecting support cannot join the nuclei.
attraction	stable-contractive-response	unbound-force-label	A force name without recurrence does not establish a bond.
occupation	exclusion-preserving-channels	duplicate-identical-cell	Duplicated identical occupation violates exclusion.
record	nuclei-channel-sharing-trace	covalent-label-only	A label cannot reconstruct sharing.
extension	no-extra-rule	free-orbital-rule	An imported orbital choice can select a desired bond.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f59fe4de319dd05bd3b6b6d48f1de3d15a39d2deee4bc65cbf5214affe909a3c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 92ed58420e5d68434c41c74d2cb5eb78fb3782d79080388c4cf7c4efdbed92d0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ee7b74ae479c8c65c71393fac1927bf6dc7eb709d54e5f3a9f14167debde57f7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a6c12d5e1a6339f3a4c4318629455f975ea16a517e11d0bb1cf857097839ea88.

Independent reconstruction. Implementation hash

sha256: 4b68316cb8fb10965e70db523fc6335b89af5a692c612d7eaa9634157368aa0f; independent certificate sha256: 3544a1af89d8576edf7c999b7ab8c70fada8cf2ad63bc3b0bef03541d0c4992c; external-validation hash sha256: 65299913bf8ed63d2e7b7ecb64db0cad8cd0ab3e9fc9e155451e9c248442ebd. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01384-2026

Comparison records:

- covalent-bond-iupac-c01384: predicted internuclear-electron-density__electron-sharing__attractive-characteristic-distance; source-derived internuclear-electron-density__electron-sharing__attractive-characteristic-distance; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks internuclear electron density, sharing or attractive characteristic distance, or if a changed row is accepted.

Measurement receipt: sha256:4389207d46667075d3136e495b54799342ddc50516f726bb0bb1caf5d15e65d8.

Isolation certificate: sha256:7660d942fb6500602f9f53a9e426844d631d14a08448c79c63029be8d6ad69e7.

Custody certificate: sha256:e1fde5da0ecc8b811a1d6525d4e18503f6be7fe623ef7f168ff148198764fe1c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01384/json> (sha256:5106a40f5b1c372a570fc42f3cb1641b63a270454e4a7138699c584754294622)

Registered target identities:

- covalent-bond-iupac-c01384 from IUPAC-GOLD-BOOK-C01384-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

Immutable identities. Source manifest

sha256:ac32a8598b14c48ec075df48f45bb5ddd1d5b331e64209e8e28ceb477190e775; derivation seal

sha256:6c8f4e6687ca434ec1572a13444e6c53ce0bc6bc359b802d3b77e511cdf152e; engine receipt

sha256:b1efa88f0ebfe5c562e52a5a39673b1242da71b91d7d3c24b66356bfd462399b at

receipts/engine/model_admitted/SFT-CHEM-BOND-COVALENT-001-b1efa88f0ebfe5c5.json; empirical

hash sha256:dd9f285006ea268f1df83b9432b26b249b9ca4ebaf0551f66b8407058f35b160; measurement

receipt sha256:4389207d46667075d3136e495b54799342ddc50516f726bb0bb1caf5d15e65d8.

28. Ionic transferred-label bond

Claim identity: SFT-CHEM-BOND-IONIC-001

Question and theorem. An ionic bond is stable joint recurrence of opposed cation/anion held charge fibres produced by closed charge transfer, with electrostatic attraction and both chemical identities retained.

cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001

Generated grammar. Generate the literal product of the registered ionic carrier, formation, orientation, interaction, identity, character, record and extension choices.

Boundary: Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support__chemical-identities-retained__exact-support-composition__ion-transfer-interaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	cation-anion-carriers	neutral-atom-pair-only	The ionic class requires retained ion carriers.
formation	closed-charge-transfer	unpaired-charge-creation	Charge cannot appear without conserved transfer.
orientation	opposed-held-charge-fibres	signed-charge-magnitudes	Negative proof magnitude is inadmissible.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
interaction	electrostatic-attraction-support	generic-proximity	Proximity does not explain ionic stability.
identity	chemical-identities-retained	element-identity-erased	Ion formation preserves nuclear element identity.
character	exact-support-composition	pure-binary-dogma	Observed bonds may retain both ionic and shared support.
record	ion-transfer-interaction-trace	ionic-label-only	A label cannot reconstruct transfer and attraction.
extension	no-extra-rule	free-electronegativity-cutoff	A fitted cutoff would import the classification.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9cbd2a42338ef908d2f97d639e4e163a85f7c01e8f842a1027ea1bd7d82156f7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e01b7d4d4f8d957a884aca8560cad6aa53ce0add2f825323d75db608900ab55d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2259224cc1c5e1e8fe927ed597b35a75ef8ace609b8f28dab7e2a23e6537117b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:92462b6af0cedacb2f75734bf5743e5454e360fc260e71256956609f5be56c39.

Independent reconstruction. Implementation hash

sha256:d343bcfb4e46576e97186ef06aba3aa0ae1e3d63bb00cafe938cc6172a3320bb; independent certificate sha256:3fb249ae42a0f7cfe14ee71adccafd0dc851d4d7cc205a8b438663482a103ce; external-validation hash sha256:49d1839c410e0bcc7ebd5cce71525f7368d8e9dfb6c89edc7c27674d1e3e8ba1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-IT07058-2026

Comparison records:

- ionic-bond-iupac-it07058: predicted cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; source-derived cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks cation/anion carriers, electrostatic attraction or the ionic-character boundary, or if a changed row is accepted.

Measurement receipt: sha256:9bbd734ddfabcb307e4f027b045a6d1f138551f43f77d142dc696342197b561.

Isolation certificate: sha256:dc95344f2d54eab75cd6be7cc5c73ff8ba72e09117a2fedb144fc96a1e4b7ca2.

Custody certificate: sha256:267fd55f24526b1cb9fed32621e9f52bb9ae246a1f02a27bf45b0b404149ea6e.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/IT07058/json>
(sha256:85d726bab39480e944dfd95e31889931459f4436480d751d73df058c68ad232e)

Registered target identities:

- ionic-bond-iupac-it07058 from IUPAC-GOLD-BOOK-IT07058-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

Immutable identities. Source manifest

sha256:27f04cbd8c0b331ccc2f8e03a5124b7017dcf58c649dfa8b3c45670b658a5c45; derivation seal
 sha256:c24c59d4b1e5603d40c6787bad3b65dcd122ccce662d17071066ba2961539f16; engine receipt
 sha256:f0effeef99b4caec396b299ab13e5d4dffe97ed7d7e85abb035e0068ad4468fc at
 receipts/engine/model_admitted/SFT-CHEM-BOND-IONIC-001-f0effeef99b4caec.json; empirical hash
 sha256:197102353edfda54d286fa8743c8ab004c3f854dcb96ca6fe8aac78159d8127b; measurement receipt
 sha256:9bbd734ddfabcbcb307e4f027b045a6d1f138551f43f77d142dc696342197b561.

29. Metallic collective-support bond

Claim identity: SFT-CHEM-BOND-METALLIC-001

Question and theorem. Metallic bonding is stable collective recurrence of a connected atomic network supported by electron-labelled paths delocalized across multiple cells of the complete finite network.

connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001

- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001

Generated grammar. Generate the literal product of the registered metallic carrier, electron, extent, joining, recurrence, identity, record and extension choices.

Boundary: Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support__stable-collective-recurrence__atomic-and-network-identities__network-electron-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	connected-atomic-network	isolated-atom-pair	Metallic support is collective rather than one fixed pair.
electron	delocalized-electron-support	localized-pair-only	A fixed pair cannot mediate the complete network.
extent	finite-whole-network-extent	unrecorded-infinite-lattice	A completed infinite lattice is not generated.
joining	collective-shared-support	independent-local-bonds-only	Independent local pairs omit collective redistribution.
recurrence	stable-collective-recurrence	transient-conduction-only	Transport alone does not establish stable bonding.
identity	atomic-and-network-identities	anonymous-metal-label	A metal label cannot create the law.
record	network-electron-recurrence-trace	metallic-label-only	A label cannot reproduce topology and support.
extension	no-extra-rule	free-electron-sea-premise	An imported metaphor cannot select the structure.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f274ec7ff0584a9b1aa9ba3b9c7ecc7e0bc0a0c9e2a02f0280cc33fb7cf4b0d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9e188caae50ec6514eba65a8d667abb1076a7ded34f9e0a45f6ec4ae466c1f27`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6fe108c3748d6af8f85ab8790147389a9628a359676b39b909121c43441035ff.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 1801270185f1d5190300b4a9bd1f172f07b407b7f606e915414dc3da23914cd1.

Independent reconstruction. Implementation hash

sha256: f2495ddd58f17bc3d5859581fa2a310c528cd32e38b9434b51221ada9d9b22db; independent certificate
sha256: 32d15d7feb44f6a46c04372c2c1dd7eb026023f09f119126adbf729397408be7; external-validation
hash sha256: 83e8ceb16a401fd1ff76690a24d5de32bf14c372ee131916bc2a49af9ffd3f7a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08789-2026

Comparison records:

- metallic-delocalization-iupac-08789: predicted not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals;
source-derived not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks nonlocalized electrons, extended whole-lattice support or explicit metal correspondence, or if a changed row is accepted.

Measurement receipt: sha256: 27487aabfa2003834b1c4f418af9a100be24bf33823fc49e8f7a13cc38ad35ae.

Isolation certificate: sha256: 7c7859fa1b0eef37bfd3a2b78bded47555ca2c7c7152352f6a4892a5b23e3db0.

Custody certificate: sha256: 8ca4fb05f88c7606938f7920cba6007b185604ba7d434da49753d1494ec25c17.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08789/json>
(sha256:570755940f01bfa32741b03b6b2f22b02742101605a2263e57369966ea433abd)

Registered target identities:

- metallic-delocalization-iupac-08789 from IUPAC-GOLD-BOOK-08789-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

Immutable identities. Source manifest

sha256: fbe017c1cc8fe61261c1d93e072d0bd206d106665fc8061c65bf50af62984c6a; derivation seal

sha256: 4d1014a3eb98cd2461b3adfae4fa284c1b3de5d4b34ca484380f448f31e7f595; engine receipt

sha256: 2f6e0fa6b523ce81d7c4ab48b0413da06863c074cdae580e39ef2f507771498 at

receipts/engine/model_admitted/SFT-CHEM-BOND-METALLIC-001-2f6e0fa6b523ce81.json; empirical hash sha256:9eb7097b10ed3df8d25bce50b38739628d4ad8f55b5339f4db3bb51d976c768e; measurement receipt sha256:27487aabfa2003834b1c4f418af9a100be24bf33823fc49e8f7a13cc38ad35ae.

30. Bond order as retained joining multiplicity

Claim identity: SFT-CHEM-BOND-ORDER-001

Question and theorem. Bond order is the exact positive rational degree of complete joining-channel support for one identified bond relative to one localized electron-pair joining unit, with the observation partition retained.

identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001

Generated grammar. Generate the literal product of the registered bond-order carrier, support, reference, quantity, delocalization, method, record and extension choices.

Boundary: Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree__rational-support-sharing-retained__partition-bound-index__bond-channel-reference-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-bond-carrier	index-without-bond	A number detached from a bond has no meaning.
support	complete-joining-channel-support	selected-channel-count	Omitting channels can force the degree.
reference	single-localized-pair-reference	arbitrary-scale	A free scale can select any order.
quantity	exact-positive-rational-degree	floating-index	Floating arithmetic cannot certify the exact support ratio.
delocalization	rational-support-sharing-retained	integer-only-order	Delocalized support can distribute a joining unit across links.
method	partition-bound-index	one-partition-as-absolute	Different observational partitions can report different indices.
record	bond-channel-reference-trace	order-answer-only	An index alone cannot reproduce support.
extension	no-extra-rule	free-order-correction	A correction can force a conventional integer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e2db93d7acdd9cf1733eff52d8fd837ae587de58815c0f9e206eff130e6bfc39`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0730cf4b586be0e8feb19dc34f293a1d2078cf398ce3ca253c97db2a0c352073`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:60a2093f539375e58bac443e95b50dbf98cd187090e8d91adb8100ffbc1a0d87`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f104e993203597e06f02f2080cf9f28583704674402ade6ceb7e601696867a7`.

Independent reconstruction. Implementation hash

`sha256:c65f2049ad12f2d618472090042a4f0a56dda8e31fabea284b65ecece6942c09`; independent certificate `sha256:cc744913cdce2a2bb18a73bddf76ea9c4e37f2ca0e0069b43d07769c5174fd49`; external-validation hash `sha256:159aa70869792c4cf42278a5f8d708222b95204bb90bec05f8a8502cf8097e2c`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00707-2026

Comparison records:

- `bond-order-iupac-b00707`: predicted `bonding-degree-index__single-bond-reference__localized-electron-pair`; source-derived `bonding-degree-index__single-bond-reference__localized-electron-pair`; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks a bonding-degree index, single-bond reference or localized electron-pair reference, or if a changed row is accepted.

Measurement receipt: sha256:a120f4b897afe2d75d9c56645abd96804f50c3006662c01685120d99d0dd0d05.

Isolation certificate: sha256:f2e6d4e1b7cb627ef32b40549779b2c5a63e70d3f4477c60c16f1133333bfff48.

Custody certificate: sha256:9ad5ecdc91597061cdb61a59b8148a89e9bbb210dee0fcc3bd8817f61aa4c5e9.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/B00707/json> (sha256:f8056279c1ae14cc144d184f01797de9aa2d8fb69ed868b9753ba2e63255dccb)

Registered target identities:

- bond-order-iupac-b00707 from IUPAC-GOLD-BOOK-B00707-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

Immutable identities. Source manifest

sha256:d871f24eb935e7f08a3195bec4a6482219ee569a0333b2f00807c71660bfff421; derivation seal

sha256:7c78f302c8b6954108b1ad8c1255e7799e7ada125732370731e21f77f16ef140; engine receipt

sha256:94f413630420c4d031cc217386f564020c765b8af20b9e107e42a15248ee076c at

receipts/engine/model_admitted/SFT-CHEM-BOND-ORDER-001-94f413630420c4d0.json; empirical hash

sha256:5eb9dba0e7fc2af3173b31cacbb28ef297d39be81679fa51200553ca76b34c10; measurement receipt

sha256:a120f4b897afe2d75d9c56645abd96804f50c3006662c01685120d99d0dd0d05.

31. Bond length, energy and identity correspondence

Claim identity: SFT-CHEM-BOND-LENGTH-STRENGTH-001

Question and theorem. Bond length and strength are separate source-bounded records paired by one specific bond identity: inter-centre distance and positive dissociation transfer under retained methods and conditions; no universal formula is imported.

specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__common-bond-identity-pairing

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-ORDER-001

Generated grammar. Generate the literal product of the registered length/strength carrier, length, strength, conditions, correspondence, measurement, record and extension choices.

Boundary: Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__measurement-conditions-retained__common-bond-identity-pairing__separate-source-bound-records__bond-length-strength-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	specific-identified-bond	generic-bond-type-only	A type average can erase the specific molecular carrier.
length	method-bounded-centre-distance	context-free-distance	Measured bond length depends on method and state.
strength	positive-dissociation-transfer	qualitative-strong-weak-label	A qualitative label cannot reproduce dissociation.
conditions	measurement-conditions-retained	conditions-erased	Phase, state and cleavage path affect measured records.
correspondence	common-bond-identity-pairing	universal-inverse-equation	A categorical relation does not force one universal formula.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
measurement	separate-source-bound-records	decimal-as-proof-scalar	Measured decimals cannot enter derivation.
record	bond-length-strength-source-trace	paired-values-without-provenance	A pair without methods cannot be compared.
extension	no-extra-rule	free-length-strength-fit	A fitted curve can force measured examples.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:7e1b3cf5cdff81e62b593729ae380c6b6151fbb768b02e8bf683fd10d67c2c8f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:6d7e7989df91ca411877aaff4507ef5c4000a87e17468950721c429d0230f768.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:fd7a974bad4e601cb6c1b60bb8f0389b4d00b597b8529173ea28367aedbff8f9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1026b70bc7c6d6d1fcaac20815b08bbbd0f66f656c3ee48a5cb01d4a3d22f86b.

Independent reconstruction. Implementation hash

sha256:1ab7ee39098a0f17e5dc2489b48970f1f57d1444593a8701101beed99fc8fd04; independent certificate sha256:93ff0d4b30b1dc3cadf664a6d1271083634771cc896a074b5af4ad60dc8dc0b2; external-validation hash sha256:2322cd378ddc66143720e6084d3f91d582d9d9801cbff53a85e6fb1a0ae48b75. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00702-2026

Comparison records:

- bond-length-strength-iupac-b00702: predicted bond-length__bond-dissociation-energy__bond-order-empirical-relationship; source-derived bond-length__bond-dissociation-energy__bond-order-empirical-relationship; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks the registered empirical relationship among bond length, dissociation energy and order, or if a changed row is accepted; it does not fail merely because no universal monotone formula exists.

Measurement receipt: sha256:bd6ee84cfb1fb9de9052f1412a4dd673ea49b85210da8a90d81a346348716a5f.

Isolation certificate: sha256:8e095fef1f0d2c5f638d7cb5830d7a85081e2dca40418e5022d8f83adf882cea.

Custody certificate: sha256:970841320705bb9ec75dc76317352a76cb270336e008f70cd9cf03ea8ec416b2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/B00702/json>
(sha256:f53dadb6357406a52d41d9d4be1f7698d4db3b8954b04f798ba823b18c40c821)

Registered target identities:

- bond-length-strength-iupac-b00702 from IUPAC-GOLD-BOOK-B00702-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

Immutable identities. Source manifest

sha256:2fb0972cd501ccaab9ccf8eb29080835cb64a954472f1f57fc6057a55dda2863; derivation seal
 sha256:600aa8b89c38a43f5c825c0f4550ff6465e06160764917147abe3a813823ad49; engine receipt
 sha256:83ea09203b02f1e1f9872073bbb65b2e2609785c15d465de4d006276ee54aada at
 receipts/engine/model_admitted/SFT-CHEM-BOND-LENGTH-STRENGTH-001-83ea09203b02f1e1.json;
 empirical hash sha256:20fe93c8b49082a6682be04e1c9c0e2034c5846046d7133e1d05dda320a16586;
 measurement receipt sha256:bd6ee84cfb1fb9de9052f1412a4dd673ea49b85210da8a90d81a346348716a5f.

32. Molecular identity and complete bonded carrier

Claim identity: SFT-CHEM-MOL-MOLECULE-001

Question and theorem. A molecule is one finite neutral multi-atomic carrier whose complete bonded graph is connected and whose constituent occurrence and element identities remain retained.

multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001

Generated grammar. Generate the literal product of the registered molecule cardinality, bonding, connectivity, charge, composition, identity, record and extension choices.

Boundary: Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained__complete-structure-identity__complete-carrier-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
cardinality	multiple-atomic-occurrences	single-atomic-occurrence	One occurrence has no internal molecular bond graph.
bonding	complete-bond-support	unbonded-collection	A collection without joining support is not one molecule.
connectivity	one-connected-component	disconnected-components	Disconnected components are separate carriers.
charge	closed-charge-boundary	open-net-charge-fibre	An open net fibre describes a molecular ion rather than the neutral molecular class.
composition	constituent-identities-retained	anonymous-atom-count	Counts without element identities cannot reproduce the carrier.
identity	complete-structure-identity	formula-alone	Equal composition need not imply one molecular structure.
record	complete-carrier-trace	molecule-label-only	A label cannot reconstruct atoms and bonds.
extension	no-extra-rule	free-molecule-exception	An exception can admit arbitrary aggregates.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bbed887164a606acb5cca576ab0d7da79e9254ab958d14e368f2d1b686224b7c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d21385a0ac8db65b285a1e5cf9b46eb58fb7d306b196d5c08fd900001d7916b1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:811a93ff2e246131afd67a2e53ecba9ba2d2c083c686188c278f8e537ee69785`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:809480cc84ffd38484c8ef07fe549a9a9e44f9c25a0943073cc937513b52886d`.

Independent reconstruction. Implementation hash

sha256: 22db6e9a5a35d9e670f83677a119a13cb1eb94927c93ed2e9e10636410408ce1; independent certificate sha256: 94b03b38ed013a00a089d3cb57cc45f51fba0e3cacc8ccc0b6697a98eb70c366; external-validation hash sha256: 82e27bfe0c7c5647a8c165bb3d152c7d144675b2faad6e0d9b5f99b0cdf71104. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M04002-2026

Comparison records:

- molecule-iupac-m04002: predicted
multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained;
source-derived
multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular record lacks multiple atomic constituents, one bonded chemical entity or retained constituent identity, or if a changed row is accepted.

Measurement receipt: sha256: 0a83e619481da1819ecc2cbf502875fd3e8bfdd67c1206492aebbed89d05e64a.

Isolation certificate: sha256: c1841e6b99452e3af115174b89db9fe0f70ba921bdb7f1797d4a8462b303f047.

Custody certificate: sha256: 873154cb8a5e98c359916e137641c6ca06699e1fb54d05c94f001d0deca9a5ed.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M04002/json>
(sha256:a5058dad716a0233d1a49fb8869d30b581cff6f483317ffdaaf10a3293c52e36)

Registered target identities:

- molecule-iupac-m04002 from IUPAC-GOLD-BOOK-M04002-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

Immutable identities. Source manifest

sha256: 8c55560f345efdaf63aac199a5026cf893b0402107d6bf9325f71ea39711da44; derivation seal

sha256: 53751882f0e9e131f435d27edd8e68a879348afcc6e857d5acf8372250f84f66; engine receipt

sha256: 5aa379d0b14c341e04c6a041483415f94a04f45833ccb11af5277e53257ee7ce at

receipts/engine/model_admitted/SFT-CHEM-MOL-MOLECULE-001-5aa379d0b14c341e.json; empirical

hash sha256: 34affa520b3d0bf26b1d70594cb1a9060663e08cebcdb01f57e08ae5c40d7b89; measurement

receipt sha256: 0a83e619481da1819ecc2cbf502875fd3e8bfdd67c1206492aebbed89d05e64a.

33. Molecular geometry from held adjacency and orientation

Claim identity: SFT-CHEM-MOL-GEOMETRY-001

Question and theorem. Molecular geometry is the complete relative adjacency-and-orientation form of one identified finite molecular carrier, invariant under a global rigid host-frame relabelling and separate from measured coordinate values.

identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid-rel-abel-equivalence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered geometry carrier, adjacency, orientation, completeness, equivalence, measurement, record and extension choices.

Boundary: Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid*

-relabel-equivalence__source-bounded-geometry-record__adjacency-orientation-source-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-molecular-carrier	geometry-without-molecule	A shape detached from an identified carrier is not molecular geometry.
adjacency	complete-atom-adjacency	atom-list-only	An atom list omits which occurrences are joined.
orientation	held-relative-orientation	distance-only-record	Distances alone can omit handed or ordered orientation.
completeness	complete-pair-support	selected-coordinate-subset	A favorable subset cannot establish the whole geometry.
equivalence	rigid-relabel-equivalence	absolute-host-frame	A host coordinate frame cannot define chemical identity.
measurement	source-bounded-geometry-record	coordinate-as-proof-premise	A measured coordinate cannot select the law.
record	adjacency-orientation-source-trace	shape-name-only	A shape name cannot reproduce the relations.
extension	no-extra-rule	free-angle-fit	A fitted angle can force a target geometry.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 37ade72544361890f8b377d0be9fb8246aad5cccc01e85702f531df5a43806cf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2eef597bf64b1918b1dfd70e164ce86c8560f1d6cfe455e285fb5744b445b4a.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 42d88d1105e6572412e6725c928b48cf33f934003dcae249480f642068779791.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: b52da0b02b3272f33b9bbfb53e964d411909027aac6b11fac7589e1149d05c24.

Independent reconstruction. Implementation hash

sha256: 4f780754facfb7667f6db5ee9eedb8ae99aaf3ea8043383e94de88c4647caafc; independent certificate
sha256: 8031cf31bbfbeb30b574d6ea6d4361d68c9fdb6cda3ce73f851af5f6bb1c512; external-validation
hash sha256: ebe9f30414581e4ed31c537b8d8140ccb10e1d98b7bf74c37a6a04bac972a36f. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06061-2026

Comparison records:

- molecular-geometry-iupac-s06061: predicted
identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record;
source-derived
identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative geometry record lacks identified atoms, relational spatial organization or source-bounded coordinates, or if a changed row is accepted.

Measurement receipt: sha256:754bd429fa783348692940e3a757fbadb588fc1cd3d825a5db48053eacf78528.

Isolation certificate: sha256:84176b1e992d227e441c27274b7af000342dc0ca13f2a2ac572f0d4bbabf7277.

Custody certificate: sha256:c08889b7bb14bf7a4c0193bb6f2461f21670d180a912b7f699ebfa877d4aad2a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S06061/json> (sha256:65cc9feaea028f9ab7bd7ed34927652255ebcc66d2f079dc94d55fe9dc7230b)

Registered target identities:

- molecular-geometry-iupac-s06061 from IUPAC-GOLD-BOOK-S06061-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

Immutable identities. Source manifest

sha256:1f5e5f023b5b2dc5d4a9e4125a3642632913cdd52ce054a352ddc9b02c38c380; derivation seal

sha256:7c7c69d3817c8e880c5621d6516d1773cf534ef508972d15409ca78c271f8cc2; engine receipt

sha256:32416047fc77a3215966466b4a829e4658bcfb50c14652b3425b9aa1e8603a89 at

receipts/engine/model_admitted/SFT-CHEM-MOL-GEOMETRY-001-32416047fc77a321.json; empirical

hash sha256:e57949262e42f976fe560cebebf3e96e91920b87aaaba582d0688c8fd102d961; measurement

receipt sha256:754bd429fa783348692940e3a757fbadb588fc1cd3d825a5db48053eacf78528.

34. Isomer distinction under equal composition

Claim identity: SFT-CHEM-MOL-ISOMER-001

Question and theorem. Isomers have exactly equal retained composition but inequivalent complete connectivity or orientation signatures after every identity-preserving finite relabelling.

two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings__connectivity-or-orientation-difference

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-GEOMETRY-001

Generated grammar. Generate the literal product of the registered isomer pair, composition, structure, equivalence, distinction, identity, record and extension choices.

Boundary: Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings__connectivity-or-orientation-difference__separate-chemical-identities__composition-and-structure-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
pair	two-identified-carriers	single-carrier-description	Isomerism compares at least two chemical carriers.
composition	equal-exact-composition	different-compositions	Different composition is not isomerism.
structure	complete-structure-signatures	formula-equated-identity	A formula does not exhaust bonded organization.
equivalence	all-identity-preserving-relabelings	name-string-inequality	Different names can denote the same structure.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
distinction	connectivity-or-orientation-difference	equivalent-signatures	Equivalent complete signatures denote the same isomeric form.
identity	separate-chemical-identities	distinction-erased	Erasing the difference destroys the chemical question.
record	composition-and-structure-trace	isomer-label-only	A label cannot reproduce equality and difference.
extension	no-extra-rule	free-isomer-class	A free class label can manufacture a distinction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:cff91e3ddb362ea4243a3a751487636d8cbe5db830e215e91e1e8691584f0dcf.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c2dbbbc0bbb52a72785bc5da5d04267b55b28aa26d80002fe05bd5ed7543bbcc.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dc161f389ebcdec14659c79d57ce7f512f7f193d1ed845be8b7ae8da6361f5e3.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ab3dfb81b50681603de1f62d122810cad4e7ad1bfa16ac3d1abc9d1abe830290.

Independent reconstruction. Implementation hash

sha256:7aa0f2f5ddb4f83e66d9ad7ac46a812691bd7707808f34962af813d170388c67; independent certificate sha256:c465beb4071d7c4bbdf65b73e04f4e05d9f999142100dbbc350ac265c948d679; external-validation hash sha256:2a1fb4a835a698f1634a646f736dc6c10fb8fc4818de777f3920c7f3f9eab559. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03289-2026

Comparison records:

- isomer-iupac-i03289: predicted
equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity;
source-derived
equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official isomer record does not require equal composition with distinct structure or spatial arrangement, or if a changed row is accepted.

Measurement receipt: sha256:98b7ffae634ea86206b13f92518e34e5f2d1b65a4f5e2ace3b00f8cdf2895d62.

Isolation certificate: sha256:fcdd26096b1379c7435d33ba9361489fb6708319418afb7f7a4915d4833ca0e7.

Custody certificate: sha256:b9b37e84c0cb3f62e42776f51416789146cca89a56f2da537e7507f90175b448.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03289/json>
(sha256:64fa2c9cc58b18b399d22481ee7f37692c7f9c904e4d5fff27c65dd3325393e7)

Registered target identities:

- isomer-iupac-i03289 from IUPAC-GOLD-BOOK-I03289-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

Immutable identities. Source manifest

sha256:daf9b16232a55caa127373480a5fa56e69c990c5c78385a0ce31c76ca3f61746; derivation seal
 sha256:3b90b3058db11528d99a744d04912eddc94fdb979eb6305c822300143f77e849; engine receipt
 sha256:9a9604de7b5433fa0f6630c3840f91143bed30acc853ff8f4ad3495544536f3 at
 receipts/engine/model_admitted/SFT-CHEM-MOL-ISOMER-001-9a9604de7b5433fa.json; empirical hash
 sha256:ab05cee07b44a2b1a227414f5f17a4c3639320518f0c1d84ee4507a78eb22ce7; measurement receipt
 sha256:98b7ffae634ea86206b13f92518e34e5f2d1b65a4f5e2ace3b00f8cdf2895d62.

35. Intermolecular interaction and residual joining

Claim identity: SFT-CHEM-MOL-INTERMOLECULAR-001

Question and theorem. An intermolecular interaction is a registered response channel between already complete molecular carriers that retains each constituent molecular identity and supports collective recurrence.

distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001

- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered intermolecular carrier, support, identity, recurrence, boundary, strength, record and extension choices.

Boundary: Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary__source-bounded-interaction-record__carrier-channel-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carriers	distinct-molecular-carriers	one-molecule-internal-edge	An internal edge is intramolecular support.
support	between-whole-interaction-support	proximity-without-channel	Proximity alone supplies no interaction trace.
identity	constituent-identities-retained	constituents-merged-erased	Erasing each molecule changes the class to a new bonded carrier.
recurrence	collective-recurrence	transient-coincidence	A coincidence without response cannot organize an assembly.
boundary	explicit-molecular-boundary	inter-intra-conflation	Conflation makes every covalent edge intermolecular.
strength	source-bounded-interaction-record	universal-weaker-law	Not every context forces one strength ordering.
record	carrier-channel-recurrence-trace	force-name-only	A force name cannot reconstruct endpoints and response.
extension	no-extra-rule	free-force-family	An imported family can select desired interactions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8c200aa3e761e0e8e583df86cea1e401ef6ac0666629e7e42bf01790e147cfaa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:66170d1985dd0b0b805f940cb21d00d932c3bc1d354a0ee4387e3f3b857262c0`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : d5722e6b4ee1020dd3d52b8d3223fe482d257809ca683e78114888c2684038a1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : fa996e1ad0ab9e877ebe04f8ca07a8deb28146632f4b5b5e8a2ee67cc8a7cb34.

Independent reconstruction. Implementation hash

sha256 : d215c873502192ac1d7219b514b59406b547ce9b6be65c68c8902105a89cde0f; independent certificate sha256 : 1eb83e9304bfe476b0b2cd840e20e5867dd505fb5a069a8f8f95d35db08aeb49; external-validation hash sha256 : 0cd07555765f7f509d1845af1df851a092d3f265ba0815ec0d761fa9716f186. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03098-2026

Comparison records:

- intermolecular-iupac-i03098: predicted
distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence;
source-derived
distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official record lacks interaction between distinct molecules or preserved molecular endpoints, or if a changed row is accepted.

Measurement receipt: sha256 : 65d7ac7e6aaed6f7c27647ba3d5d7dfca452b9e02cefabea2145fe4bc57d0126.

Isolation certificate: sha256 : 0e5f0f13353f306255f7a322f5a2d28d5208caa83bbdc70c629a7b3375a6a760.

Custody certificate: sha256 : d12f4da2ed33d511ff213173300012d603acba468e9ab374f7b8fbb17a41d447.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03098/json>
(sha256:ba7264c36447a463893ffdf66e0bce7478e764db59b928911a8f82b59af4b3e)

Registered target identities:

- intermolecular-iupac-i03098 from IUPAC-GOLD-BOOK-I03098-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

Immutable identities. Source manifest

sha256:09a428c01beb574236e6d875f85e3f30281d2afa9c24e3ca3ca41b8d10365f77; derivation seal
 sha256:b3845294f802bd5cbb48345120d9351b6f86e33dc0fc7a3e5abf10bd4588749c; engine receipt
 sha256:ba1eabd232574b380b85aff50567257eb0b4c5318ba79e238410a737c38ea7ef at
 receipts/engine/model_admitted/SFT-CHEM-MOL-INTERMOLECULAR-001-ba1eabd232574b38.json;
 empirical hash sha256:de6ce0f2148ee584439dbc21a4562e84e201f9fb05b120a83ee9ed54b4435a3d;
 measurement receipt sha256:65d7ac7e6aaed6f7c27647ba3d5d7dfca452b9e02cefabea2145fe4bc57d0126.

36. Supramolecular organization by reversible recognition

Claim identity: SFT-CHEM-MOL-SUPRAMOLECULAR-001

Question and theorem. A supramolecular whole is a finite assembly of multiple intact molecular carriers joined by reversible complementary recognition support, with all components recoverable.

multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001

- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

Generated grammar. Generate the literal product of the registered supramolecular components, joining, recognition, reversal, closure, scope, record and extension choices.

Boundary: Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity__finite-declared-assembly__component-recognition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
components	multiple-molecular-components	single-molecule-only	A supramolecular assembly contains multiple molecular components.
joining	noncovalent-interaction-support	new-internal-covalent-graph	Replacing components by one covalent graph erases the supramolecular boundary.
recognition	complementary-recognition	arbitrary-collision	A collision has no complementary selection trace.
reversal	reversible-component-release	components-unrecoverable	If components cannot be recovered, their identities were not retained.
closure	assembled-whole-identity	components-only	Separate components do not form an assembled whole.
scope	finite-declared-assembly	completed-infinite-assembly	A completed infinity is not generated.
record	component-recognition-trace	assembly-name-only	A name cannot reconstruct recognition and release.
extension	no-extra-rule	free-host-guest-rule	An exception can force any desired assembly.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4624527ef8a225f350b93ce389704a0ca9f1c3f27a4fea4781d818ed9fd8b1d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:39aa04f9f35d247da547a24dd03d6d7994e5812a23cd1563bb5fd6a73ee2f846`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:af9101491f206ffe81a9ab14550635b603e45d1c4ee776f501cb4326b4556fdb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d14f7bef34eb9b65bd346b19ba3fc422b78107c054db310315f56e20fd7e11ec`.

Independent reconstruction. Implementation hash

`sha256:0a59a72d662c2a0a8f04c59801c5cf4384a68ee42c609bc133443b8c3589b0e4`; independent certificate `sha256:1d0d5cadf95fd4ea413ff2a3d3fcb215d96a64f1e82e125012659bd464ffa85c`; external-validation hash `sha256:f7abcbdbd1385de8ccb0fc0c71462f91c8a4f9fa6839e5181c2e26d080baa871b`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-13867-2026

Comparison records:

- supramolecular-association-iupac-13867: predicted
multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole;
source-derived
multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole; exact
match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official supramolecular record lacks multiple molecular components, noncovalent organization or retained component identities, or if a changed row is accepted.

Measurement receipt: sha256: fe72bd84f6bd8a08a2daf4fe24ebf8643641c4f623bfec35a75fe37fb77ef829.

Isolation certificate: sha256: 9e8fac94eeb9b3319898781caf37f407c79baf98ca43f6fd46b87451325bc9f4.

Custody certificate: sha256: 4c43539c1e8c358ca5ddc4f8c4b60c1bf035777d3d1936c3dcc87d58340ff926.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/13867/json>
(sha256:34dd0e2a436c762eefbb6ddf8ab84b16b716b252632fc0835ec083d4d4a2551a)

Registered target identities:

- supramolecular-association-iupac-13867 from IUPAC-GOLD-BOOK-13867-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

Immutable identities. Source manifest

sha256: ef01da7f61edfa7a0bb00febaa3923c6f9494745cb856e97390422195c613802; derivation seal

sha256: d600cd2296f751be3bbbc930c434a149828066f7103a0cf968ce8a0eedc1aaef; engine receipt

sha256: 03bc9e8b4e9ac2b3ef0a87f81bafeba98a31955ed55b226936c51f4f6dea045b at

receipts/engine/model_admitted/SFT-CHEM-MOL-SUPRAMOLECULAR-001-03bc9e8b4e9ac2b3.json;

empirical hash sha256: 24a3138fd3845b956857932d3beef77d0dd313d332bfb1a1ecec9413382ccf83;

measurement receipt sha256: fe72bd84f6bd8a08a2daf4fe24ebf8643641c4f623bfec35a75fe37fb77ef829.

37. Molecular network and connected chemical whole

Claim identity: SFT-CHEM-MOL-NETWORK-001

Question and theorem. A molecular network is one finite maximal connected component of named molecular nodes and registered interaction edges with complete reachability and provenance.

finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-census__finite-maximal-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001

Generated grammar. Generate the literal product of the registered molecular-network nodes, edges, connectivity, paths, boundary, composition, record and extension choices.

Boundary: Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: *finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-census__finite-maximal-boundary__node-identities-retained__complete-network-trace__no-extra-rule*. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
nodes	finite-molecular-node-set	anonymous-node-count	A count cannot recover molecular identities.
edges	registered-interaction-edges	implicit-relatedness	An implicit relation cannot be audited.
connectivity	connected-maximal-component	disconnected-union-called-one	Disconnected components are not one connected whole.
paths	complete-reachability-census	selected-path-only	One favorable path cannot establish network connectivity.
boundary	finite-maximal-boundary	unbounded-network-name	An unbounded name does not enumerate a finite object.
composition	node-identities-retained	network-erases-molecules	Nodes remain chemical carriers, not anonymous points.
record	complete-network-trace	network-answer-only	A connectivity answer cannot reproduce the network.
extension	no-extra-rule	free-connectivity-edge	Adding an ungenerated edge can force connectedness.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: aecb155adfee2fa5c2569fcce2754efa066481a4569263794de20d763d1eb430.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 88618a1b00901ee0c581709ea8f5ed231b4d9e37c51f0c9d5a4ec7d4bcb94730.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 0d5086ed6a62b7e51f3ae5bef5470bf27fc01435c48ec2fd806f412699448e52.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 134c1b01b41fb77ae299a443f0fe0be5edc83e08dd803e6331c14bdef0cef7e.

Independent reconstruction. Implementation hash

sha256: 4803b87a071efe0f05f28748a53f932a80b12b5025d108a6bceb53444e66f1cf; independent certificate sha256: 1c37d683d2ac8ff6c01fb6d7e2afe60fa7ea9d0dc2d0ba54915c360000eaacdf; external-validation hash sha256: 800745f1353e9c562fa7aaa3fcf1e36bea7c949cc9b19d1da62c601ed3d10222. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-N04112-2026

Comparison records:

- molecular-network-iupac-n04112: predicted
finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace;
source-derived
finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular-network record lacks molecular nodes, interaction links or connected organization, or if a changed row is accepted.

Measurement receipt: sha256:712805a35f631a8453bbd1d28c9949b1f18f3cd8bb0701420342432ef5e6367c.
 Isolation certificate: sha256:f2ed1f3080498cff0b42b999c2dc09f5abdeae96f3b6ce9644412ef5d07f541b.
 Custody certificate: sha256:e28b6e1bb189bf2708b659fe46f141792eea9ea12591fb2bbbed521a5d0e33cc1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/N04112/json> (sha256:a146d63f8340f0cf5043e1f9fe2f1cfa8e5fa7c3bf27b8a409e665f40ae2c046)

Registered target identities:

- molecular-network-iupac-n04112 from IUPAC-GOLD-BOOK-N04112-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

Immutable identities. Source manifest

sha256:3643fb6d040ae6d9e29e381e91403a69074a6e3d83482714d345f4e1f54e1bfff; derivation seal
 sha256:7485b6cd62d05d1076ddd8a17d89baef6ded9be36adf7a1022549417fddc76be; engine receipt
 sha256:b8aca3860e24266b5d87c32e1ed0dff589097626ee821b8472a8849132ca0041 at
 receipts/engine/model_admitted/SFT-CHEM-MOL-NETWORK-001-b8aca3860e24266b.json; empirical
 hash sha256:ce8ed0c73af4cc4a3e2a77988b055e034813888ba52cbaec158fcc452573363c; measurement
 receipt sha256:712805a35f631a8453bbd1d28c9949b1f18f3cd8bb0701420342432ef5e6367c.

13. Acid Base Redox

Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier.

38. Conjugate acid-base partition

Claim identity: SFT-CHEM-AB-ACID-BASE-001

Question and theorem. A conjugate acid/base relation is a pair of retained chemical identities differing by exactly one named proton carrier: the protonated member donates it and the deprotonated member accepts it.

two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

Generated grammar. Generate the literal product of the registered conjugate pair, carrier, acid role, base role, difference, orientation, record and extension choices.

Boundary: Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton__held-donor-acceptor-orientation__pair-and-proton-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
pair	two-conjugate-identities	one-species-label	A conjugate relation compares two carrier identities.
carrier	one-held-proton-occurrence	anonymous-hydrogen-count	A count cannot trace the transferred occurrence.
acid-role	proton-donor-role	acid-by-name	A name cannot establish a transition.
base-role	proton-acceptor-role	base-by-name	A name cannot establish a transition.
difference	differ-by-one-proton	unbounded-composition-change	Additional changes destroy conjugacy.
orientation	held-donor-acceptor-orientation	signed-acidity-scalar	A signed scalar is not a carrier trace.
record	pair-and-proton-trace	acid-base-answer-only	An answer cannot reconstruct conjugacy.
extension	no-extra-rule	free-acid-base-exception	An exception can label arbitrary species.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c636c0c317cb207c4839f718a739e5a683b6040155125536f9280ca737ac0e4c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 05b2c6729d885da3646df12d9ee9a76888e057a0da289866cd518789a4558085.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : f316b770b291518f418e84e5b494397d05ba8c8ec8525bd5a01ff8d30b72f537.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 408b29d1a7a54536d715a8da4f7f9947a771419db6481093b17a405f5ae716f4.

Independent reconstruction. Implementation hash

sha256 : 69b4921f1868f68bf465f91a281a6c5f67e8da2de9f06b518a49325116c84b52; independent certificate sha256 : 88152d90d6de4349d732338892b209665ac0f6dff91ab5a7099a95bf1bfd25c4; external-validation hash sha256 : 0a138d9393293cbf97515a96c0640e5006a047eae2c093cae2fcce0201fff7775. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Comparison records:

- conjugate-acid-base-iupac-composite: predicted
proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; source-derived
proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative conjugate acid/base record lacks proton donation, proton acceptance or a one-proton conjugate relation, or if a changed row is accepted.

Measurement receipt: sha256 : bfd39f473b2a824e2dc08fb73a6b5953e41112556bf93ecce09cd63dc43c3583.

Isolation certificate: sha256 : 06b0a1e27a4cee94ced3a9a3a2c26e6f30c32a4b580c7822559f02a105ca5dae.

Custody certificate: sha256 : 4108990961c157bd1e0b13333bc3ad0a9cbf8374f9e750c1da09abc435da09af.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01266>
(sha256:1ac0d6138e316d6230d10b2e7f99d4837175fcb640e824a8cb2249100221a504)

Registered target identities:

- conjugate-acid-base-iupac-composite from IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value

- external target content remains inaccessible until the prediction is sealed
- Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

Immutable identities. Source manifest

sha256:08459031c560cd3795677b3e9f2daac98746438dfef7bdc06ff3f87853fdac63; derivation seal sha256:76cfcdb1dedf829c2eb2b856cf3b614fe25284574d69d330651d2913b426dd53; engine receipt sha256:2f52c0900674b54387bc0168fc5c57ecc9ced3819c0423e31aa7bb0f39b9fdf7 at receipts/engine/model_admitted/SFT-CHEM-AB-ACID-BASE-001-2f52c0900674b543.json; empirical hash sha256:66b231bbb4bc7172986aafbe91d31f8caecafa83ee356ca5b9ffae7c5378e1b5; measurement receipt sha256:bfd39f473b2a824e2dc08fb73a6b5953e41112556bf93ecee09cd63dc43c3583.

39. Proton-transfer acid-base relation

Claim identity: SFT-CHEM-AB-PROTON-TRANSFER-001

Question and theorem. Proton transfer moves one named proton carrier from one registered donor occurrence to one distinct acceptor occurrence while conserving carrier identity and both endpoint records.

one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001

Generated grammar. Generate the literal product of the registered proton carrier, source, destination, transition, identity, closure, record and extension choices.

Boundary: Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after__complete-endpoint-closure__carrier-endpoint-transition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-named-proton-carrier	proton-count-only	A count cannot establish conservation of an occurrence.
source	registered-donor	source-erased	Without a donor, the carrier could be created freely.
destination	registered-acceptor	destination-erased	Without an acceptor, the carrier could be lost.
transition	donor-to-acceptor-transfer	copy-or-delete	Copying or deleting violates carrier conservation.
identity	same-carrier-before-after	before-after-carrier-change	A changed carrier is not transfer conservation.
closure	complete-endpoint-closure	unpaired-composition-change	Unpaired change leaves the reaction record open.
record	carrier-endpoint-transition-trace	product-label-only	A product label cannot reconstruct the path.
extension	no-extra-rule	free-proton-source	A free source can force any reaction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 69777e7c94f864618f23bb8c316cb4755aa69932bbd5024824f7d651970eb1ed.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: c4d8f6608680b8ea69ff9a3d1aff3134aeb226b21fc7553992373580dbbb6b56.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 9e07bea854076486d00d939b92c32227e48d5576783837be7dc0119c913f1f5f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7183352ddde61f7554681ce6e9790f2a1b8e9c929bdf68cb50845f22b1c10ca3.

Independent reconstruction. Implementation hash

sha256: 4256a0b14274123ec7c79959d029bf6359f031402bce163da90a11f367076c4d; independent certificate sha256: 59456ef8b2f6dbb57ed62c61f0c0384384fd8aa61d956eb93ad1981bf8a4102a; external-validation hash sha256: 9fffb40fdfac3a2ca4d200b00134403d075d06a6967abe711854e93ae994710af. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-P04915-2026

Comparison records:

- **proton-transfer-iupac-p04915:** predicted
one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; source-derived
one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative proton-transfer record lacks a proton donor, proton acceptor or conserved transfer relation, or if a changed row is accepted.

Measurement receipt: sha256:27a7bad3dc157b9bbb078e311faa01b52b090e49db53b82e626cb7f4ed513d1.

Isolation certificate: sha256:da496511cabd9493a6c94231218cf200d4cc1da32cc5c24a6b92ff8ed59269e2.

Custody certificate: sha256:a0c0faa1c58b19f307ca52e35599f0e7c27616c591c14e8e095ea14c744f2b3b.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04915>
(sha256:72d9b1431f3daa6b0190094d8400bc73b31e786b7d69b4052632297b4b709d33)

Registered target identities:

- proton-transfer-iupac-p04915 from IUPAC-GOLD-BOOK-P04915-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

Immutable identities. Source manifest

sha256:98afb03693336d7bab7a816aad595a967794e469dd77fb106019d9a07b0125c5; derivation seal

sha256:97df51a71af6e82a16c67b9cb05ebce3b37ab689e84cbaf2676b164f69a65fb6; engine receipt

sha256:bc5be8027f456ae522262e597279b16717f0831b2edf5008d8c268bc58d9814d at

receipts/engine/model_admitted/SFT-CHEM-AB-PROTON-TRANSFER-001-bc5be8027f456ae5.json;

empirical hash sha256:b376664238e671b494333ebee4d4e794574a78dce1eaa40f5b7b7b50cde4670;

measurement receipt sha256:27a7bad3dc157b9bbb078e311faa01b52b090e49db53b82e626cb7f4ed513d1.

40. Electron-pair donor-acceptor relation

Claim identity: SFT-CHEM-AB-LEWIS-001

Question and theorem. A Lewis acid/base relation joins a distinct electron-pair donor and acceptor through the same two exclusion-preserving electron occurrences, with both identities retained.

*distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transfer
red-pair*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001

Generated grammar. Generate the literal product of the registered Lewis endpoints, support, donor, acceptor, joining, conservation, record and extension choices.

Boundary: Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transferred-pair__both-electron-identities-retained__endpoint-pair-joining-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
endpoints	distinct-donor-acceptor	one-unoriented-species	The relation requires donor and acceptor roles.
support	two-electron-occurrences	single-electron-only	The registered relation concerns paired support.
donor	pair-donor-support	electron-pair-name-only	A name cannot demonstrate available support.
acceptor	pair-acceptor-support	unregistered-vacancy	An unregistered destination cannot close the interaction.
joining	shared-or-transferred-pair	copied-electron-pair	Copying violates exclusion and conservation.
conservation	both-electron-identities-retained	electron-identity-erased	Erasure cannot reverse the relation.
record	endpoint-pair-joining-trace	Lewis-label-only	A label cannot reconstruct pair support.
extension	no-extra-rule	free-orbital-premise	An imported orbital rule can select a desired classification.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2cc98aa40593481481ad39b0cb07b7ce428222ea723733f562c37b1394b2b6b6.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:95cc34d98ca5a838b98c1074a1db31ed70571fbb0c387b70c5e910d134f18cab.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c1fa64533d715cd9e3911fa10b34f9b093332ff954c844029a150f2f90ab4fcf.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:abc13b877a1bebe4f04b54c9618bdc2137ed324f325e4d0f3499a6153c8680f6.

Independent reconstruction. Implementation hash

sha256:5e16152a66733442a6bde36d7de224f715f0f64f53c671e108eb36c57deef79f; independent certificate sha256:8bfe72a6239326e94c2ec7358fbed27e047702cb5ccf66ba4f41343be1deaa26; external-validation hash sha256:2107b13390cccd4a90480875a9e463b2c1f442858271c49024c3b68661f97025. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Comparison records:

- lewis-acid-base-iupac-composite: predicted
electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; source-derived
electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative Lewis record lacks an electron-pair donor, electron-pair acceptor or conserved pair relation, or if a changed row is accepted.

Measurement receipt: sha256:7e1da91cd604bf1a3d0c5b196dfe2a0f0f7632670886763eeef0ab88649f93e7.

Isolation certificate: sha256:d404805c02d76dbdba92d562da70a1c79b219d2e185760709ba33901f23d6332.

Custody certificate: sha256:88b6adbf5b0ca93e411fde367c64a519f74c24348f4b5ef93993eb9223a8a223.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/L03508>
(sha256:9fa013b12d6d0e3f305883a0db53dac7484e292e0288e2c3d0ba9625cfb2011c)

Registered target identities:

- lewis-acid-base-iupac-composite from IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier

- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

Immutable identities. Source manifest

sha256:e98cb4175058cfc641fd6d022f278c4f0098bd15d9d42e2e8c69982ecc03352c; derivation seal
 sha256:75683df9ae25bb776f74e5fb9971972ccd137cea37b5c50ab35c08ca4545cace; engine receipt
 sha256:05e8f96b150320e2809549b58d4e7f47862b5714153a8382309f958e1cc31959 at
 receipts/engine/model_admitted/SFT-CHEM-AB-LEWIS-001-05e8f96b150320e2.json; empirical hash
 sha256:05d6954874a144897c0cdc4eb2e2bf0a304e0fb1f9a8f6a1dcb0840527ed22f7; measurement receipt
 sha256:7e1da91cd604bf1a3d0c5b196dfe2a0f0f7632670886763eeef0ab88649f93e7.

41. Amphoteric dual-role relation

Claim identity: SFT-CHEM-AB-AMPHOTERIC-001

Question and theorem. An amphoteric species is the same retained chemical carrier appearing in two separately held contexts, once in a valid acid role and once in a valid base role.

same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001

- SFT-CHEM-AB-LEWIS-001

Generated grammar. Generate the literal product of the registered amphoteric species, acid context, base context, context separation, duality, identity, record and extension choices.

Boundary: Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized__species-identity-retained__dual-event-context-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
species	same-species-identity	two-unrelated-species	Dual behavior must belong to one retained carrier identity.
acid-context	valid-donor-event	acid-label-without-event	A label does not establish donation.
base-context	valid-acceptor-event	base-label-without-event	A label does not establish acceptance.
context	contexts-separately-held	roles-conflated	Conflation can assert incompatible events simultaneously.
duality	both-roles-realized	one-role-only	One role does not close amphoteric behavior.
identity	species-identity-retained	species-changed-between-roles	A changed carrier does not establish dual capacity.
record	dual-event-context-trace	amphoteric-label-only	A label cannot reproduce both contexts.
extension	no-extra-rule	free-dual-role-exception	An exception can admit any species.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 5ade57913bbd3c0150acc9c9fb75df8d67ed170ae255e8b89209829a95841595.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f922ab1bf0af3b619c30a84ce5dabf9646cfe4aeedce383f3029f901c58cac3d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 1ad519e6b1b5619447797c273c1f06084e45dfa0283bef8c23618549db219f5e.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 8ea2d3cfbcf557a1aa8f9c6a6400a0ebca9d6921702a109600abd4a47e87c374.

Independent reconstruction. Implementation hash

sha256: ac8f026dbc2ed95ba69ffdfaacc29bd5e7de7d5699e902f37d1b09665cf4528db; independent certificate sha256: 48312bf23953149b2f7e9b86e4955f1fa4c4a2d0c1fc994bb1035905b0bdc8ac; external-validation hash sha256: a85ed53689c1d3bb8a41d06a081216e29135af2bc234686d8bceba3f0c6db119. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00306-2026

Comparison records:

- amphoteric-iupac-a00306: predicted
same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; source-derived
same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative amphoteric record lacks acid and base behavior by one species under retained contexts, or if a changed row is accepted.

Measurement receipt: sha256: ebe83d8610812de30385fc9a306ef23658a714603f04385dd7932e754b661651.

Isolation certificate: sha256: 0e3e33ee179d3f0922e56c3784d3ab91b7760b0e2e0dd4c829daa39378872377.

Custody certificate: sha256: 5c071a5b14f36df3201ef1f1255aad9d9738746d79e213533d4aa1936f0ab4a4.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00306>
(sha256:70ab5021338400d9f02cbecc5c45b9f4271792e919d880aafff81684739022ab)

Registered target identities:

- amphoteric-iupac-a00306 from IUPAC-GOLD-BOOK-A00306-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

Immutable identities. Source manifest

sha256: 4d6bf3fa0ddc8b638be3957bf07a5aa1fd7235ea9fea906a160ee523e387db6a; derivation seal

sha256: 62fa5affb388260891509ffa90941ac314c52fad8f8ab8db3791c8525e6421f5; engine receipt

sha256: 9633c37ce51efb4c0d0109e6cecd161e45d21de52ce7c1d6b732235e7f9837dd at

receipts/engine/model_admitted/SFT-CHEM-AB-AMPHOTERIC-001-9633c37ce51efb4c.json; empirical

hash sha256: 8e94f3de67b0214a8db493d5f07e34b30b581905d05489027544d930cba7c6d1; measurement

receipt sha256: ebe83d8610812de30385fc9a306ef23658a714603f04385dd7932e754b661651.

42. Buffer response and conjugate-pair retention

Claim identity: SFT-CHEM-AB-BUFFER-001

Question and theorem. A buffer is a finite positive reservoir of both members of a conjugate pair whose complementary paths consume added proton support or supply proton support to added base while retaining pair identity and finite capacity.

conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-AMPHOTERIC-001

Generated grammar. Generate the literal product of the registered buffer components, support, acid response, base response, retention, capacity, record and extension choices.

Boundary: Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained__finite-source-bounded-capacity__pair-count-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
components	conjugate-pair-reservoir	single-unpaired-species	One species cannot supply both complementary responses.
support	positive-support-of-both	one-component-absent	An absent member leaves one response unavailable.
acid-response	base-member-accepts-proton	added-proton-unanswered	No response permits the proton distinction to remain open.
base-response	acid-member-supplies-proton	added-base-unanswered	No response permits proton removal without paired change.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
retention	conjugate-pair-retained	reservoir-identity-erased	A response without conjugate tracking cannot be reversed.
capacity	finite-source-bounded-capacity	unlimited-buffer-claim	A finite reservoir cannot have unlimited capacity.
record	pair-count-response-trace	pH-answer-only	A pH value cannot reconstruct composition and response.
extension	no-extra-rule	free-equilibrium-fit	A fitted equation can force a measured response.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:15d9bacd7e2b1a04d502efb2b9edf19cc831eb2248f12f176669be2ed55b7727.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7e69a9d59c2e2894ca3ea5c30cf7467cf5128645d30c60e368cc800a0eb067ed.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1f4d2e4ef0767fc1a44344a407756603a283dad0ee6c5407c8604ac5338bc788.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:696c34f1a9d474e9875c4d03a58e6b5215831e6504ed23e52477dd7d20c84fe8.

Independent reconstruction. Implementation hash

sha256:8646664e19c6946f77ce8eeb44b31ed57ac3ef2d67c745f9aa666f3a3246d09; independent certificate sha256:393f76310d7a38322171e44145f2cfcc0652668cdd4f8bdca9cd8eeeaeb112d6; external-validation hash sha256:13ee1a405b692d5e6debbba2ed178b73f55528d90fd46e95bb980242be0927389. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-DIDAC-E15-2026

Comparison records:

- buffer-iupac-didac-e15: predicted
weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity;
source-derived
weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative buffer record lacks a weak acid/conjugate base pair, response to added acid/base or finite composition-dependent capacity, or if a changed row is accepted.

Measurement receipt: sha256:a7d7a39d4e3f869e1e0a7ab1fe909342503280e66423babd9d3452fb85a56b03.

Isolation certificate: sha256:25359ccbb35e3a8a10125c2aa86ee066ac906e9e415ad3382e014cba6c4df7cc.

Custody certificate: sha256:2c18aab35d14550dd2373ed5059fd1ff351d566c29a23a930a9587cd5a9b91ff.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://old.iupac.org/didac/Didac%20Eng/Didac02/Content/E15.htm> (sha256:570a91d1a7de35b22a8469c117f10af9e4bed3012ae61f54362fe45534b3d01e)

Registered target identities:

- buffer-iupac-didac-e15 from IUPAC-DIDAC-E15-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

Immutable identities. Source manifest

sha256:dfd3b4141fab3d6c698fe25ad20f011d5eda39d7c3c557a6610576736d955c69; derivation seal
 sha256:d42c6cc2ed4c8a44fd95cd60a82c910ac9062dfd1eed14e9a1b426f59ecc0bc3; engine receipt
 sha256:c1503c569172a5cf32d266206aed2383ef3ca4998987c70907e138c93d930444 at
 receipts/engine/model_admitted/SFT-CHEM-AB-BUFFER-001-c1503c569172a5cf.json; empirical hash
 sha256:b4ac25bc6410610189cbd06724947ec31532bd88a8b1b341b9a74a6d8c12633b; measurement receipt
 sha256:a7d7a39d4e3f869e1e0a7ab1fe909342503280e66423babd9d3452fb85a56b03.

43. Electronegativity ordering

Claim identity: SFT-CHEM-ELECTRONEGATIVITY-001

Question and theorem. Electronegativity order is the transitive preorder of complete pairwise endpoint recurrence for identical shared electron support; equal traces remain an exact equivalence class rather than receiving an arbitrary rank.

*two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-pref
 erred-endpoint__transitive-preorder-closure*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

Generated grammar. Generate the literal product of carrier, support, comparison, orientation, tie, extension, record and extra-rule choices.

Boundary: Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-preferred-endpoint__trace-equivalence-class__transitive-preorder-closure__complete-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carriers	two-named-atomic-occurrences	anonymous-element-values	Values erase endpoint identity.
support	one-shared-electron-support	free-affinity-scalar	A scalar imports the ordering.
comparison	complete-endpoint-recurrence-comparison	name-selected-order	Names cannot select preference.
orientation	held-preferred-endpoint	signed-difference	A signed magnitude violates the proof domain.
ties	trace-equivalence-class	forced-strict-order	Equal complete traces cannot be ordered uniquely.
extension	transitive-preorder-closure	pairwise-cycle-admitted	A cycle destroys lawful ordering.
record	complete-comparison-trace	rank-answer-only	A rank cannot reconstruct comparisons.
extra	no-extra-rule	free-scale-constant	A scale constant can tune any ordering.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0b4cdc1f4aee52d241ee6c50286036754d52606fb34b3123efdf8788925569`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:75b4db050fe85f16506da2c636dce9c265cf2318fc57636085303aa2888a7b34`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e328e529b6c738adc4b931a31da3aa43513455bfb2797dff915c4b9c5f39bc88.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : c1c883deac6b9242a292b10d7a1e05a83ca81c2276fad9e8e34732e35f6f959f.

Independent reconstruction. Implementation hash

sha256 : a07b6ba2f87119929254b9ab82389f69c9e26a2659a799af6cfd6fdee3b42622; independent certificate sha256 : f09179fb7686c4719baaceff6b28302171cbb14392ab3eb079be703c9f8acc70; external-validation hash sha256 : 6c345e0ee142a0f452732d235c9dc4fab7bcb10bb8959db65495d75368eacb24. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Comparison records:

- electronegativity-iupac-composite: predicted
atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; source-derived atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks atomic attraction for shared electrons or permits a scale value to replace the retained comparison relation, or if a changed row is accepted.

Measurement receipt: sha256 : bd86b140ac84feb20fcd1503c6284630317056d1b172d1063aab0832cf6f2deb.

Isolation certificate: sha256 : 7728d84dacaf8a2d81d017ead378e76b352da93fc4b8106ad8b7845188118103.

Custody certificate: sha256 : 6c14f366bf01e75f03c76d99e2b637c938cc5618785b6f4aae53054700cc92eb.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/E01990>
(sha256:bc2888f84a2a3efc53f89738797c88528af75cd0564295e2a1d4138cd1c778e3)

Registered target identities:

- electronegativity-iupac-composite from IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

Immutable identities. Source manifest

sha256 : ddb0747ac6471850417d28ff5ad8ad3decea2fead388f80b3fc664fde4a4dc5f; derivation seal

sha256:9efe41b7f2f880257c5ad610b1b1cf5cc75636df410afe3c16d3fb518922611c; engine receipt
 sha256:eb149468f0bf1746f85bd081caeabc00bf72c241c04ef4039e78a8a1d9fc735cb at
 receipts/engine/model_admitted/SFT-CHEM-ELECTRONEGATIVITY-001-eb149468f0bf1746.json;
 empirical hash sha256:665ff48f0d66dd7a54ef47601152879d5487449acdca892dc041056ad8eda8e9;
 measurement receipt sha256:bd86b140ac84feb20fcd1503c6284630317056d1b172d1063aab0832cf6f2deb.

44. Bond polarity from unequal held affinity

Claim identity: SFT-CHEM-BOND-POLARITY-001

Question and theorem. Unequal endpoint affinity for the identical shared support forces an oriented but still shared bond partition, with opposed partial-charge fibres held as labels rather than signed proof numbers.

registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001

Generated grammar. Generate the literal product of bond, affinity, support, partition, orientation, closure, record and extra-rule choices.

Boundary: Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres__paired-endpoint-response__bond-affinity-partition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
bond	registered-covalent-bond	two-unjoined-atoms	Polarity belongs to a shared bond.
affinity	unequal-complete-affinity	equal-or-unknown-affinity	No direction follows from equality or missing evidence.
support	same-shared-electron-support	different-electrons-compare	Different support cannot orient one bond.
partition	unequal-shared-partition	full-transfer	Full transfer changes the bonding class.
orientation	held-partial-charge-fibres	negative-charge-number	A negative scalar is unnecessary.
closure	paired-endpoint-response	one-endpoint-only	One endpoint leaves the bond record open.
record	bond-affinity-partition-trace	polar-label-only	A label cannot reproduce the comparison.
extra	no-extra-rule	free-polarity-threshold	A threshold imports a parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bc9919833ca65448defb66482960c2ac35503f2445d3cf46e5a2a8d745c1a6d6`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c24cafbcb7496d10a8ec0cedbe90979502eb498af4dd5e8e8ee9dd9837b28933d`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:81b2675f657711b9a4a484f076fe8dfb4d12779fae46fbd14f41a9f3aa753391`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:496e405f3e97843fe901246d4f831feb336c6e91ebb49dc12bf17fa6ca42503c`.

Independent reconstruction. Implementation hash

`sha256:4c6ba5a086ff250256dee1c270231597286030abf637798a6138e036820db6fc`; independent certificate `sha256:97626f4d4ea6cc78c33c2a67460285297e027e25af4c7c7cec3f822cf909ff36`; external-validation hash `sha256:189db76256e2381a4b7dd2bba3182648b78e51a339d8507be86f53728cba2625`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Comparison records:

- bond-polarity-iupac-composite: predicted
unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; source-derived
unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks unequal attraction, polar covalent sharing or partial charge separation, or if a changed row is accepted.

Measurement receipt: sha256:a88c1a4d6fc6f7e9271bf40141d884b3fec4148c2129033170b5e82b3dd03b05.

Isolation certificate: sha256:2e00cdeb8f6e343c929edf4ade2b58c3664a6d05c05d6424d27f128cb8cbcb3.

Custody certificate: sha256:1779fac6f4228e3daf6fa32d38d89148c85956f622b7adb121fa243b3939fceb.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08195>
(sha256:a907133200cc3296e7ecdb7c8970ae8ce35af7b6d3b44d6746f7d18ad5086d41)

Registered target identities:

- bond-polarity-iupac-composite from IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

Immutable identities. Source manifest

sha256:e6017cc25cd31727ee3d2f5b4d0a4be83094d10b9b4ce160a9fbb2a3dc55a07; derivation seal

sha256:bb37eaabf4cfe67424c181ad8adda039704a087602340374f4c07061edeaedfd; engine receipt

sha256:b0a872705523d4edddb27b335602aa73685ead906bb0c466c1b0df5a1de6ab52 at

receipts/engine/model_admitted/SFT-CHEM-BOND-POLARITY-001-b0a872705523d4ed.json; empirical

hash sha256:86ba63a8b53a6f70676c4d7ea2c6e5f1b84d92a1bdb52b59d156446a471343e5; measurement

receipt sha256:a88c1a4d6fc6f7e9271bf40141d884b3fec4148c2129033170b5e82b3dd03b05.

45. Oxidation-state accounting

Claim identity: SFT-CHEM-REDOX-OXIDATION-STATE-001

Question and theorem. Oxidation state is a complete source-bound ionic-approximation accounting; each bond-electron occurrence is assigned exactly once by endpoint affinity, with magnitude positive and orientation separately held.

complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__each-electron-assigned-once

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001

Generated grammar. Generate the literal product of whole, partition, unequal-bond, equal-bond, count, conservation, record and extra-rule choices.

Boundary: Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__positive-count-plus-held-orientation__each-electron-assigned-once__atom-bond-assignment-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
whole	complete-bonded-chemical-whole	isolated-atom-answer	Oxidation accounting is composition-relative.
partition	declared-ionic-approximation	unstated-electron-rule	An unstated rule permits arbitrary answers.
unequal-bond	assign-to-preferred-endpoint	split-despite-preference	It contradicts complete affinity order.
equal-bond	equal-endpoint-division	arbitrary-endpoint	Equal traces cannot choose an endpoint.
count	positive-count-plus-held-orientation	signed-integer-primitive	A signed number hides orientation.
conservation	each-electron-assigned-once	electron-created-or-lost	Open assignment violates the whole.
record	atom-bond-assignment-trace	oxidation-number-only	A number cannot reproduce the assignment.
extra	no-extra-rule	free-state-table	A table can select desired values.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: eec9720b659074465f3bedf55d78324ab1d030d36810d96e0261e9ae158f95a9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b38971bae9d17c53c09af666594f47d248c37c82c5e5ac2f821cfea4eccf1a2b.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: b7ff129c257d37fbaeb72d84a6aca6cf5a240ecff39ca00dfee215ef9ba230ab.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: f9451d77ea5492dbc9190e7320c55355c41a38dfc67711cf3c06479836059344.

Independent reconstruction. Implementation hash

sha256: dde5adfebf754c437df1cfb00ac8e89e98023da3859ecc8e189227024cf5f072; independent certificate sha256: 4c303af97dc7aa5fa8ab7803de872f79e74602902c10343f340eccf5f0457caa; external-validation hash sha256: 87581483c009cc3ef228fb01d87001ad94075b6c92557fdb4514a93a8890b7c4. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-004365-2026

Comparison records:

- oxidation-state-iupac-o04365: predicted ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; source-derived ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ionic approximation, electronegativity-based assignment or complete per-atom accounting, or if a changed row is accepted.

Measurement receipt: sha256: 26213121965ff8f746f36669916d9591fdf2411cf348653da49258b56621fb3d.

Isolation certificate: sha256: 94464dc36dc6570e43deeac9a15d29c8bfca7a3bf58233efcdee56000a179f3c.

Custody certificate: sha256: d9999eca3435ab84ed1a0ba8f1e32f6ebcab78a27bb191f09a52bbcea1561b00.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04365> (sha256:5413171bfaa9152ff9a6a2dee0c79a2ec3e31a4d1efbfff0f8c76804c7fdb06e)

Registered target identities:

- oxidation-state-iupac-o04365 from IUPAC-GOLD-BOOK-O04365-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

Immutable identities. Source manifest

sha256:8ba5ba8a14eb179b9dd7ca533ca5cdac2729479bca9c059c23b9ade08fb46088; derivation seal
 sha256:176aa2e998290d7f1ccffc12882274e0f19953b0ea05f527a1b0f7f875dcebda; engine receipt
 sha256:b08912e566745bea7d1a368aa201ca2f4d92c23e6a36de0c3c6e2ed0ad2482ed at receipts/engine/model_admitted/SFT-CHEM-REDOX-OXIDATION-STATE-001-b08912e566745bea.json; empirical hash
 sha256:4bc79cda405568128bebfd09f964d6f8c05a6db104ddb79254e0b676240ec141; measurement receipt
 sha256:26213121965ff8f746f36669916d9591fdf2411cf348653da49258b56621fb3d.

46. Coupled oxidation and reduction

Claim identity: SFT-CHEM-REDOX-COUPPING-001

Question and theorem. Oxidation and reduction are the opposed endpoint records of one conserved electron transfer; neither half closes without the other.

named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001

Generated grammar. Generate the literal product of carrier, donor, acceptor, coupling, identity, accounting, record and extra-rule choices.

Boundary: Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after__opposed-assignment-updates__carrier-endpoint-coupling-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	named-electron-carrier	anonymous-charge-change	It cannot establish electron conservation.
donor	registered-oxidation-endpoint	donor-erased	The carrier could be freely created.
acceptor	registered-reduction-endpoint	acceptor-erased	The carrier could be lost.
coupling	one-paired-transfer	independent-half-changes	One open half violates conservation.
identity	same-carrier-before-after	different-carrier-after	That is replacement, not transfer.
accounting	opposed-assignment-updates	one-sided-state-change	One side cannot close the assignment.
record	carrier-endpoint-coupling-trace	redox-label-only	A label cannot reverse the event.
extra	no-extra-rule	free-electron-source	A free source breaks closure.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 93dc0f0dca3a8d0637be0f8366a9c9f7efbabe274caf377dc0d68648b26162ca.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 135e22bd09a5e264309f9102ca7412c0debb155083e86f4b50edca884843fd63.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: f7c8cad06259b068ad6ad36db72aebfbdde432e2cb83719d9e7cf26d4c680e0b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: fcdd87f9ce4203ac4f9b1284a0db8ea61f1f4ab2f08bbe6176dbaf3e6d2dcd12c.

Independent reconstruction. Implementation hash

sha256: a4955d74f704c817aac676ffc3af783df9b6bb7f328576e68920563fed210eab; independent certificate sha256: 76ec599dda0c5859c14d2c306a87f6205b3a6c018234b8512be63c2f3d9c3db8; external-validation hash sha256: ba80a84a3f87be5debc6abaa7f7d33cd4860b2326c5adacbdedbf2a41913d2f3. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Comparison records:

- redox-coupling-iupac-composite: predicted
oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; source-derived
oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks paired oxidation/reduction, electron loss/gain or conserved transfer, or if a changed row is accepted.

Measurement receipt: sha256: 56a9f13bed1324bb69831ef0046928a2c757d59f10f99d89c9a384b76a3844b1.

Isolation certificate: sha256: 2c0211cda70ab9fe2c50674950761fe08cadb4808982525aae774882250d7edb.

Custody certificate: sha256: b0d147cfff0011710434513be7176ed15d8eea1dd2523dc50fe6f1360e14061b1.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04362>
(sha256: 9e367d54ae7e7f9e5a40366f9298b1c8dffb4c4677847f60458b78e7e16fc68)

Registered target identities:

- redox-coupling-iupac-composite from IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints

- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

Immutable identities. Source manifest

sha256:55db1a263d7bd172615c685ba171b340f048cc2bda81ec82083dfa5393ce639a; derivation seal sha256:9a9262fd9a3f9e90c08a41eb5c2224ea19127901a4a9399f8addcf87454888c8; engine receipt sha256:69fc6428332eb2faeb8e2f40987067e351a39b0c0138e777a2438219723b657e at receipts/engine/model_admitted/SFT-CHEM-REDOX-COUPILING-001-69fc6428332eb2fa.json; empirical hash sha256:d4468eee8b0587e279b7126d893f7101d9e9faaaa47232ea88e086e0e7702633; measurement receipt sha256:56a9f13bed1324bb69831ef0046928a2c757d59f10f99d89c9a384b76a3844b1.

47. Electrochemical cell and separated redox closure

Claim identity: SFT-CHEM-ELECTROCHEM-CELL-001

Question and theorem. An electrochemical cell is a separated coupled-redox process with complete external electron and compensating ionic paths, retaining site identities and source-bounded work transfer.

distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-transfer-cycle__source-bounded-work-transfer

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001

- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001

Generated grammar. Generate the literal product of site, electron-path, ionic-path, coupling, work, orientation, record and extra-rule choices.

Boundary: Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-transfer-cycle__source-bounded-work-transfer__held-cell-orientation__sites-paths-transfer-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
sites	distinct-oxidation-reduction-sites	one-conflated-site	Separation is required for an external path.
electron-path	complete-external-electron-path	electron-teleportation	It omits the transfer trace.
ionic-path	complete-compensating-ionic-path	charge-accumulation-unclosed	Accumulation halts sustained transfer.
coupling	one-redox-transfer-cycle	independent-half-reactions	Unpaired halves violate conservation.
work	source-bounded-work-transfer	free-energy-output	Untraced work creates support.
orientation	held-cell-orientation	negative-potential-primitive	A negative scalar is not needed.
record	sites-paths-transfer-trace	cell-label-only	A label cannot reproduce either path.
extra	no-extra-rule	free-electrode-potential	A fitted potential can select behavior.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:63371e499dce6d50287fc484b0f0cbf70c3dfd532ed6b9703aa7bf01d4945db0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8c3c9a1814cd2e0e60db3b4e95284e76c9a5eb2f3500e44588cc6f80b50c61cb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:815ea97bdb3eea0fe85ee56fef6342a6e546c2c71f10bcfba66baee6eb6247ce`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:458f2a1f78edf5913a5538be6612662eb14b0c60edf88ce2f52524182f1a2383`.

Independent reconstruction. Implementation hash

`sha256:4e3a0f2270afaa2bf36345a962acbb96bc9e41cdf8f268cc01a75f3ff5854c3f`; independent certificate `sha256:ce4e3f99a3026cab0531efad71c9733fc92a01eab2650628c4ed9bcce6636279`; external-validation

hash sha256:c0aaf6da4ea7ab9fabf07342aada8550cd487cf511d1a64b11957d6f2e8ecc58. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Comparison records:

- electrochemical-cell-iupac-composite: predicted
separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; source-derived
separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks separated redox sites, electron and ionic paths or chemical/electrical work conversion, or if a changed row is accepted.

Measurement receipt: sha256:80b09725552c16e8044130c94315dde3c334ece64d749a2cca81d5375c5070e9.

Isolation certificate: sha256:ecb151d1a3352b9c4412df686d5bdc3e89a457047d8b3294983704eea7f38a22.

Custody certificate: sha256:9b84f7968931f9dc9b0ede41d8b1caf0328c45586668affee2b7869964848a78.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/09058>
(sha256:0016d0976a51c44b0208458810ad455a459f988c7573d186293edff0aea33d68)

Registered target identities:

- electrochemical-cell-iupac-composite from IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

Immutable identities. Source manifest

sha256:c361a8df2ace000382f95e6b5d5001408aabf068043d398be65fde3845230252; derivation seal

sha256:1de4b3b109b17025ece085f31ca01dd775867525a773cb277451718c2dcb247e; engine receipt

sha256:05740565b573149c2c748a6c30271add9304a26fd1086aa6ffbd2c097d8374e5 at

receipts/engine/model_admitted/SFT-CHEM-ELECTROCHEM-CELL-001-05740565b573149c.json;

empirical hash sha256:b2ef2d6f2c6542cea7535df4bdc260ab8bb5bdf08ad5a9d81300e9663ea624e3;

measurement receipt sha256:80b09725552c16e8044130c94315dde3c334ece64d749a2cca81d5375c5070e9.

14. Reaction Kinetics Thermodynamics

Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries.

48. Chemical reaction as source-bound identity transformation

Claim identity: SFT-CHEM-RXN-IDENTITY-001

Question and theorem. A chemical reaction is a complete source-bound reactant-to-product transformation conserving every elemental carrier while changing at least one chemical relation.

source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001

• SFT-CHEM-ELECTROCHEM-CELL-001

Generated grammar. Generate the literal product of source, input, output, carrier, change, orientation, record and extension choices.

Boundary: Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed__held-reactant-product-orientation__complete-endpoint-carrier-relation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
source	source-bound-transition	unregistered-transformation	No chemical provenance is retained.
input	complete-reactant-organization	reactants-erased	A product alone cannot define change.
output	complete-product-organization	products-erased	An input alone cannot define change.
carriers	all-elemental-carriers-conserved	element-created-or-lost	It violates chemical closure.
change	at-least-one-relation-changed	identity-transition	No changed relation means no reaction event.
orientation	held-reactant-product-orientation	unordered-endpoints	Direction cannot be audited.
record	complete-endpoint-carrier-relation-trace	reaction-name-only	A name cannot reproduce the transition.
extra	no-extra-rule	free-reaction-exception	An exception can admit arbitrary change.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fb754a6029d272793d88a9743f1a5ca1c2e28481c465d2910713413cb2b900f3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:959bb4b2950fa5b8ca9f5ac5db294ffbcf0d889230a15c3bfffaf0026b4ccc46`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:767e576739cb6dcdc5eaa7f0d4ae141f26bf618676e521133b66f4407821b1e7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:15b247c519ea516c0b538768e82bebe17813c244712eb2495d2664222da5cd88`.

Independent reconstruction. Implementation hash

`sha256:15c43d7930b426b88179e32fba7fbf6289d9c438b043a1e571c14243f6f908a2`; independent certificate `sha256:f250baeb1f01ff51426dcf57f3edc484ff4bfa408801c5e5aa6dd631239b1d9d`; external-validation hash `sha256:90e6ba56dc2f4993855f74d01bb735893ea0d60af9bb83a73576ac33728d789d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Comparison records:

- chemical-reaction-iupac-composite: predicted
reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record;
source-derived
reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record;
exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant/product transformation, chemical change or carrier conservation, or if a changed row is accepted.

Measurement receipt: sha256:185372e20b90007da19bb1bb3d5007eccdd0313e05633693564d0cb92cd4f26e.

Isolation certificate: sha256:9d31d049b62ad20873e79e82b014c747e29dff2f06eb52af633d7585b241200b.

Custody certificate: sha256:1075f201a9d80dad816867b668546d316ea96c788cd9c55b5d4d7dc6688380df.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01033>
(sha256:83d7891114ad65c012ce2e35f4021beaee6a37d4473c416170d6643db4dc7347)

Registered target identities:

- chemical-reaction-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

Immutable identities. Source manifest

sha256:69c8d1b58730966ecc0addff5b0380c42fcec1a75f2140c33824d136f9fe1126; derivation seal

sha256:8d997c8416d20ae98d74d99e1be40c6f8c64acd2c113caf785c19c4702bb9864; engine receipt

sha256:554bb7fe234a60461eb8d475daff205fb91fa08c51da4e9fd598c1feefd0c212 at

receipts/engine/model_admitted/SFT-CHEM-RXN-IDENTITY-001-554bb7fe234a6046.json; empirical

hash sha256:cb4ca23d7af4dee184fd5904c38247d456df7dba5e35a8bf43ed0dcf243e92dd; measurement

receipt sha256:185372e20b90007da19bb1bb3d5007eccdd0313e05633693564d0cb92cd4f26e.

49. Reaction mechanism as complete elementary-step trace

Claim identity: SFT-CHEM-RXN-MECHANISM-001

Question and theorem. A reaction mechanism is the complete ordered composition of elementary chemical transitions whose internal states join exactly and whose external endpoints equal the overall reaction.

ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__same-overall-endpoints

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

Generated grammar. Generate the literal product of step, adjacency, elementarity, internal-state, endpoint, conservation, record and extension choices.

Boundary: Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__sa`

me-overall-endpoints__carrier-conservation-each-step__complete-ordered-step-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
steps	ordered-elementary-step-word	unordered-step-list	Unordered changes do not form a path.
adjacency	adjacent-state-composition	unmatched-step-endpoints	The proposed path is discontinuous.
elementarity	declared-indivisible-steps	hidden-substeps	The trace is incomplete at its declared boundary.
internal	internal-states-held	internal-species-erased	Cancellation cannot be audited.
endpoints	same-overall-endpoints	different-overall-reaction	The path does not explain the claim.
conservation	carrier-conservation-each-step	carrier-loss-between-steps	Local openness breaks global closure.
record	complete-ordered-step-trace	mechanism-label-only	A label cannot reproduce the path.
extra	no-extra-rule	free-hidden-path	An unseen path can explain any endpoint.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 2b08b15144e7448122e88a2a305aad7c8ac006ff06273c79a20de192a4534887.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 5a8ce953ed1770bff5b7b4cc8d8b8b44826a0eb87215bec2ebdbef36dc82ecdf.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 81abefbb5f378e8edd9b45f331410f29b0fc2b044732df627c2030f63501c9b9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: cca8d7ea96b3dd5513a244338663507233b8fa000c8323f450f93ae9113130ea.

Independent reconstruction. Implementation hash

sha256: 52cf78c584d78eea0045c48685d0bca81995b54d84f0a39dc59fa91be71cec1b; independent certificate sha256: e015218ecd7e1982bb5a476c05f79d02779995e38b0a668982c9638121c14058; external-validation hash sha256: 3b7e30bc294fcac5ff8ed82c7d095f7cbc43a4570cf0bd89c3f8689d8edf2f2f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Comparison records:

- reaction-mechanism-iupac-composite: predicted overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; source-derived overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ordered elementary steps, retained intermediates or composition to the overall reaction, or if a changed row is accepted.

Measurement receipt: sha256:ce957ea02d3aa439297fa6c68e7585380101e9094fb5a5cd68ac8860bee79323.

Isolation certificate: sha256:9cc3021837608f8cc64939c78ada877af842ad07d6af4edb555a19b168817182.

Custody certificate: sha256:72738e2e206fb1adbf46a53f6aa2de242e2064d6c4e7bd0a34f576c081e65876.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03804> (sha256:b84ddab1f13dbc71482ed3f7653eeab3e2cfe5b7ffba64c30ef9d08513d6be33)

Registered target identities:

- reaction-mechanism-iupac-composite from IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

Immutable identities. Source manifest

sha256:a99f963a1966c398b026249b9ff679cb2136238f538b2f8b574a8891c38305ff; derivation seal

sha256:c020425e337f94fab0d38deec7533c1110d9f0f5ccd814bf1435c7c189c98cac; engine receipt

sha256:0735442bcc4a807fe1c867dc97063a4fafea6963688fe9f02b3dff8f89897982 at

receipts/engine/model_admitted/SFT-CHEM-RXN-MECHANISM-001-0735442bcc4a807f.json; empirical

hash sha256:71b3b85bbd94f28a469840d4fb578c4013df607b68a5b77d5987afde0e99add6; measurement

receipt sha256:ce957ea02d3aa439297fa6c68e7585380101e9094fb5a5cd68ac8860bee79323.

50. Reaction intermediate and retained path identity

Claim identity: SFT-CHEM-RXN-INTERMEDIATE-001

Question and theorem. A reaction intermediate is a named internal species produced by one mechanism step and consumed by a later step, canceling from net endpoints while remaining in the complete path trace.

*named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancel
s-from-net-endpoints*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

Generated grammar. Generate the literal product of identity, production, consumption, order, net, retention, record and extension choices.

Boundary: Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancels-from-net-endpoints__intermediate-path-identity-held__production-consumption-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
identity	named-internal-species	anonymous-transient	Its recurrence cannot be checked.
production	produced-by-one-step	not-produced	It is not generated by the mechanism.
consumption	consumed-by-later-step	not-consumed	It remains a product, not an intermediate.
order	production-precedes-consumption	consumed-before-produced	The trace is causally open.
net	Cancels-from-net-endpoints	present-in-net-products	It is then an endpoint product.
retention	intermediate-path-identity-held	intermediate-erased	Erasure destroys the mechanism proof.
record	production-consumption-trace	intermediate-name-only	A name lacks step witnesses.
extra	no-extra-rule	free-short-lived-premise	Lifetime alone cannot define path role.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a2752d09725cb5de86a3b147b38a4775c0d0becc9b26791e17fd354ce5226e48.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : d64901b4b0f43026487180a4ab4442dca557b6ac66e399ebaa765e1d14e9028f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 2fdca90259d1e28cb742f7dc20ae1ba5f976a83c45c3f10a0c1e382fb943b0a7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f2b5675a4b4d7b517d7b472aec706efc56e4d23acf3987a148244f3fbccde8f9.

Independent reconstruction. Implementation hash

sha256 : c903ed412ee204f58fc66bb48dd011c89b5c4b3b5fe31abbd90c28a67b8d4473; independent certificate sha256 : 068785565e96bdc6f82dc2adf032fa8837690bfab3da2ea0a7b9ccbddb92901d; external-validation hash sha256 : 4acb5fdf7c79667e0c65b328079b6a6937b3f641ca2f36b8e5e76d810375369a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03096-2026

Comparison records:

- reaction-intermediate-iupac-i03096: predicted
formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace;
source-derived
formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks produced/consumed internal species behavior or net cancellation, or if a changed row is accepted.

Measurement receipt: sha256 : c92e03ebda1a8f51b0eb7c92c323ff29908c9edb55982ffa634603239131a2ca.
Isolation certificate: sha256 : 20ee286750cb36b467a8df990444e0e08b4fcfcdb29bf513162d0fd3d06f9201.
Custody certificate: sha256 : c6f3cd23f5cdf1cf1ff8a689c62a2381d4acfd12aeca3ac90d51a589d4dbda9.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/I03096> (sha256:9de39353e9af29cbe1b9dcb7f7972e2125378129e82924805f5d97696d93fd4a)

Registered target identities:

- reaction-intermediate-iupac-i03096 from IUPAC-GOLD-BOOK-I03096-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

Immutable identities. Source manifest

sha256:9c76d49c1d41800d93c513b98d52abca22f94cad8d7c03b733b6edb1d16f582e; derivation seal sha256:60419f37d0f2f2af2ab227df24f07cdf1492dea83d28ea32a32eaa52de479425; engine receipt sha256:d3d526c8adf59a9b3ec0a7c57e737f1d92396a72901748f64e0d35c426d28a4b at receipts/engine/model_admitted/SFT-CHEM-RXN-INTERMEDIATE-001-d3d526c8adf59a9b.json; empirical hash sha256:dd800138a2834ab9693b6852fab4a4d7951a54709ce78c59b2264deedc2f9219; measurement receipt sha256:c92e03ebda1a8f51b0eb7c92c323ff29908c9edb55982ffa634603239131a2ca.

51. Activation barrier and transition boundary

Claim identity: SFT-CHEM-KIN-ACTIVATION-001

Question and theorem. Activation support is the least source-bound addition required for a generated reaction path to reach its highest transition boundary relative to the reactant boundary.

complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

Generated grammar. Generate the literal product of path, source, transition, reference, minimality, orientation, record and extension choices.

Boundary: Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support__held-reactant-to-transition-orientation__path-source-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
path	complete-generated-reaction-path	endpoint-only	No crossing requirement is represented.
source	source-bound-support-account	free-energy-number	An imported number selects the barrier.
transition	highest-path-boundary	arbitrary-high-state	It need not lie on the path.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	reactant-relative-support	absolute-energy-origin	An arbitrary origin imports a parameter.
minimality	least-crossing-support	nonminimal-added-support	Extra support does not define activation.
orientation	held-reactant-to-transition-orientation	signed-barrier	A negative scalar is outside the proof domain.
record	path-source-boundary-trace	activation-value-only	A value cannot reproduce the path.
extra	no-extra-rule	free-temperature-fit	A fitted input can tune the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:660d0bba8291ba95de30f0439f20507f8a51eac2f38cbaf90287d61f61dd0645.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b027b4fdc5b38f0c0ef3c9a3232b0eabfa547393ee6c41f228abbb05e86328ad.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:94281f85f5d6d9d09d06e622c59d30d07f5246431edbbba2ab67d03b2cc131be4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4784092078887c2ffacaa9f8423c814871eb6d69a86186db2664831d2e22cac0.

Independent reconstruction. Implementation hash

sha256:e383c27e7a4bb6acf54b280cfc8dbc77ee6df11b4dd68d4495893046a0b8c5d9; independent certificate sha256:2549feed7fb6748985ea7bfc80cd168b8d37152be45c05d441f5fc7a39972385; external-validation hash sha256:81b003fa87a790174213b42f0c5cdfc301b58cce5b0ab2e6bb5be25149bb3edb. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Comparison records:

- activation-barrier-iupac-composite: predicted
reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded;
source-derived
reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks a transition-state energy difference or activation barrier relative to reactants, or if a changed row is accepted.

Measurement receipt: sha256:d322b1255c7d22f4a88b1ce7ef963ab0e3c0ed4f7c06fc3442f5474faa2980fc.

Isolation certificate: sha256:f96a8486f9a212c48347a26e2794743b280f23151aef7dd473cb5f77a3c41fde.

Custody certificate: sha256:97533f43d037a6e4fb53c50c27aea335f3312632f49bcc66a8f1dbd768179531.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/G02631>
(sha256:c51e2c7167603bd633c54e11d238728c7313aa8584662ea4bfc3c3a4bf97a04c)

Registered target identities:

- activation-barrier-iupac-composite from IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

Immutable identities. Source manifest

sha256:ba9ef75073520a218b3d9b2733c17d66f85deb93f63bbbc4c6ff2d27a050bb4b; derivation seal sha256:5a6a90df0542817d3bb5ac872ebef9abcd3ce5a5a89126e41ba94d91f0ac73b1; engine receipt sha256:00815de40099c443e96ea38652bf6728bf86197c5ecbb5239f0886df392109a6 at receipts/engine/model_admitted/SFT-CHEM-KIN-ACTIVATION-001-00815de40099c443.json; empirical hash sha256:923e4d7894816358f0e86e2de5bc28295e718de596a7b63ca28d3f6b0ca40591; measurement receipt sha256:d322b1255c7d22f4a88b1ce7ef963ab0e3c0ed4f7c06fc3442f5474faa2980fc.

52. Reaction rate from counted transitions

Claim identity: SFT-CHEM-KIN-RATE-001

Question and theorem. Reaction rate is the exact positive count of completed instances of one registered reaction per positive reference recurrence and retained observation boundary, with conditions source-bound.

one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

Generated grammar. Generate the literal product of event, count, duration, boundary, relation, condition, record and extension choices.

Boundary: Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence__source-bound-condition-record__event-duration-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
event	one-registered-reaction-event	unidentified-change	Events cannot be counted consistently.
count	positive-completed-event-count	continuous-rate-primitive	It imports a continuum quantity.
duration	positive-reference-recurrence-count	clock-free-count	No rate relation follows.
boundary	registered-species-observation-boundary	unbounded-observation	Amount change is undefined.
relation	event-count-per-recurrence	fitted-rate-value	A fitted value is target-derived.
conditions	source-bound-condition-record	conditions-erased	Rates cannot be compared reproducibly.
record	event-duration-boundary-trace	rate-answer-only	It cannot reconstruct events or time.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extra	no-extra-rule	free-rate-constant	A constant can tune any target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2ca5149a0037bdc6b6c5af664a16fbe871bd044417c8461d3d1799bf414c4b85a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1ab78e2ba0ff39f6f97b554e3c8e84baeaf11592a3e925622243241c756f0513.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a570e4838fecc768effd3446ebfe6dd2165ad3b10676081251be393901f4813f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8c1067c6d6dd67c38da69b8631a81a9466230c9781a06506a5d3b2b95bcf7ab7.

Independent reconstruction. Implementation hash

sha256:e6152457338984a6274e730a62ca2fe591bc2e4483bd7e7da9993b3f7eebef18; independent certificate sha256:6f370ecdb33bbca8628cb1164fc257a347847ae3548c54d3b7cb185ac3d21225; external-validation hash sha256:9610b18821a426fd436888599258a591f378fa1291f3ac3aaa9d85c96db2caf0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Comparison records:

- reaction-rate-iupac-composite: predicted
extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded;
source-derived
extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction change per time or an observation/condition boundary, or if a changed row is accepted.

Measurement receipt: sha256:926af1d645efd5da4142a2cb0c7271c25017b27a229092ad0c1d7c7920515683.

Isolation certificate: sha256:c62c0366edd418c631e6b7a5c183aa562107e90bc15e9322b4d8e1a7a1f197b2.

Custody certificate: sha256:21609c1c1336495c948a13baac0c314b671dae43e321fd90a97bc28ee611e98c.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05147>
(sha256:4f61cb4a9d72b0266dd464f84a2349aee57aec161521fb2466bcf8febed361fe)

Registered target identities:

- reaction-rate-iupac-composite from IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

Immutable identities. Source manifest

sha256:5b37c081ee7916fb4c1590180a31e5c3ff8150c407f248c8ed34eb45df6b69a6; derivation seal
 sha256:a3bf9aa4b92f014a8469b1efd9dd85aecc078e380a2b8843b4e54c1845405364b; engine receipt
 sha256:99c96fd6528df9b183cd7eaa7eefbb6ba0706cb3bc423712f861fcceae8cbc66 at
 receipts/engine/model_admitted/SFT-CHEM-KIN-RATE-001-99c96fd6528df9b1.json; empirical hash
 sha256:35a59511424767f7bf92ceaa0c27aad4c89bcbe5c6839dd697703c255c3ae958; measurement receipt
 sha256:926af1d645efd5da4142a2cb0c7271c25017b27a229092ad0c1d7c7920515683.

53. Kinetic order and dependency multiplicity

Claim identity: SFT-CHEM-KIN-ORDER-001

Question and theorem. Kinetic order is the exact condition-bound multiplicity of a named reactant coordinate in the complete intervention-to-rate dependency trace; independence is Empty rather than numerical zero.

named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reactant, intervention, response, multiplicity, independence, scope, record and extension choices.

Boundary: Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form__condition-and-mechanism-bound-order__reactant-intervention-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reactant	named-reactant-coordinate	anonymous-concentration	The dependency coordinate is lost.
intervention	source-bound-coordinate-intervention	passive-correlation	It cannot establish dependence.
response	complete-rate-response-census	single-rate-answer	Multiplicity cannot be recovered.
multiplicity	forced-dependency-occurrence-count	stoichiometric-coefficient-assumed	Kinetic and endpoint counts need not coincide.
independence	empty-dependency-form	numerical-zero-order	Zero is prohibited as proof quantity.
scope	condition-and-mechanism-bound-order	universal-condition-free-order	Observed order can be condition-specific.
record	reactant-intervention-response-trace	order-number-only	It cannot reproduce the intervention.
extra	no-extra-rule	free-power-law-fit	A fit can select the exponent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:b2ec76b3e1f64ef6bbe60439d7c492f27614b93516d4c65092e90f2f94fc8985`.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 295e33bd061e7b035038ac143c6ff58b9c0d6055338ea690a7eac1f7ef087903.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: b1000a7635db0efabd0fbd160210eef648866a5cf90229cc0f5b0bdd55efeb31.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: dc3f77164ac93ae31817f766ae36505994210688eb8237c8a701a7f67084bc5e.

Independent reconstruction. Implementation hash

sha256: ef4271771caeb278a9789ab2a7d72ab743de0a0cae928d1bec475ef3bdb934a2; independent certificate sha256: 7d4b1f0011c6d5b0bce0edadeb270489f2a5924c3f7758d972ed5c640a79e26d; external-validation hash sha256: 0230a4cc76598b16b09d36909165acbf8a4725fa7fb2e012d26b14b2eecd5284. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-O04322-2026

Comparison records:

- reaction-order-iupac-o04322: predicted
rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry;
source-derived
rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant-specific rate dependence, experimental determination or separation from stoichiometry, or if a changed row is accepted.

Measurement receipt: sha256: b7890e640e17be8ec0a7adfabd3206761a25e02aaa84c609a674c4b107b1f4a4.

Isolation certificate: sha256: 81bbea60df23b9c7aa725798e54e899c18c5c86b16a7b1c954c251e14d855e4b.

Custody certificate: sha256: 54ed67554f342a9d0318a7bc538d3d62d96a410e35bb44df450049d24531bb7a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/O04322>
(sha256: acb0c1bcc39fca71b9fad77cfbc8813c6ffdc1728be3943d56dcd7631c90b9f4)

Registered target identities:

- reaction-order-iupac-o04322 from IUPAC-GOLD-BOOK-O04322-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

Immutable identities. Source manifest

sha256:069ae7a975d9f3116f2ee9a8203a90d4fda441c301daa5fbdab95d409558459b; derivation seal sha256:3c7cfff10bb4f18a24b259412017485981e62d7e11e5076b6053681e820b6d74; engine receipt sha256:e8260df43aa9bdf7fbdce385d508c2727ecc65a933199419d2a1dc2dce1e2807 at receipts/engine/model_admitted/SFT-CHEM-KIN-ORDER-001-e8260df43aa9bdf7.json; empirical hash sha256:46440e2c4707141567a275ba3c3801f79be741177ed049dd4ea1ed5eaf3882c1; measurement receipt sha256:b7890e640e17be8ec0a7adfabd3206761a25e02aaa84c609a674c4b107b1f4a4.

54. Chemical equilibrium as balanced reversible support

Claim identity: SFT-CHEM-EQ-CHEMICAL-001

Question and theorem. Chemical equilibrium is persistent forward and reverse reaction recurrence on a closed boundary whose paired carrier changes preserve the macroscopic composition class.

forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001

- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reversibility, activity, balance, composition, boundary, condition, record and extension choices.

Boundary: Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary__source-bound-condition-record__forward-reverse-composition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reversibility	forward-and-reverse-transitions	one-way-reaction	No reverse support exists.
activity	persistent-opposed-events	static-no-events	Chemical equilibrium is dynamically supported.
balance	paired-forward-reverse-support	unpaired-event-counts	Composition would drift.
composition	macroscopic-composition-class-recurs	microscopic-state-fixed	Individual events need not stop.
boundary	closed-observation-boundary	open-material-flow	External flow can imitate balance.
conditions	source-bound-condition-record	condition-free-equilibrium	The class can change with conditions.
record	forward-reverse-composition-trace	equilibrium-constant-only	A number cannot reconstruct dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fitted constant can force a target.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:efbdf446dafef76afa9d768e8ab86b366db99bb763a88014f1559f5279b7a246b.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:101db4c738b823c2aaa66bf158c163a9c9f0b6dff0ebcaa24e4f669f0e8b77b6.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bef1a0371237a62a4e9f2ccebde1c97119c7797967f5081c7d6014b377fdd4fc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:1d3e059d0e008a62401bd5313bb3466cef844adb001f118d0cdd58a821fa6dae.

Independent reconstruction. Implementation hash

sha256:9795f2eafcfe4afe0c107b84963aeab091d2859dec0229c033bea11e3d13b0c6; independent certificate sha256:83220324729c01a4a2bdb47ec45c4b6bfc93606c5266a3cdca91c5e3165d6c21; external-validation hash sha256:d17f66994f5cd49c91591a208e5efdb6aa2ccfee607a6e268efda1c87908c4ad. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- chemical-equilibrium-iupac-composite: predicted forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; source-derived forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks continuing opposed reactions, equal rates or constant macroscopic composition under fixed conditions, or if a changed row is accepted.

Measurement receipt: sha256:ff84b4286b37cf2810cbfe7e7df69ba44d3cd578188ab16641520059df284dcb.

Isolation certificate: sha256:0259897be117fe27df8f7f19f3d31b3a2d62560d635f780fe0ef4e45f4f01197.

Custody certificate: sha256:b553447a99205c4bd732e6551f68ff5a2c8f9c3ecffff9052885f96f68432eb2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01023> (sha256:8720b60c684dd7869500da1577be0f0a91f2362e02420d755fbbd1abbad6582b)

Registered target identities:

- chemical-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

Immutable identities. Source manifest

sha256:d2a8ad0114607185b74f98b10700b4398001277e7b8f7e87b74d344375914eec; derivation seal

sha256:018d95e3da5aa30562b76c4a00f079db3ef1eb3e0286a4d91808db3dbb450239; engine receipt

sha256:978807f747d3d6af21f13d5b83308e9bde9ad276651aadffcd721abd3edcfd8 at

receipts/engine/model_admitted/SFT-CHEM-EQ-CHEMICAL-001-978807f747d3d6af.json; empirical

hash sha256:d2e8df7e234a5777b25e5724a4e5f69e063a6a4cf89409ab2fdb0436f09e05ef; measurement

receipt sha256:ff84b4286b37cf2810cbfe7e7df69ba44d3cd578188ab16641520059df284dcb.

55. Reaction thermochemical accounting

Claim identity: SFT-CHEM-THERMO-REACTION-001

Question and theorem. Reaction thermochemistry is the complete source-bound energy ledger across one conserved chemical transformation, with transfer orientation, phase, conditions and reference state retained.

registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001

Generated grammar. Generate the literal product of reaction, input, output, conservation, orientation, condition, record and extension choices.

Boundary: Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation__phase-condition-reference-held__reaction-energy-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reaction	registered-reaction-trace	energy-value-without-reaction	No chemical identity is retained.
input	complete-reactant-energy-support	reactant-energy-erased	The ledger cannot close.
output	complete-product-energy-support	product-energy-erased	The ledger cannot close.
conservation	energy-transfer-balanced	free-energy-creation	It violates closure.
orientation	held-absorbed-or-released-orientation	signed-heat-number	A signed scalar is unnecessary.
conditions	phase-condition-reference-held	condition-free-value	Thermochemical quantities are state-bound.
record	reaction-energy-source-trace	enthalpy-answer-only	It cannot reproduce accounting.
extra	no-extra-rule	free-calibration-fit	A fit can select any value.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f0d09549fc7aba9e3e959cdcd632a04ccd046cf9645b3a55925f0779021368d1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:57872b69512c1c9f80e2e7c241245dd8bd4bed71f6fdb2ff6b57b3ebc2f931a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:76d290aaa533e03e6abe301a16e9471b1e7120820c210f8187c6bef6d0a0a7d0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fdb60292a3bddf4eb85038e4d768a41a44dc18bdc79863a07913318ce58e8db3`.

Independent reconstruction. Implementation hash

sha256: f04a8631eb33adf1005011b19ed41ebf2b14be6ce7d9ebffcda79c8026cbd7fb; independent certificate sha256: 3701008c6ea7b1b78f1f9909e75c5d5edac9a3877f7772c9607399a0232932d9; external-validation hash sha256: a880f1c747e82bd929353b86c60c814e7ec5e6be52ee739b47ed8b955a119dd6. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Comparison records:

- reaction-thermochemistry-iupac-composite: predicted
reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded;
source-derived
reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction energy difference, transfer direction or state/condition boundary, or if a changed row is accepted.

Measurement receipt: sha256: ac1d842030c78560b84d78d2a9b4b9cad4644c9643f26fe814c8075d13ce1435.

Isolation certificate: sha256: 48b44c40284e3c5eafcbcd74672e91eb8116112abd94aba93e528cf9e4fe810.

Custody certificate: sha256: b6b35c656d07aac110dda579955013faa44e5baccee8eb0863786d1e1a2475a2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05923>
(sha256:e976fc5090d3344af00af1a64fea3bc564f126f53d920a0387c1fbf56ba07bf6)

Registered target identities:

- reaction-thermochemistry-iupac-composite from
IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

Immutable identities. Source manifest

sha256: 69e956035fb491afd25f4dca581426cb39ea6281bb212b3208b43858ec22c968; derivation seal

sha256: 4ee0be5635e2212cfbf412062acaf3dfb7f7efb4b5eb4e72e007156bed858800; engine receipt

sha256: 840194dda329e635a6f566c8611fa14c264341b2dcb3df8c5023ed8d1b0a0ddc at

receipts/engine/model_admitted/SFT-CHEM-THERMO-REACTION-001-840194dda329e635.json;

empirical hash sha256: 0925b392e7ec1d0556d5a245934271f2e02009aec55dc249873fcaa8a0e22d70;

measurement receipt sha256 : ac1d842030c78560b84d78d2a9b4b9cad4644c9643f26fe814c8075d13ce1435.

56. Reaction direction under complete energy and distinction accounting

Claim identity: SFT-CHEM-THERMO-DIRECTION-001

Question and theorem. Reaction direction is the held orientation selected by strict complete support order between forward and reverse paths; equality yields no selected direction.

forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001

Generated grammar. Generate the literal product of path, energy, distinction, comparison, orientation, equilibrium, record and extension choices.

Boundary: Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction__empty-direction-at-equality__path-energy-distinction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
paths	forward-and-reverse-paths	forward-path-only	No directional comparison exists.
energy	complete-energy-transfer-ledgers	energy-change-erased	Available support is unknown.
distinctions	complete-accessible-distinction-census	entropy-label-only	A label cannot count support.
comparison	strict-support-order	imported-spontaneity-sign	A sign assumes the answer.
orientation	held-favored-direction	negative-Gibbs-primitive	A negative scalar is prohibited.
equilibrium	empty-direction-at-equality	forced-direction-at-tie	Equal support cannot select.
record	path-energy-distinction-trace	direction-answer-only	It cannot reproduce comparison.
extra	no-extra-rule	free-temperature-threshold	A parameter can force a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:daf33949c5cd054da35a9146169943e1dda63e5891e02fb4492534265b797eb8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bf934a0d381cf363f0d6e66be8f507d530b60942e2322d5230310374c465f779`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b87f78bc4cf618f78510d9a021bb6ce4de8289791c3af93cd66300979fbf8e65`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 5343842832622e21645afc61d2c5a53702e7ee8f0ebee61619e03653bbb8d1a.

Independent reconstruction. Implementation hash

sha256 : 6576a2864e07da87d8d37adce128d68cdf2776c2fcfcc90f2b7e5deef9925c4e; independent certificate
sha256 : 2189dca95b759411e57de2a862e881e2ded04255a51f0b74b44c3d7eb15631d0; external-validation
hash sha256 : 5f99d50d30440c5391246981f586984022055369a8134aeee5ec6fc890c78d4f. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Comparison records:

- reaction-direction-iupac-composite: predicted
forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded;
source-derived forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks condition-bound forward/reverse tendency and equilibrium at no driving force, or if a changed row is accepted.

Measurement receipt: sha256 : abdeeeef3cbbfbba3e06d870ecee641c90d50e615c6db77864a6c5e636e370.

Isolation certificate: sha256 : ca15d06d7c7be40abbf8aec72cd6d8abb2f312492c19bb46f036c50b83a2f4f5.

Custody certificate: sha256 : 5921bd0bc5554b2c92ef339ace23e7c271de63e287eae04f4638d21837d051c3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00178>
(sha256:29e64f1b53aba905340496145e5393f853b118f32684689de8e79c573e2642ef)

Registered target identities:

- reaction-direction-iupac-composite from IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

Immutable identities. Source manifest

sha256 : 73936f608dd0627a40b89cad3a1ed88af0d90ee17fd67bdc2406b020057ce4f6; derivation seal

sha256 : a5aa5ba02e12d483b2334877b1363d1c76e1146fda259c551bb7685d9aade456; engine receipt

sha256 : 44288b966ad2be73f14444d40e340c5ae7b84124a568089710041c4e71fccf64 at

receipts/engine/model_admitted/SFT-CHEM-THERMO-DIRECTION-001-44288b966ad2be73.json;
 empirical hash sha256:f5dff69383067d491f8058244253651f77cf7658d9ff9468d604ebd952745865;
 measurement receipt sha256:abdeee3cbbffbbba3e06d870ecee641c90d50e615c6db77864a6c5e636e370.

57. Chemical phase and coexistence identity

Claim identity: SFT-CHEM-PHASE-CHEMICAL-001

Question and theorem. A chemical phase is a maximal connected region of complete intensive-trace equivalence; coexistence retains two phase identities and their recurrent interface under fixed conditions.

maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interface__paired-interface-recurrence

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001

- SFT-CHEM-REDOX-COUPPING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001

Generated grammar. Generate the literal product of region, homogeneity, extent, boundary, coexistence, condition, record and extension choices.

Boundary: Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interface__paired-interface-recurrence__temperature-pressure-composition-held__region-trace-interface-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
region	maximal-connected-region	arbitrary-sample-label	It does not define phase identity.
homogeneity	complete-intensive-trace-equivalence	single-point-observation	It cannot establish uniformity.
extent	extent-independent-class	size-dependent-phase	Phase identity is intensive.
boundary	retained-phase-interface	interface-erased	Distinct phases cannot be separated.
coexistence	paired-interface-recurrence	one-phase-only	No coexistence relation exists.
conditions	temperature-pressure-composition-held	condition-free-phase	Phase depends on source conditions.
record	region-trace-interface-record	phase-name-only	It cannot reproduce homogeneity.
extra	no-extra-rule	free-phase-diagram	An imported diagram selects the class.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: eaad5bde6903bc5cb65e88d04f3ee3e8bd4a9421c6af7f9751047698beeb46b0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: 71c4e3eb21b71d48f16e94127cbf7c5b2bee89e466b76e2db5d2d8ee43e98db1`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: bf9dcdbde12718e36cc70d466631dfe841fa2855c168100de4cc9fc351f596967.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: ae5241a8b94116a4a4db862b70868b989231718eb7d846d5376fdb5cacee158c.

Independent reconstruction. Implementation hash

sha256: 2b5773927b802554089e42a4a081ec5460410a5064e57aece41893d4a9408eab; independent certificate sha256: 6c954b256c8c36582c6097aacb9c944371cfd17d01731f9fc6a336517e6cef23; external-validation hash sha256: 62d3208c01b4e308cedebc0e3f314c5bfdaad58b61b2dab26edb9a7dbf7a9388. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Comparison records:

- chemical-phase-iupac-composite: predicted
homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded;
source-derived
homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks homogeneous intensive properties, a phase boundary or condition-bound coexistence, or if a changed row is accepted.

Measurement receipt: sha256: 378e8c8a6055fa9ac51adbe2baedf501aea62e6a5b2ba476f2b22c20ea4666e8.

Isolation certificate: sha256: 6733cbc835038977074b56c9c85c585e3e5019620263c9251af4b8975350f6c7.

Custody certificate: sha256: 3256de62bbb1d0bdad6104e6e0db7b32ca48fa69be264bf83368a58738732c8f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04528>
(sha256:c903c27903b67f47a61403a4de408c23bb21c50100ee37e5d05e9a8aa168c4eb)

Registered target identities:

- chemical-phase-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

Immutable identities. Source manifest

sha256:d7047f0c37d679103c43e499b2a4ae61c751a601d38019c2c1d77d71d497cffd; derivation seal sha256:a966aa7be9fcab6078680631122c59e3596902f95a476543a43bd54e3a2a51d7; engine receipt sha256:5e82a4bf1e2a7cafa2e2016288ccc4389039c406cb713d9258817d01690cadb6 at receipts/engine/model_admitted/SFT-CHEM-PHASE-CHEMICAL-001-5e82a4bf1e2a7caf.json; empirical hash sha256:d22585e67a829aaaf7d93acb4d3f5ee960f472e8d4986c3a93c47e704e35d8fa; measurement receipt sha256:378e8c8a6055fa9ac51adbe2baedf501aea62e6a5b2ba476f2b22c20ea4666e8.

58. Solution and solubility equilibrium

Claim identity: SFT-CHEM-SOLUTION-EQUILIBRIUM-001

Question and theorem. Solubility equilibrium is paired dissolution and return recurrence on a retained solution boundary, preserving a condition-bound macroscopic dissolved-composition class at finite saturation.

retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of solution, dissolution, balance, saturation, composition, condition, record and extension choices.

Boundary: Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs__phase-temperature-pressure-held__solution-event-composition-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
solution	retained-solute-solvent-organization	unidentified-mixture	Solute and solvent roles are lost.
dissolution	dissolution-and-return-transitions	one-way-dissolution-only	No equilibrium can close.
balance	paired-opposed-event-support	unpaired-event-support	Composition would drift.
saturation	source-bounded-capacity-boundary	free-solubility-number	A measured number imports the target.
composition	macroscopic-dissolved-class-recurs	microscopic-state-fixed	Individual events may continue.
conditions	phase-temperature-pressure-held	condition-free-solubility	Solubility changes with conditions.
record	solution-event-composition-trace	solubility-answer-only	It cannot reproduce the dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fit can select any saturation.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:759effeb3a46056715c37ecb42ca70aab80ba89291af7b19bc00e39f9ff790ad.

- **tampered_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1444804b3af5b2334fab3e15a71486349ff605b48105b8b37a8e3f7916b0cbc.
- **tampered_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:0794da732271434dc54f50e6cf7985ead0f54ea16e2550bd49c249b7947af18f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:941cae2d85c4715f190d7e25bf31cc19ff04e0c64b3c5b8f0ec3d6218ced6651.

Independent reconstruction. Implementation hash

sha256:04f14dc7cddb5249ae6e469a5f9a7d95e1cdd028e8618ba000ecfe236c7bbda1; independent certificate sha256:7016e64075d4ce9a6ea43e2ca21334b72162eac1201bf3ca2634f086cff9ec6d; external-validation hash sha256:285f027c0ee4d166d12027761c3ac52d0f033fdf7f8e575f636324ec6ec138e3. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- solution-equilibrium-iupac-composite: predicted dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; source-derived dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks opposed dissolution/crystallization, saturation balance or condition-bounded solubility, or if a changed row is accepted.

Measurement receipt: sha256:e05637d5d7eb545895db3b52bea9e512eb68bd36a6a08dae932dd29b33c18ebf.

Isolation certificate: sha256:582aa0e7333c4629b606aaa1764ab3315c6df9b6014f596fda80a3e9999d7b15.

Custody certificate: sha256:f32ab21383a5df840a6c9c40887f1841e6ad05b8d5f6f9cc2f8fc51644097379.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/14656> (sha256:bd07661ff6120f150e0a53e67a72b1e212293bf96bfad41628c1e8f7ec9ffdcdb)

Registered target identities:

- solution-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

Immutable identities. Source manifest

sha256:9ce1a2df72de781c8a0e63ca571e8f76a12f4005a3897da83e2bf88ce50fd03b; derivation seal sha256:dbf569566576f95f51a09112cb46de35d86bc76ee5e6b0eefe1c41edd9b87a8d; engine receipt sha256:bf4b6867d9bdc697d9105e06e2774b8634ccc97bf843faf21b1ad37e6ddea9f6 at receipts/engine/model_admitted/SFT-CHEM-SOLUTION-EQUILIBRIUM-001-bf4b6867d9bdc697.json; empirical hash sha256:37417706e059873afb2f0ce6b42c19164259c4dee15e5cfc4b416c3d9b44c09f; measurement receipt sha256:e05637d5d7eb545895db3b52bea9e512eb68bd36a6a08dae932dd29b33c18ebf.

59. Photochemical excitation and reaction channel

Claim identity: SFT-CHEM-PHOTOCHEM-001

Question and theorem. A photochemical event absorbs one retained photon carrier into a named molecular carrier, produces distinct excited support and closes through a generated reaction or relaxation channel with complete energy accounting.

one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-RXN-INTERMEDIATE-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-KIN-ORDER-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001

Generated grammar. Generate the literal product of radiation, absorption, state, channel, conservation, selection, record and extension choices.

Boundary: Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger__generated-channel-census__photon-state-channel-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
radiation	one-retained-photon-carrier	anonymous-energy-input	It cannot establish carrier identity.
absorption	carrier-absorbed-once	photon-copied	Copying violates conservation.
state	distinct-excited-state-support	ground-state-unchanged	No excitation occurs.
channel	source-bound-reaction-or-relaxation-channel	unregistered-outcome	No chemical path is retained.
conservation	complete-photon-molecule-energy-ledger	energy-erased	The path cannot close.
selection	generated-channel-census	free-quantum-yield-fit	A fit can force outcomes.
record	photon-state-channel-trace	photochemical-label-only	It cannot reproduce excitation.
extra	no-extra-rule	free-spectral-rule	An imported rule selects a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:604c53c9f236ae0f1bbe2349ede60a046b166a3e5310ea1874f6118f092477a4.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:db644f9f56463d159d682018a30e2c3e1b261e3080722afb120458ce503a11ee.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:bbd4f835c874e51ff9de61a5de9b060ed22e3cc5400b77f7b6a1e19cda121d35.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:79ae19af5ff3de24c52ed6a09785a6d314a2b60b931535b8373fee3ff3743582.

Independent reconstruction. Implementation hash

sha256:15de61fa6b9a5576d3eef1e3535ff6b4376d8041b93d8757fda8751a81ad1abe; independent certificate sha256:ff70f8241cc8f564e7d87df9ebdc5621ebd377a57b73bda7386a5ffdd1749376; external-validation hash sha256:7ddb6f6ef6b9a799d6f25bcb7c7522d96d304aa9d3857d408cc2158b28e2d858. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Comparison records:

- photochemistry-iupac-composite: predicted
light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted;
source-derived light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks light absorption, excited molecular state, reaction/relaxation pathways or photon-energy accounting, or if a changed row is accepted.

Measurement receipt: sha256:8036e055d4e492ed9f5c5aa0c6ba35f582b0f505d4562be11c072be5343519d5.

Isolation certificate: sha256:6b3830ac094fd71628d2cc484dc7beed556cf80b902333697ebcb5a88dab3c60.

Custody certificate: sha256:391fef0779e04135985b95dabecf4e4ad3fa00a0e8e244577d9c093fdbb7600b.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04585>
(sha256:5e052ba4dda35e0152ec2de28c4d90a748933bd99a82481d5b0b31d9f0c0ed3d)

Registered target identities:

- photochemistry-iupac-composite from IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

Immutable identities. Source manifest

sha256:eeeeef76938bbd5739b71a584660a924d9d29c5b3a23d4ea929825109a5e52de5; derivation seal
 sha256:e87d7d9d897f34a9de12c4f818d9b863c1e91a394eadbe8786df7aee2d48ccbb; engine receipt
 sha256:c3d62a2cb062977f327a9d9bf8db2ede9bd15e72a8b12b5cbbd690c8539f89af at
 receipts/engine/model_admitted/SFT-CHEM-PHOTOCHEM-001-c3d62a2cb062977f.json; empirical hash
 sha256:e20a75360518123d270b5a9eb3be2226df57fac0e9ffe44dd492e077d312c5a8; measurement receipt
 sha256:8036e055d4e492ed9f5c5aa0c6ba35f582b0f505d4562be11c072be5343519d5.

15. Catalysis Networks Interfaces

Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates.

60. Catalyst identity conserved through a reaction cycle

Claim identity: SFT-CHEM-CAT-CATALYST-001

Question and theorem. A catalyst is a retained chemical carrier that opens an alternative closed reaction path and returns with exact identity while the net reaction endpoints are unchanged.

*reaction-cycle-participant__exact-catalyst-identity-retained__alternative-step-path-retained__reaction-rate-increased__
 _net-reaction-unchanged*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `reaction-cycle-participant__exact-catalyst-identity-returned__alternative-step-path-retained__reaction-rate-increased__net-reaction-unchanged__both-directions-share-path__complete-cycle-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
role	reaction-cycle-participant	ordinary-reactant-only	The carrier is consumed.
identity	exact-catalyst-identity-returned	carrier-identity-changed	It is not recovered.
path	alternative-step-path-retained	overall-reaction-only	No catalytic path is distinguished.
rate	reaction-rate-increased	rate-unchanged-by-cycle	No catalytic action occurs.
net	net-reaction-unchanged	catalyst-in-net-products	The carrier was consumed or produced.
direction	both-directions-share-path	one-direction-rule	A reversible cycle is incomplete.
record	complete-cycle-trace	catalyst-name-only	Identity return cannot be checked.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extra	no-extra-rule	free-catalytic-fit	A fit can select the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a0ae5548ab64a5314989e01db584c23471703055d234de86e8ecf9285c346a9e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:fc5ab89d3cfab7add282f8d230b65bbc434f7ef691e76e8ed714e1ca4f5f4c71.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9c78d99f45777e5234b96e68742b6f38d3807dddab51f0149c8aa228825b70df.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0f4686e4369b29d1c58e4bad23b4d7954f59c0dbfe99122ffaddef061229f026.

Independent reconstruction. Implementation hash

sha256:58ec12d7c67ad1fd917c9639d1ff8ced462e348d4c471e56192d4f5b8a7a314a; independent certificate
sha256:f573cafbd6724ed8c57b766c376536ba7562110459be982c9209dae132e385f8; external-validation
hash sha256:ef91777583f0538821a1e265b7a626a150d14043bb13afc9d614706e5205913a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Comparison records:

- catalyst-authority-composite: predicted
reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; source-derived
reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if an official authority does not retain changed rate, an alternative path, regenerated catalyst identity and unchanged net reaction, or if a tampered row is accepted.

Measurement receipt: sha256:96d8e97195cd37027b33af2b192ef23d00925f57b66bff2ab20ba127440643ac.

Isolation certificate: sha256:bf6cf0fee16d5c387c00fcb0ca8e6c284dcd727845c7addf275e9d825667d15.

Custody certificate: sha256:712e07f3fc11a1f7a8494cf407b8f6630363cbe3a16a95cbb3de52dcd03a266d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C00876>
(sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalyst-authority-composite from CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

Immutable identities. Source manifest

sha256:f271c54c6bf485dbd04582f78a97baca117a1518aa8221b753dc87b84ffd1323; derivation seal sha256:dff7a7af9f63f2ce6f3acb8ebba8a8daef932c3556723ea419109b456507103a; engine receipt sha256:239ac25a2ef81dbb6f4a5a2664ee3310f3fb0e9c6388aa4a5fc6a80aaa214b35 at receipts/engine/model_admitted/SFT-CHEM-CAT-CATALYST-001-239ac25a2ef81dbb.json; empirical hash sha256:5b97ac2db40346dd17a6fbd79f10d8f7ee129c5c3aac5cf36fce2fb428f3067b; measurement receipt sha256:96d8e97195cd37027b33af2b192ef23d00925f57b66bfff2ab20ba127440643ac.

61. Catalytic alternative-path relation

Claim identity: SFT-CHEM-CAT-PATHWAY-001

Question and theorem. A catalytic pathway is the condition-bound complete step trace sharing overall reaction endpoints with an uncatalysed path, ordered by activation support and closing the catalyst cycle.

same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001

- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced__conditions-held__pathwise-energy-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
endpoints	same-overall-reaction	different-overall-products	The paths are not alternatives.
mechanism	alternative-reaction-mechanism	step-trace-erased	No pathway comparison exists.
barrier	lower-activation-support	imported-energy-number	A target value is not forced.
carrier	catalyst-cycle-retained	catalyst-consumed	The cycle does not close.
conservation	all-intermediates-balanced	free-intermediate	Matter is unaccounted.
conditions	conditions-held	condition-free-path	Path availability is unbounded.
record	pathwise-energy-trace	endpoint-answer-only	The alternative path is unreproducible.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extra	no-extra-rule	free-mechanism-choice	A choice can force any path.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d56c7c24dd710906db456ed73dff4648c5cbacb903334859dbc6eaedd87247d5.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:b1a3c0ce3bbe224fe6adfac32428de47fcd2130bf09026d2d09617fb085506f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:62a01744198cbb8eb2e61f83caf3adc1b591dcfe2bfd037895bec0be9f32c9e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a54f92031e4559f61065b6a954f4941b2298198daabf36d8b3930933d0279811.

Independent reconstruction. Implementation hash

sha256:4d8c60ce565b3448c751cf5e52ccfefa1c9bcdd202ffd84f9e595d0d55365b4b; independent certificate
sha256:f6bb5877f01fc2c5ebd935d2d55a1fe52134d695628d90f9755638631f692193; external-validation
hash sha256:49310f8a512ebff0b4dd89b6eb4b823e25671e4a71707790cb830bce35d8158f. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Comparison records:

- catalytic-pathway-authority-composite: predicted
alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; source-derived
alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an alternative lower-activation mechanism with unchanged overall reaction and retained catalyst cycle, or if a tampered row is accepted.

Measurement receipt: sha256:61f3215190fac9070c00cea0bf3365a37660a3b7d2a45cb99b44562c31599b37.

Isolation certificate: sha256:7d484dc4ef0c775af469c9efc9f70780f7b717f839452044e0211b6dbd42a8a8.

Custody certificate: sha256:a4f0216152b5b357744336fe8e965ca0b06b612a97e474c3eb8fafeff4ebb423.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and Royal Society of Chemistry:
<https://goldbook.iupac.org/terms/view/C00876>
(sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalytic-pathway-authority-composite from CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

Immutable identities. Source manifest

sha256:99a85f1de54f702ec08c3208b7e63ec4f447fa6b3daea3dc656d4091b72a2e37; derivation seal
 sha256:5e1ee14cf850b606ce8067cc51d25b66035f367c24c5cbb3cdfb1083aea494e9; engine receipt
 sha256:ba206769a2ff14720c048a7eafd71f61561086d11dcd6af2e98e6fc1731aad8 at
 receipts/engine/model_admitted/SFT-CHEM-CAT-PATHWAY-001-ba206769a2ff1472.json; empirical
 hash sha256:b7afd3ae80af2084bbdaf49b9af00adfb3728dac118da016658cb7468f933c6e; measurement
 receipt sha256:61f3215190fac9070c00cea0bf3365a37660a3b7d2a45cb99b44562c31599b37.

62. Catalytic selectivity among generated products

Claim identity: SFT-CHEM-CAT-SELECTIVITY-001

Question and theorem. Catalytic selectivity is the exact maximal product-recurrence class over all generated competing products at one reactant, catalyst, condition and observation boundary; ties remain a cofavored class.

*all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie
 __reactant-consumption-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-CAT-PATHWAY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie__reactant-consumption-held__condition-and-method-bounded__complete-competing-product-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
products	all-accessible-products-generated	one-product-presupposed	Alternatives were not generated.
comparison	complete-product-recurrence-order	yield-number-imported	A measurement selects the answer.
selection	one-product-favored	all-products-equated	No preference is represented.
ties	cofavored-class-at-tie	arbitrary-tie-break	Equal support cannot force identity.
reactant	reactant-consumption-held	consumption-erased	Selectivity lacks a denominator.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
conditions	condition-and-method-bounded	universal-preference	Selectivity can change with context.
record	complete-competing-product-trace	selected-name-only	Competing products are hidden.
extra	no-extra-rule	free-selectivity-fit	A fit can choose any product.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5a1a0226f6360ac2b96c67319fff4be09265d2181106cd3396fc7cd908d5542e.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7a4e988ef2cfc8faa54e8492f557dc4e4d2b096938a9759bc4e90657f1ac86d0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:694038c24110a7ede1b6ebe7a63ee0b88c201fe288c773816c65416663b5d448.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2788954e8e9dccabbf66dbe5c81d681fc83223ad6cd276ebe12ff20dc1c1e337.

Independent reconstruction. Implementation hash

sha256:754cf0c2b2c7594c8784cc47def21254bbe89b6db3682727307ffad8ada6be12; independent certificate
sha256:c45ba906fb4d3c4aa0f4d308389ebb355098d0766df03a451d237d5d88866196; external-validation
hash sha256:714193c4b32267e9fc0f7f9d3ea403f7b0017e7b2ae8f1313b0d2b1ecb3a0552. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Comparison records:

- catalytic-selectivity-iupac-composite: predicted one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; source-derived one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks product preference relative to alternatives, specified reactant consumption, a bounded method/condition or retained competitors, or if a tampered row is accepted.

Measurement receipt: sha256:521c12218f929ef529d6f59320f28470cdddf57dd9eabd3c5199cb039673b03d.

Isolation certificate: sha256:76ef47150ee7c3a9bb165a1de4560555e188915a78e73cf82cd221bb9dd17c93.

Custody certificate: sha256:92f075412058de4db93764aa8125c147d44fb094629ec239e44ce347c4c17b10.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05563>
(sha256:72489fb0beb2453befdf2628aa48d7fe5a18f8a8ce4e09573c09b3bb464d7506)

Registered target identities:

- catalytic-selectivity-iupac-composite from IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

Immutable identities. Source manifest

sha256:45e34c1bd951fb10d5e97639ec24c57ea53d3b52ddb59ec2b4b0eb99c58d0fa2; derivation seal
 sha256:31e00c5a6c64c7da5f688349a84bf49340ed271f9dc47797a6440c0dbeba3c35; engine receipt
 sha256:a417163df1c29bfa35f2d677fbbc4453f6416bb9cac81d33e0a11ea22e9ea934 at
 receipts/engine/model_admitted/SFT-CHEM-CAT-SELECTIVITY-001-a417163df1c29bfa.json;
 empirical hash sha256:5c4f47ecf21879ecfcee5a88bcca5a1de4a1cef30fae345f1cac1eaed0fa796d;
 measurement receipt sha256:521c12218f929ef529d6f59320f28470cdddf57dd9eabd3c5199cb039673b03d.

63. Chemical reaction-network composition

Claim identity: SFT-CHEM-NET-REACTION-001

Question and theorem. A chemical reaction network is the complete finite directed incidence organization generated by balanced reaction steps sharing retained chemical-species identities, including branches, merges, cycles and external boundaries.

chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-PATHWAY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained__reaction-cycles-retained__complete-node-edge-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
species	chemical-species-nodes	anonymous-state-count	Chemical identities are lost.
reactions	reaction-steps-connect-species	unlabelled-adjacency	Transitions are chemically ambiguous.
composition	shared-species-compose-paths	isolated-reaction-list	Shared carriers are not composed.
boundary	source-product-boundary-held	open-unrecorded-flux	The network cannot balance.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
branching	all-generated-paths-retained	single-path-only	Network alternatives are erased.
cycles	reaction-cycles-retained	cycle-collapsed	Recurrence is lost.
record	complete-node-edge-trace	network-picture-only	The network cannot be replayed.
extra	no-extra-rule	free-network-topology	An imported graph selects the result.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c57d530691d21b8498f915c7441c1c1c725a6c7fe1e30d386eb9c97135bb20d9.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:204274c29c30cf943174df0510debd011859a5541c2f550be0a193709e490e2.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9b7d1d4cc915cbac4b24e9145301723b46be11253e0968d7d57b88349708238e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5be5d2c477de356160384c12d2baf67ce3b355ccce7adb22dec3b542764681.

Independent reconstruction. Implementation hash

sha256:ffd05d6f83262ada8319d0cb30a8e8cdc9427acf976ee02b05d8020bb1e2584d; independent certificate sha256:f848abc22cd52e77c1a07e21dc4db6dc7f9fa6577f918a27f4c4c7b1dbb82768; external-validation hash sha256:f3ce744eb51ec16543e115842875f46f2e768a1832114eb6deb3e5e5a253bf05. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Comparison records:

- reaction-network-iupac-composite: predicted
chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary;
source-derived
chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks species, reaction connections, composed network paths or a complete boundary, or if a tampered row is accepted.

Measurement receipt: sha256:a0e5902ebb2de9a2acd4fca5737e3434886bf2769fb7364158090f57226bb557.

Isolation certificate: sha256:e6b8c7fd2342ad34dbcb9921216e09113f41ffefa009f17902dc5b963a77649d.

Custody certificate: sha256:89356d5175bcf29197a8cef05b3d527da520202ae080feef574f711bba781db3.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01210>
(sha256:f7949843026a8fe090e71f1d952633b8c63eb8277ad6b067b4bcdf5babf3598d)

Registered target identities:

- reaction-network-iupac-composite from IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

Immutable identities. Source manifest

sha256:7e74fcc6796358d9b6922a7a9556f0037463586d4f95343216f78ec33aab630b; derivation seal sha256:58e90881b731b6595a516720f137da24f2032970bc515232fe3eeb8c5a52b831; engine receipt sha256:750b66a0cd8a1a3298b488e222fca2863ae173c09b92a75bde9515044d6b1259 at receipts/engine/model_admitted/SFT-CHEM-NET-REACTION-001-750b66a0cd8a1a32.json; empirical hash sha256:c3a5a2cb271c5a164e83fe7f3f22489239e3a582478859528c166dcfee668741; measurement receipt sha256:a0e5902ebb2de9a2acd4fca5737e3434886bf2769fb7364158090f57226bb557.

64. Autocatalytic closure and self-amplifying path

Claim identity: SFT-CHEM-NET-AUTOCATALYSIS-001

Question and theorem. Autocatalysis is a finite resource-bounded reaction-network cycle in which a retained product carrier re-enters as catalyst for another production cycle of the same identity.

reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001

- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle__finite-resource-bounded__complete-feedback-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
product	reaction-product-is-catalyst	external-catalyst-only	Self-production is absent.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
feedback	product-accelerate s-own-formation	product-inert	Output cannot affect its formation.
identity	same-identity-feed back	product-role-erased	The feedback carrier is unknown.
growth	resource-consuming -amplification	unbounded-free-copy	Creation violates closure.
cycle	positive-feedback- cycle	linear-step-only	No feedback path closes.
limit	finite-resource-bo unded	infinite-support-assumed	Resources are not counted.
record	complete-feedback- trace	growth-label-only	Carrier balance cannot be checked.
extra	no-extra-rule	free-growth-constant	A fitted rate can force behavior.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0cee95067a1a5431ddc8780c8864ae551f96a26bb2c001743642747736cac51d.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2b89cecb7863ccd95d6a1d95bf55533e2c1ad530a14eed634a88ca8534be5bac.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b95e37ad7b16c81b5d7997fd345b2151209a5f626569efa1032891f153b64d4f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6f3686eed26f288c5bc85889d39921ff67795676ea2ba9075b06efdc5e039bfe.

Independent reconstruction. Implementation hash

sha256:fdd0d83368dd716f7302b78183a3d93e151ae640703569d3fc20ae81a2299ebc; independent certificate
sha256:dc9fca524512a204e305f1f2262f8ffc25587cb8c349b09c19b137519e8a8b88; external-validation
hash sha256:2156c92b02cecc376144e9f781437a7d29137e858280b03cad69228218c99c0d. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Comparison records:

- autocatalysis-iupac-composite: predicted
reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded;
source-derived
reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded; exact
match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a product acting catalytically in its own formation, feedback closure or the declared finite boundary, or if a tampered row is accepted.

Measurement receipt: sha256:66d2251aab1445f74aa98b13c768c89e3cfe69b569e73821b7a0b9badb0400ba.

Isolation certificate: sha256:f19eba414c114b3b34ec93cb821f88c4f04138d4ae921704e00edd80ecbd2948.

Custody certificate: sha256:e77eb7d19e8706602456f42c2dcfb7a442240a9a1af18a9acf79fdd9457868.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00525> (sha256:cd6e2b68b8eefaa71fd7212cb577bd9996810a0d64771508748f7d3599469b06)

Registered target identities:

- autocatalysis-iupac-composite from IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

Immutable identities. Source manifest

sha256:3f7036a451de8e876c5ace7871e00b7d78cca3400001b7c8ec288b4d8c7d3489; derivation seal
 sha256:ae8a19e7cbe8513f19168ebdf9b53cb0d926dd9a5541f1c416569e853e96d472; engine receipt
 sha256:c84301ced7057f2e50d0535f4e7fa1b52543773aebdd8dfe367fd8dacbf03c63 at
 receipts/engine/model_admitted/SFT-CHEM-NET-AUTOCATALYSIS-001-c84301ced7057f2e.json;
 empirical hash sha256:1ee148345c085815d08e6ee3bb71c9f7c033bf9c2c8dc3f0146cbf27139fd632;
 measurement receipt sha256:66d2251aab1445f74aa98b13c768c89e3cfe69b569e73821b7a0b9badb0400ba.

65. Adsorption at a retained interface

Claim identity: SFT-CHEM-SURFACE-ADSORPTION-001

Question and theorem. Adsorption is reversible localization of retained adsorbate carriers at a retained adsorbent interface, producing surface support distinct from bulk support under declared conditions.

retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary__condition-and-surface-bounded__complete-interface-event-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
boundary	retained-interface	bulk-only	No interface is present.
roles	adsorbate-and-adsorbent-retained	carrier-identity-erased	Surface membership is ambiguous.
location	accumulation-at-interface	uniform-bulk-distribution	No surface accumulation occurs.
composition	surface-concentration-differs	surface-equals-bulk	The interface adds no distinction.
binding	reversible-desorption-boundary	irreversible-erasure	The adsorbate cannot be recovered.
conditions	condition-and-surface-bounded	condition-free-coverage	Surface support is context-bound.
record	complete-interface-event-trace	adsorption-name-only	Localization cannot be reproduced.
extra	no-extra-rule	free-isotherm-fit	A fit can select coverage.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1222829e939050aadd0f8822802fd259e38a34efc4956031923d6901d9376f77.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5e79aced536f06e06417b9bd97561298a94bae3a6ed69a8edfc2238aa255bd21.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4208793f4c9b2043c17f721fc4db84dacd13580da5eb86b7480bbabfd24e615b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:59eae4ccdfd45006bede3138562ce34a170fd3ccea0c58c81440c9b92f26b70.

Independent reconstruction. Implementation hash

sha256:594768dbdef5e5e68cea02238fda64b8a1d762d3f6bf2af4706a08291bc6db7c; independent certificate sha256:083528ba959417a9c14e7a719aef5b1cc340b3bfbcb296a7ef03ad2d78b7542f7; external-validation hash sha256:f11677f2823ddad32937c31ae3ade62fd9f54f6e158e169f378444f5d3238964. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Comparison records:

- adsorption-iupac-composite: predicted accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; source-derived accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks interfacial accumulation, retained adsorbate/adsorbent roles, surface/bulk distinction or a desorption boundary, or if a tampered row is accepted.

Measurement receipt: sha256:d550367fcdee479c0559ff1fc1276a6a6a52308771fa6d22b3fd5528ecffdbd4.

Isolation certificate: sha256:239b6b47fac5ed2eaa7fde95097f7fe600c9e051004544809afe5be723d1f1c6.

Custody certificate: sha256:d2aae0aeb061606cb0227ab1ea72028d3fd79e6b4a6e513d4167c933a9ca27fb.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/A00155> (sha256:8d8d32d4fad6f36f8be4aeed8056bac1efb3ef5f9d03c3ec26fddaade7cce582)

Registered target identities:

- adsorption-iupac-composite from IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

Immutable identities. Source manifest

sha256:0be55d631dee35d8972bab503e9a130d958537fa6235088f72d7cc4722b693db; derivation seal sha256:1012d07cc5d979d4401c997892ff6aec336c1b6976dff62a08d214a7d3d35b27; engine receipt sha256:b89566d1d2c1059b72ea9b65da47b81b1c89d56c93678c6ebadc655bdde4ba34 at receipts/engine/model_admitted/SFT-CHEM-SURFACE-ADSORPTION-001-b89566d1d2c1059b.json; empirical hash sha256:ad2d7c764bb84113a8b69aa87b15e8f0bfcef87a653fc39cc60418d7716ccceb0; measurement receipt sha256:d550367fcdee479c0559ff1fc1276a6a52308771fa6d22b3fd5528ecffdbd4.

66. Colloidal dispersion and phase-interface support

Claim identity: SFT-CHEM-COLLOID-DISPERSION-001

Question and theorem. A colloidal dispersion is a source-bounded population of finite dispersed-phase regions distributed through a retained continuous phase, each with a retained interface and observation-scale boundary.

dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001

- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded__dispersion-condition-bounded__particle-interface-distribution-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
phases	dispersed-and-cont inuous-phases	single-uniform-phase	No dispersion exists.
extent	finite-dispersed-p articles	molecular-solution-only	No mesoscopic carrier is retained.
interface	phase-interface-re tained	phase-boundary-erased	Particle identity is lost.
distribution	particles-distribu ted-through-medium	one-settled-region	Dispersion is absent.
scale	particle-size-sour ce-bounded	size-free-label	Colloid identity is unbounded.
stability	dispersion-conditi on-bounded	eternal-stability-assumed	Persistence is condition-dependent.
record	particle-interface -distribution-trac e	colloid-name-only	Phase support cannot be checked.
extra	no-extra-rule	free-dispersion-fit	A fit can force stability.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : edd8f0455d2f299a3ad9630f6622b49d804706b79d355ae6319912a942000e0c.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 723ba74dce8f73823e5842be52b5b1dd8ecf3cde9c4f11da3b779a472b975234.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 083d54c83dfc2538aee457363b236556741bb91855fe1a35c5554550b3e837b5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : f01f5eab860c62ac8679d4555e9eb9ce1c8f72eb35560ce05eff3a3858e0ae2a.

Independent reconstruction. Implementation hash

sha256 : 1733d436397c2812555cf24e9c2225cf28f9d2e77c81560dbd60b58f74ae99aa; independent certificate
sha256 : 455bdd9c0d4dc97eb1652ea2880c1a40fd3a2f25097e9e24371f13a601ed227a; external-validation
hash sha256 : a22012b42ae63a9b80400636e84a46866c7411fddd22ca8ade6d045a9f67fda2. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Comparison records:

- colloidal-dispersion-iupac-composite: predicted microscopic-phase-dispersed-in-continuous-phase__phase-interface-retained__p
article-size-boundary__dispersion-condition-bounded; source-derived microscopic-phase-dispersed-in-continuous-phase__phase
-interface-retained__particle-size-boundary__dispersion-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks dispersed/continuous phases, retained interfaces, a particle-size boundary or condition-bounded dispersion, or if a tampered row is accepted.

Measurement receipt: sha256 : 48e5f39d9e7a9213bfd4187c1cd0692968678126b908e4d8578b98c99b634dac.

Isolation certificate: sha256 : f2d6fd9dd14de0627cb79e6bc2c71f006c4d27bbd9a8fb4b86a1d330fced508a.

Custody certificate: sha256 : 244dd247b747fc920add450fdffc9e02fd293551a67738e69db705f7bc8fea5.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01174> (sha256:a8261e1400d58c847f37722d7c1cb39f782a97844d847648156bf01dfad7a636)

Registered target identities:

- colloidal-dispersion-iupac-composite from IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

Immutable identities. Source manifest

sha256:1071e1711cfa513a50971c9cdbae95c642ed212f8f9e099613c69a42d6f3c0ca; derivation seal sha256:3f5b7909acd18202b17542290f3695277eebc733c7f4fe91354d45459b5c2d06; engine receipt sha256:b342599b09b46b62c530e7b466cc4a61a64a2727202f164053826c0058819a97 at receipts/engine/model_admitted/SFT-CHEM-COLL0ID-DISPERSION-001-b342599b09b46b62.json; empirical hash sha256:d8ae80a4d9f2c5742b5e4b9128f4506d4b0b567dcf1c7a4a9c9f0a3035ea053b; measurement receipt sha256:48e5f39d9e7a9213bfd4187c1cd0692968678126b908e4d8578b98c99b634dac.

67. Chemical transfer across an interface

Claim identity: SFT-CHEM-INTERFACE-TRANSFER-001

Question and theorem. Interfacial chemical transfer is the counted, oriented crossing of a retained chemical-species carrier from a named donor phase to a distinct receiver phase, pairing departure and arrival under declared conditions.

two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-SURFACE-ADSORPTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition__transfer-rate-condition-bounded__complete-donor-interface-receiver-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
interface	two-phase-boundary	one-phase-only	No crossing boundary exists.
carrier	chemical-species-retained	anonymous-flux	Chemical identity is lost.
orientation	held-donor-to-receiver-orientation	unsigned-transfer-number	Direction is ambiguous.
balance	mass-balance-preserved	carrier-created-at-boundary	Conservation fails.
process	boundary-crossing-transition	instantaneous-relabel	No transfer event is represented.
rate	transfer-rate-condition-bounded	universal-transfer-value	Rate depends on context.
record	complete-donor-interface-receiver-trace	flux-answer-only	The crossing cannot be replayed.
extra	no-extra-rule	free-transfer-coefficient	A fitted coefficient can select any value.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : c930e6944f1a09f686a08d3c96eb6287aba3fb827d5bad341ba8d8f4b7548113.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f24b244547dcfdb4e01c3b88b91ca4cb460bee0b24c241efe4f14e01f1406e96.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 72fe92a1dbf643723166962831da07bfc71712dda65fa11d6fd07ceb9cc88d78.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 3d25aea62ec6d8302fefb5738b84969989268a5917e5d8a22d080a65215cd0b3.

Independent reconstruction. Implementation hash

sha256 : f9ce306e7d93983fb7c043ef150e796b853c53283ed531a9c34d2332848d44a9; independent certificate sha256 : 85ceb7f7d1bf236cc83718fb44d1f95b727ee51b66f36d86ba5e4608540032f44; external-validation hash sha256 : bcc0c0c51ac48d5b57682cc8879e0ca7d4347b1f270d773d7d06296280a6ac77. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Comparison records:

- interface-transfer-iupac-composite: predicted chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; source-derived chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a species crossing, two retained phases, a condition-bounded rate or mass balance, or if a tampered row is accepted.

Measurement receipt: sha256 : 96e6ca08d7e05fe76facc2f4cc855b2564a0d7f597ccf589a7342bde233489f7.

Isolation certificate: sha256 : e38df0c603e3e727043aa36f198fe5c2b56c661009262d1db8656ca668659ff1.

Custody certificate: sha256:d74fcd2a35755fe07cd20835e3be116db9f53629849b9acd27645f0fa845b294.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/P04536>
(sha256:b60eba64e32f00b5bcb7ecf818b25abc53208476546555ae05349a52cbd7e7b6)

Registered target identities:

- interface-transfer-iupac-composite from IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

Immutable identities. Source manifest

sha256:a3ac97ebcb8ff635bf61442b1a8d0dfbbd8f52f5ac2c3a6b5276174a2e891d39; derivation seal
sha256:4fdc50af81ae5440e6c0d7e6d9762ab95019915dc8c2dca15aa99c9ef5915cad; engine receipt
sha256:6a6f5dde7f37947d6ea43b84dee4726844e0397ddf9f34667254e4c796bc1c4a at
receipts/engine/model_admitted/SFT-CHEM-INTERFACE-TRANSFER-001-6a6f5dde7f37947d.json;
empirical hash sha256:0e0e49c608e196154bfce21e5735870fd3340be1aa5c83e9ae2f0d175e723bbe;
measurement receipt sha256:96e6ca08d7e05fe76facc2f4cc855b2564a0d7f597ccf589a7342bde233489f7.

16. Stereochemistry Organic Polymer

Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition.

68. Chirality and non-superposable held orientation

Claim identity: SFT-CHEM-STEREO-CHIRALITY-001

Question and theorem. Chirality is the retained distinction between a complete oriented molecular carrier and its exact reflection when exhaustive identity-preserving proper superposition admits no equivalence.

complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__no-proper-superposition

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__all-proper-finite-relabelings-enumerated__no-proper-superposition__orientation-and-map-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-three-dimensional-object	formula-only	A formula has no spatial orientation.
reflection	exact-mirror-image-generated	mirror-image-not-generated	Chirality cannot be decided.
composition	composition-preserved	mirror-composition-changed	The comparison is not reflective identity.
adjacency	adjacency-preserved	mirror-adjacency-changed	The comparison changes constitution.
superposition	all-proper-finite-relabelings-enumerated	visual-guess	A guess cannot establish equivalence.
result	no-proper-superposition	arbitrary-handedness-name	A name cannot force distinction.
record	orientation-and-map-trace	chirality-label-only	The result cannot be reproduced.
extra	no-extra-rule	free-chirality-rule	An extra rule can classify anything.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : a90fecae5d272bfff9642b6c1cbb2246da4a5e5ea36336d27de3e3f38c61b0842.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 2f4df44ccf7ea37f290000c67426ecfaea88c1d5dc75908b21ddeba196961327.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : e30626a49e357397a5b503d28c26a3b0fb8439bd2b6284caa64566cde654d7d2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 2dc9b28c06452315912058f81e49812f2abb24e5d4a343fce7464529a67affe1.

Independent reconstruction. Implementation hash

sha256 : c58e29aeec8dd7045a2a6a0a35bac932bb4ebe42211a40aa3378176cb79256d6; independent certificate sha256 : c5dba95a5d23f251698226727ab9cad978861019221b4f9eeeff049ad2237d87; external-validation hash sha256 : 77217828c3c566132373f558fb9b133c20e0e7bd631c4e7945069e06aef99672. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Comparison records:

- chirality-iupac-composite: predicted
object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held;
source-derived
object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held; exact match
True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks reflection, non-superposability, retained chirality or an observation boundary, or if a tampered row is accepted.

Measurement receipt: sha256:2f858469a86812414592d37f8c551eb2cab9aeefa37411935749a5a0ce5fbec0.

Isolation certificate: sha256:d81385cebada96333bdc67b2222c699f3cab100ecc5a0aad4bd27fd5223f9bf6.

Custody certificate: sha256:7d44cc5ed9943c5e98870d49e914e1630ded60d1a79629b9a8e716a8558d8a15.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C01058>
(sha256:a787109fa5a2c606b0628013669773016b0245479ed6387584ba027d43e5f01b)

Registered target identities:

- chirality-iupac-composite from IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

Immutable identities. Source manifest

sha256:b81da2d3eb9742af6442f41fee14e6564d3d67853348cd4ecb1b8f2b37aec59d; derivation seal

sha256:2273f6d6e92d30bb607b5e7a6e75b44c980946bc18cf3630bf0b06c70ebdf659; engine receipt

sha256:00ee8b785fd197058e6a47b394fd8fd0b9de6f6108599558989f4c190d45de07 at

receipts/engine/model_admitted/SFT-CHEM-STEREO-CHIRALITY-001-00ee8b785fd19705.json;

empirical hash sha256:d4d540b31689d011277f61ffb31004d673d3eeea320a7e8b44eacf812429c28f;

measurement receipt sha256:2f858469a86812414592d37f8c551eb2cab9aeefa37411935749a5a0ce5fbec0.

69. Enantiomer pair and chiral observation boundary

Claim identity: SFT-CHEM-STEREO-ENANTIOMER-001

Question and theorem. Enantiomers are the paired complete stereoisomer carriers related by exact reflection and by no identity-preserving proper superposition, with opposed configurations retained.

paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-STEREO-CHIRALITY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held__chiral-observation-boundary__paired-map-and-observation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	paired-stereoisomers	one-structure-only	No pair exists.
constitution	same-composition-and-connectivity	different-connectivity	They are constitutional isomers.
mirror	exact-mirror-relation	unrelated-orientations	The pair is not enantiomeric.
superposition	non-superposable-pair	superposable-pair	The reflected structures are identical.
orientation	opposed-configurations-held	orientation-erased	Opposed configurations merge.
environment	chiral-observation-boundary	universal-property-difference	Response depends on observation context.
record	paired-map-and-observation-trace	pair-name-only	The mirror test cannot be checked.
extra	no-extra-rule	free-enantiomer-rule	A discretionary label can force the pair.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:78874adb9051f2cda593f82c2378adc94592159d0fce0287299f8bc1501a94dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:80898deff1518c63ea5d791407d62d7bbb29dc25b5cbfc6477a8240f1135088d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d892bdf9ea15e6523b592d1b7b9439d6d5022e55032d91d191b7bdc843c221d5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f8cd4ae6999f6643eb7752e0d261eb4389e0a29ea01b45e51c0da078e636c67`.

Independent reconstruction. Implementation hash

`sha256:b9e0407b93ac2315487b7c6dcda9bc34f65c4a439fa518cbdc0f817c39a0321d`; independent certificate `sha256:2c44dff49dba2fb9f7f9259ec34e3ed7e75956aab0a1073f1a5df99c26b52d23`; external-validation hash `sha256:036249ae0933ab4c85a240207a4ff934ea7dea61a49ea5ee614624e486148ad4`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Comparison records:

- enantiomer-iupac-composite: predicted
molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; source-derived
molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks molecular mirror relation, non-superposability, opposed configuration or chiral observation boundary, or if a tampered row is accepted.

Measurement receipt: sha256:dc426d9ff89781d78446229965847bd3cf8b3a077a7c62418cbe9319c8d5f5d8.

Isolation certificate: sha256:58dbd4c37e772e1bf231b949816368e91ce331d05bbd3d9c03b2111ab4b9997c.

Custody certificate: sha256:c44b309051d6d5348249284b7a529dbc8f0dce32e6dbd0b46c0db61ff49b6441.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/E02069>
(sha256:833912292d5eba7d3528d2a31a748a95f4cd4d28b92e896f6e1b5d9ef914935d)

Registered target identities:

- enantiomer-iupac-composite from IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

Immutable identities. Source manifest

sha256:f416c38a39383152bd2d37cec4f40adf10e1161504ca7b9274af54f9004eee94; derivation seal

sha256:e6500f21f86255a34516550e01be6b61647115c7c539be982e874b43360113ba; engine receipt

sha256:41db74ec3403458e6a269d9904e6768351dc7229b25ed7a50e0bd088e95a05f2 at

receipts/engine/model_admitted/SFT-CHEM-STEREO-ENANTIOMER-001-41db74ec3403458e.json;

empirical hash sha256:d086045f16b868dea7b858347fd148adfe876e1e89bed2f596ab8ae6385bfe36;

measurement receipt sha256:dc426d9ff89781d78446229965847bd3cf8b3a077a7c62418cbe9319c8d5f5d8.

70. Diastereomer distinction

Claim identity: SFT-CHEM-STEREO-DIASTEREOMER-001

Question and theorem. Diastereomers are paired non-superposable stereoisomers of equal composition and connectivity that are not related as exact mirror images.

paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001

- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-STEREO-CHIRALITY-001
- SFT-CHEM-STEREO-ENANTIOMER-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held__property-distinction-open__complete-pair-classification-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	paired-stereoisomers	one-structure-only	No pair exists.
constitution	same-composition-and-connectivity	different-composition-or-connectivity	They are not stereoisomers.
superposition	non-superposable-pair	superposable-pair	No stereochemical distinction remains.
mirror	not-mirror-related	exact-mirror-pair	That pair is enantiomeric.
configuration	complete-relative-configurations-held	all-centres-erased	The difference is unreproducible.
properties	property-distinction-open	properties-forced-equal	Non-mirror structures need not share response.
record	complete-pair-classification-trace	diastereomer-label-only	The exclusions cannot be checked.
extra	no-extra-rule	free-diastereomer-rule	A taxonomy can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8682d8dcee95265f2e77c106b5f8b4cea6d08860c097fe761367e7042ae95966.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 3a570f720ae4e97c27103ded8bc92baa50d7d553202e4053e17e83a4c0f7aebd.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: d351449dc0617402486df5ae3d907c79a641ac271c17fb87bcb906f884a6a24.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:ad5e7ea3c443b07d374848fb6d542ad5d4a5351ccb2ed46fb83b0be1f3bb9318.

Independent reconstruction. Implementation hash

sha256:3936122354bc8e32351cbd45cb5e1b9835a7bb5b3a12546e2893154c9daf820c; independent certificate sha256:2c1b89fe37e89c64402624627281e4a28c22f9a9305246322c89e5ec3b433d5a; external-validation hash sha256:386d2400e7c726fa214c4d322547d02963af42c972567d28bf0bf539df4b4159. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Comparison records:

- diastereomer-iupac-composite: predicted
stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; source-derived
stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks stereoisomer identity, non-mirror relation, configurational distinction or possible property distinction, or if a tampered row is accepted.

Measurement receipt: sha256:6b0cf84493406f61a95dad059b2dd6dd1657cfae00dfc4243550381a52452606.

Isolation certificate: sha256:130abe45fb5e7ec3816e683b7ebc3ee72dd6d89bc2801a17aa2c5e777f342065.

Custody certificate: sha256:45e736ef7791936d3786802d3d45a2aae0513047f1f81e90c58e8bf39818b96b.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/D01679>
(sha256:4fa09c83930088c3eeea3fd428d370918e6bf98c361673287c0fc8ef9d394078)

Registered target identities:

- diastereomer-iupac-composite from IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

Immutable identities. Source manifest

sha256:4c27425f692f832a454a838d286dbed27f7f84abc5d5162c23df40333f3479d8; derivation seal

sha256:5fce9da04cde73ee2735571bad53578d233459db70b035d84bd266cdb7f8d3f2; engine receipt

sha256:661fb2a1428aae2c60554c1101fb05f81fe985b7d8fbf5eb3877134c439aa00a at

receipts/engine/model_admitted/SFT-CHEM-STEREO-DIASTEREOMER-001-661fb2a1428aae2c.json;

empirical hash sha256:7ec2cac6638c1561a917b26cd83c4d5a34fe22251c807c77be745662e0ae089f;

measurement receipt sha256:6b0cf84493406f61a95dad059b2dd6dd1657cfae00dfc4243550381a52452606.

71. Functional-group recurrence and reaction role

Claim identity: SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Question and theorem. A functional group is a retained atom-or-group subcarrier whose complete local structure recurs across molecular embeddings and carries a characteristic generated reaction role within a declared context.

named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001

- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained__molecular-context-bounded__substructure-and-reaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
support	named-atom-or-group-support	whole-molecule-only	No recurrent substructure is identified.
embedding	molecular-embedding-held	isolated-fragment	A functional group acts in a molecule.
recurrence	recurs-across-molecules	one-molecule-only	No group class is established.
reaction	characteristic-reaction-role	shape-without-transition-role	Function requires chemical behavior.
identity	group-identity-retained	context-erases-group	The group cannot be followed.
context	molecular-context-bounded	universal-context-free-response	Neighboring structure can modify response.
record	substructure-and-reaction-trace	group-name-only	Recurrence cannot be reproduced.
extra	no-extra-rule	free-functional-taxonomy	An imported list selects the group.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c188cf693c44cb41b1a98a4cac9a337f8c0045f92bc18d237416427b14b2f689`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3727abd174787d207fb36092b3e137e2f11ec5878725228dbc473d2ad541183b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bd73c500dd3af7832d2bc4f118d46f20ceda9568f900c6b69d93ab2c7001b32b`.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : 72208840975e03044131c459563f053a4d989014820fecebf1b28b8d6f418397.

Independent reconstruction. Implementation hash

sha256 : 600186025be20195d13b50b988a6c1454172503b1e3c9af181c2c3579850101b; independent certificate sha256 : 93cb8ec0892c4409a11bf050cc142e7c068898f82bfeb071fcc325d3df3b5f39; external-validation hash sha256 : 26d535307714a6d4c78a06397ac3b9b746b85da168b6e540e61fea7586c9fd78. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Comparison records:

- functional-group-iupac-composite: predicted
atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded;
source-derived
atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an atom/group, characteristic behavior, recurrence, reaction role or context boundary, or if a tampered row is accepted.

Measurement receipt: sha256 : f0b1699714ef6972d4d8895729103d4adde9a22bf84b29f24f56cfa15fbd3c94.

Isolation certificate: sha256 : 465ae863c10469dbc677f8c13c18a0075fa9c5abf9b0a0b575fc77f547dfd0b3.

Custody certificate: sha256 : db6f6803ea248e418b3f3459559cdda8568041733155229399e273d96d9c1451.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/F02555>
(sha256:1fc0f254f488921a762c25454615ea9ec4f40d54736be524fa9f4ff8d79869d1)

Registered target identities:

- functional-group-iupac-composite from IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

Immutable identities. Source manifest

sha256 : 298e6a9e7080ce74fd46a9c94cce0d11d896a24cc7edb03f29e882a412f113b0; derivation seal

sha256 : 3e542a809254cdb62a1e8758b3205b2c139c674b3490f26cc9684bbbcc344dc5; engine receipt

sha256 : 35355b53a7cfcfb013366ae51b0830d9ba686362b669f5feb9cb16895d9eb502 at receipts/engine/model_admitted/SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001-35355b53a7cfcfb0.json; empirical hash

sha256 : dfa2a272245ab4d917ccf3e8cb92b0614e0da545db9afeb73661b8447229c0da; measurement receipt

sha256:f0b1699714ef6972d4d8895729103d4adde9a22bf84b29f24f56cfa15fbd3c94.

72. Organic reaction-family compositional law

Claim identity: SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Question and theorem. An organic reaction family is the equivalence class of complete balanced reactant-product maps sharing one minimal bond-change pattern and role correspondence without assuming a common mechanism.

multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001

- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced__mechanism-not-assumed__member-and-pattern-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
members	multiple-generated-reactions	one-reaction-only	No family is formed.
encoding	complete-reactant-product-maps	reaction-names-only	Names do not establish common structure.
pattern	shared-bond-change-pattern	unrelated-bond-changes	No shared family relation exists.
roles	reactant-product-role-correspondence	reactant-product-roles-erased	Pattern orientation is ambiguous.
composition	elemental-carriers-balanced	carrier-balance-erased	The transformations cannot close.
mechanism	mechanism-not-assumed	mechanism-assumed-from-pattern	Equal net changes need not share a mechanism.
record	member-and-pattern-trace	family-label-only	Membership cannot be audited.
extra	no-extra-rule	free-reaction-taxonomy	An imported list can force membership.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256: 2794e829a8c2cd1c9b2bf0a50c8294a70400ffe90be534b840b1363dec4b90c6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256: 35977ac840a0e2d15e7593c90926ed8ba21498624ab5ca81adce3b34fbc24762`.

- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 94e838e1e737d341840056cf2b4c397e9745eaed535e4a80dd9e9b7e65e12e93.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: db912a2311b33366ee25d520989015f8231a0f4959c831e03149d92dad4561c3.

Independent reconstruction. Implementation hash

sha256: 1af395b5f6f38579aaffb318538190387ebb8f470c3d46f7881ec9ad2c3a6cf6; independent certificate
sha256: ff5a8998035aab50f52a103e7a6886cfcf88a42d9f1049d282fd448408ba77c7; external-validation
hash sha256: 8a804b572fd73c3596d472ae7977e57d5008e5a60a3ab8d82dfaa1878d3e5e98. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Comparison records:

- organic-reaction-family-iupac-composite: predicted
shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded;
source-derived
shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded;
exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks shared reaction pattern, reactant/product roles, separation from mechanism or condition boundary, or if a tampered row is accepted.

Measurement receipt: sha256: 4c0191c6b4bb019cc609b2a3a976737f275d49e25f6f56a832d96644e636f42f.

Isolation certificate: sha256: 735875b659aed84e0f9ff94dbe41f4f49063103f4a42bf6d886ba7b390a9cde2.

Custody certificate: sha256: 27927674882d94957fa8d8a5a595f358bc93d37a0011dc7c822ab202255d5411.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/R05164>
(sha256:e7253285c30faeb217a73b62d4ca4cea7636a626c1306e2eebe8b71960abf6eb)

Registered target identities:

- organic-reaction-family-iupac-composite from IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

Immutable identities. Source manifest

sha256:7a5883890cde64ed26c6bfb0ca4038354f1366b346a20ef2b3ae9eadb667f346; derivation seal
 sha256:ae06f0a1fc25bb0951b6f2f2662ab64d4d21b5c8ca404ea8be82f4bce944d8d8; engine receipt
 sha256:73b8569eaabf2253d83ba94bf9f729a0ff89709bcb6c64500bfff2856abc64672 at receipts/engine/model_admitted/SFT-CHEM-ORGANIC-REACTION-FAMILY-001-73b8569eaabf2253.json; empirical hash
 sha256:d855f814b6d2d5332bba692d17dabf1ae9a1b6f4d69d37242571e31c74673f0f; measurement receipt
 sha256:4c0191c6b4bb019cc609b2a3a976737f275d49e25f6f56a832d96644e636f42f.

73. Polymer chain from repeated monomer identity

Claim identity: SFT-CHEM-POLYMER-CHAIN-001

Question and theorem. A polymer chain is one finite connected macromolecular carrier formed by covalently ordered occurrences of a retained repeat-unit identity with exact positive extent and explicit ends.

connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chain-incidence__positive-finite-repeat-count

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chain-incidence__positive-finite-repeat-count__chain-end-boundary-retained__repeat-connection-end-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	connected-macromolecular-carrier	unconnected-monomer-mixture	No chain exists.
repeat	repeated-low-mass-unit-identity	anonymous-segments	Repeat identity is lost.
connection	covalent-sequence-connections	noncovalent-coincidence	The sequence is not a polymer chain.
order	ordered-chain-incidence	occurrence-order-erased	Chain topology is lost.
extent	positive-finite-repeat-count	fixed-universal-length	Length is population- and synthesis-bound.
ends	chain-end-boundary-retained	infinite-chain-assumed	A finite carrier needs a boundary.
record	repeat-connection-end-trace	polymer-name-only	The chain cannot be reconstructed.
extra	no-extra-rule	free-polymer-rule	A threshold can arbitrarily select chains.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 8726c5ae36f1d1dda15c6109697ba8f767017f4b5806a911dc62ff6d46f82db7.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 4527583bc01617f404e6e917d56a214b3f33732d24bb19ab270d9e50793a8f99.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 3154c241dd7a4f1d0d151c880c2e340905664003bc6411691de8ba3be063ae7d.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7e6b7adcf2ff991f470a76322fad0d788027a04b2ee07e6087c6a331bf625206.

Independent reconstruction. Implementation hash

sha256: fc9ba42e65fefb02b61cbf5a7b32554e3c38cf5d592a5371b70f07f620fc7a16; independent certificate sha256: 8e5fd2dd33f54399c8a6a3416e84050d860244c45d6b8b202d877bd8c1de9268; external-validation hash sha256: fe77ad02dfd37c3137f2ca0b48e948bedee53aa0ba55f040722a061cceb4f99d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Comparison records:

- polymer-chain-iupac-composite: predicted
high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; source-derived
high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a macromolecular carrier, repeated units, covalent sequence or chain boundary, or if a tampered row is accepted.

Measurement receipt: sha256: 223181a892b117c6316b5ea0b1fbd98c3be3cb4a3ff1516886ebf4ec3c705b5c.

Isolation certificate: sha256: c1a94a17425a77377d1353ffc7f70eca79c658f079eaa129bc92295d9e6d57d5.

Custody certificate: sha256: 77b60028ee938c7506691af30ade2c909f7f5f7f742df34595b20b9066bf9247.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03667>
(sha256:b38c6c106e35763169204b065edaabf18df22bf97f4ea9094db3b8b72383e85a)

Registered target identities:

- polymer-chain-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero

- external target content remains inaccessible until the prediction is sealed
- Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

Immutable identities. Source manifest

sha256:820cbc89ab68ac2cefc56ec7564a14bd8e3f0d3339799384f62d303cc0e5519d; derivation seal sha256:1fcd451ad68862206f24ed642c27aa2c8aa62529d27f19183d459a0413fe7a2f; engine receipt sha256:f84c4fa8a4329d7f4d1ef445df7cdb7988d1d976c0dd68fece9701469df20cda at receipts/engine/model_admitted/SFT-CHEM-POLYMER-CHAIN-001-f84c4fa8a4329d7f.json; empirical hash sha256:730ba9addbe68f27c6fd2c64f95e545fd456286daa8ea1d0ccc065395033b02e; measurement receipt sha256:223181a892b117c6316b5ea0b1fbd98c3be3cb4a3ff1516886ebf4ec3c705b5c.

74. Polymer population and retained chain-length distribution

Claim identity: SFT-CHEM-POLYMER-DISTRIBUTION-001

Question and theorem. A polymer distribution is the complete source-bound census of resolved chain carriers by repeat count and positive multiplicity, with number and mass weightings retained separately.

all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001

- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution__method-and-sample-bounded__chain-population-on-trace__no-extra-rule`. Closure is `depth-independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
population	all-observed-chain-carriers	single-representative-chain	A sample distribution is erased.
length	exact-chain-length-counts	mean-only	A mean loses alternatives.
multiplicity	positive-chain-multiplicities	duplicate-chains-collapsed	Population frequency is lost.
measure	number-and-mass-support-separated	one-universal-average	Different summaries retain different information.
distribution	complete-molar-mass-distribution	answer-scalar-only	The population cannot be reconstructed.
method	method-and-sample-bounded	method-free-census	Resolution changes the observed population.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	chain-population-trace	distribution-name-only	The census cannot be audited.
extra	no-extra-rule	free-distribution-fit	A fitted shape can hide unfavorable chains.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256 : 88065ddd66a529ae65468a34e9385d4efa760e54ccd0c87359658a17f83ea231.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 3e9fe0202d7069000f6003a37591693635f37214510a56dd32ee5d33c3c7881c.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256 : 90d31eee582fad2b4a6b2b14cc405a233e8a7c2c0a7db750908f4137db7b3800.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256 : ee81365154d6fc7ee014cdac864f9d47eb8e41328b830b95829704096f87ed57.

Independent reconstruction. Implementation hash

sha256 : 8628fe82a0a16f901776576b549a688e90d8c857ac9004825836c20252b39bd3; independent certificate sha256 : ffe61525dbbc336a4802001ad25570d65a61f25408c301d1484cf53898dda99d; external-validation hash sha256 : 33f7aec26af0ea140a0bb4712fffb155c16954cdd88dde3c8306a0fe0c6f222fb. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Comparison records:

- polymer-distribution-iupac-composite: predicted
polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded;
source-derived
polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded; exact
match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks chain-length diversity, molar-mass distribution, distinct weightings or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256 : e3a7d9f2e424c9405dd02ac15d6c658092dd0c6088ae782d3e671ff1ad0b4fa5.

Isolation certificate: sha256 : 83629e16de4f587b5217518bff300797dca00881c165503e791ec9574970b890.

Custody certificate: sha256 : c225e04d254605315c9d7ac6be911f896e9183f41ed8732cc895c79ee4f4457f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/12215>
(sha256:98c415c9e851b41083ffc7ee3b9e1ba2d2b9e81bc6e05bb0dce170dd9733e91a)

Registered target identities:

- polymer-distribution-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

Immutable identities. Source manifest

sha256:dfb937231e96d9d92dff71bb7720da00f9ccbd2532d0cf61e14989378fa5ae2f; derivation seal
 sha256:a3fdedaaab5e27ef897deb2cb9874778e0a6843eca446bc7c148dfe75a0cf1f; engine receipt
 sha256:b7cde3ce4ce07e58a22eb8f654a9cd4074b9b41148d59e4ee9c15b73455a86a0 at
 receipts/engine/model_admitted/SFT-CHEM-POLYMER-DISTRIBUTION-001-b7cde3ce4ce07e58.json;
 empirical hash sha256:565c4e20bea7f88ee39a2977f1f1d9ce31ddd683902ec45267cefbf347fb3e39;
 measurement receipt sha256:e3a7d9f2e424c9405dd02ac15d6c658092dd0c6088ae782d3e671ff1ad0b4fa5.

75. Chemistry-to-biology molecular handoff boundary

Claim identity: SFT-CHEM-BIOMOLECULAR-BOUNDARY-001

Question and theorem. The chemistry-to-biology boundary passes only a complete source-bound molecular identity, structure and chemical state; biological function is a separately retained downstream relation and cannot select the chemical law.

chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held__chemistry-to-biology-handoff__composition-structure-state-function-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	chemical-molecular-identity-retained	biological-function-without-molecule	Chemistry requires a molecular carrier.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	composition-preserved	composition-erased-at-handoff	The object cannot be traced chemically.
structure	molecular-structure-preserved	structure-erased-at-handoff	Biological recognition loses its substrate.
state	chemical-state-boundary-held	timeless-molecule	Function acts on physical chemical states.
function	biological-function-separately-held	function-as-chemical-identity	Different functions need not change chemical identity.
boundary	chemistry-to-biology-handoff	biology-imported-into-chemistry	A downstream law cannot select chemistry.
record	composition-structure-state-function-trace	biomolecule-name-only	The handoff cannot be audited.
extra	no-extra-rule	free-life-criterion	A life label can select any molecule.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:d24fc84b3d2eb24db795ab0e5e24264436d5b446a3d0db66271770e515baf85f.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d9532c7da21be65210d77703ceafb00a1d0ed36e06cb60b83d3ed2582d4e9a9f.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:e3b5635d8c33cd0e99f9d01f3d9368a5206a4d053087c72ecc3fc2a813a78e46.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7c47d008b17ce95a41abeb1071ec1c3281cfac6d126816e7529602799b36c302.

Independent reconstruction. Implementation hash

sha256:d831eace1b996519fb40bc5b13e8897cdf6c007f87d9159e66bf77edeef4130a; independent certificate sha256:f4d1b90f1ef528e8bb267bdc296905e85dcbd67d0ea2eee3e84ff09c71e7561e; external-validation hash sha256:ce2e18c75f02c7a9ac082c3edb4160da04c0050bc6994594fdf9901c79a2f37f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Comparison records:

- biomolecular-boundary-authority-composite: predicted chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; source-derived chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence cannot separate chemical molecular identity from biological function at an explicit handoff boundary, or if a tampered row is accepted.

Measurement receipt: sha256:1fd01394c060361f41594cb2e5faeafacf6c2f2f4c6f9a57a1d93548da7df75a.

Isolation certificate: sha256:819826e2239d7b8e241efa0e71ca7e1ca417dd34a948063166a5fd66af8a9f79.

Custody certificate: sha256:3434cfb36ec513b6089ed50cea8cb2fb844fca1392239b585b2c82af1f3db026.

Registered source descriptions:

- EMBL-European Bioinformatics Institute: [https://www.ebi.ac.uk/chebi/](https://www.ebi.ac.uk/chebi/(sha256:d862deab28de555e80751d9fce3fecf4ac4de970d17d9a4c1481dc6976731f24))
(sha256:d862deab28de555e80751d9fce3fecf4ac4de970d17d9a4c1481dc6976731f24)

Registered target identities:

- biomolecular-boundary-authority-composite from EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

Immutable identities. Source manifest

sha256:ac63afa7c56f2f87a630caef127128b2787a10a4b96693836c0bb0f47d76f537; derivation seal
sha256:cc93dcd2e0b68c5cd2d497b751efc1292cb3a845fb29e86a07b8b7d9cf4c5a14; engine receipt
sha256:610945f40717478ae45485c17632b717085f1801b29cf794ae1dd1b7c008193e at receipts/engine/model_admitted/SFT-CHEM-BIOMOLECULAR-BOUNDARY-001-610945f40717478a.json; empirical hash
sha256:3070285b3783ed838c144def1aca376ee3b6a5facdaa5215b5c971ebb5ebc572; measurement receipt
sha256:1fd01394c060361f41594cb2e5faeafacf6c2f2f4c6f9a57a1d93548da7df75a.

17. Analytical Spectroscopic

Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law.

76. Analytical sample, analyte and matrix distinction

Claim identity: SFT-CHEM-ANALYTICAL-SAMPLE-001

Question and theorem. An analytical sample is a declared material portion whose complete resolved component support is partitioned by held roles into the analyte and retained matrix, together with its source-to-sample map.

declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held__component-response-map-held__sample-analyte-matrix-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
portion	<code>declared-material-portion</code>	<code>unbounded-material-name</code>	No sampled carrier is identified.
components	<code>complete-resolved-component-support</code>	<code>component-support-erased</code>	The portion cannot be compositionally audited.
analyte	<code>analyte-role-held</code>	<code>analyte-not-designated</code>	No measurand-bearing component is selected.
matrix	<code>matrix-support-held</code>	<code>non-analyte-components-discarded</code>	Interference support is hidden.
sampling	<code>source-to-sample-map-held</code>	<code>sample-equals-whole-source</code>	A portion need not reproduce the source whole.
observation	<code>component-response-map-held</code>	<code>response-without-component-map</code>	A response cannot be assigned.
record	<code>sample-analyte-matrix-trace</code>	<code>result-only</code>	The sample distinction cannot be reproduced.
extra	<code>no-extra-rule</code>	<code>free-sample-rule</code>	A discretionary rule can select any portion.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3efc490e1bb313eb3a34c6f86d105d7b0feb050e3dc07907cb22f9e557e3db2a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f6628399c538353c7fb79523e499076b0985aa2d7845782e3bab7e269ba2402e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54733cd21625ad9d2061bffcbbfc18170a824966c03c94ce4abd76a18e66cdfcb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7c171463044bd6ddcb0a3699fa40084a8b5f356e72f0015643dd009118f5dab8`.

Independent reconstruction. Implementation hash

`sha256:2e248b730cee3eeb2aeb40d4018c0beafc12e8ac5692a7b8c6ed0b72657168bc`; independent certificate `sha256:29057279273fb54b1e5c5c6f56b99df6749458dbbb89aec2895105ee80e84ac8`; external-validation hash `sha256:2f3f30c535e2eddb27a8d50f12b69be618efdb99175f206525cdf726e1b86970`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Comparison records:

- analytical-sample-iupac-composite: predicted sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; source-derived sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks sample portion, analyte, matrix or sampling boundary, or if a tampered row is accepted.

Measurement receipt: sha256:9dd5852a454c591de6380c54af7fd342025d09cd158b7f4af562639bef615080.

Isolation certificate: sha256:2b6ebe84cd1474932eb55716fa6f13b30fe3b952ecbf9af3224490cf5259286f.

Custody certificate: sha256:03c9a215a94e78ae118236a2ad453098c1dcac55244e58a3e3525578612faa7f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05451> (sha256:f2d630c6a9a8ec0ab1304a6e9e3acf1bc495b9b3f2d5f3958375aa0c3fafa184)

Registered target identities:

- analytical-sample-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

Immutable identities. Source manifest

sha256:59738f8ee3fa617ea9ee488d7328b171fff70fcf05b7a0adf148ab96a42e1f91; derivation seal

sha256:3247a2f5a28abfd5d60dab628748e12a960f6d7b406fd52e7824b34df1bd3803; engine receipt

sha256:1880410ee99b23f159fdc8775f6c1186f2094319030db175661c992b48590551 at

receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SAMPLE-001-1880410ee99b23f1.json;

empirical hash sha256:96d710f063d8db488f420012076fd14288519c901b11c1d733ae55cf4c5197f2;

measurement receipt sha256:9dd5852a454c591de6380c54af7fd342025d09cd158b7f4af562639bef615080.

77. Analytical calibration and traceable comparison

Claim identity: SFT-CHEM-ANALYTICAL-CALIBRATION-001

Question and theorem. Calibration is the complete traceable comparison of held reference input/response pairs with a sample response inside an explicit finite method domain, retaining uncertainty and every row.

traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001

- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite registered reference-response set and sample response sharing one declared observation organization.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain__reference-and-sample-uncertainty-held__reference-map-and-result-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reference	traceable-reference-input	unknown-reference-input	Comparison has no fixed support.
response	reference-response-pair-held	reference-response-erased	The comparison cannot be reconstructed.
ordering	exact-generated-comparison-order	unordered-fit	A fit can conceal incompatible rows.
sample	sample-compared-with-reference-map	sample-outside-reference-map	No calibrated comparison is possible.
range	declared-calibration-domain	universal-unbounded-map	Calibration support is finite and method-bound.
uncertainty	reference-and-sample-uncertainty-held	uncertainty-erased	Alternatives are falsely collapsed.
record	reference-map-and-result-trace	calibrated-answer-only	Traceability is lost.
extra	no-extra-rule	free-calibration-parameter	A fitted value can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:032e873ba65355fe014ee4ea11bc71860ad6fe7164823becca09a813425962ad.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:ec401bf322205d24affee40106306cd23ff06e1726aaa862af3db593374ff9bb.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:454c698f8da30d9662a41b9526c30f56193e6a98cee4296f04011056542b30dc.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f2d3b9679de44e3fec73bc19dec2b45f5bf4a9a76a71918982afc544d84b3980.

Independent reconstruction. Implementation hash

sha256:1dbf78592e06bd73224c7c6afeee79718cf93ce2936d633abbc26654f817b49b; independent certificate sha256:94a382fa81641e7ff83e6cef4cb4683a3db0e00ef1acce42986f34d6b71c5a72; external-validation hash sha256:fbc85191232ce6b12444ae069c1d3f22e0da138e5eab25fd8eee4051877efa0. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Comparison records:

- analytical-calibration-iupac-composite: predicted calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; source-derived calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks response/reference relation, traceability, domain or uncertainty/method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:3ca1469a4ebf546a35e8913fb787a87358e5be94e75d96e21717da05b0f72d72.

Isolation certificate: sha256:1a8973b8fb35f45380983bcc6b57642e32a3a2231060c37d09bb1800aefcb708.

Custody certificate: sha256:b26d8132d69aeb2dd353a3738f78de034ea5318793651caa7d3e49a55c2889db.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/C00775> (sha256:9a50ffb4e909ae78ac6ccf2cfa69db372a7ff659db4fb88b48e89e79322b69c0)

Registered target identities:

- analytical-calibration-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite registered reference-response set and sample response sharing one declared observation organization.

Immutable identities. Source manifest

sha256:6fc05197ce55d2639e3bb3f6cfeeda3802c9003ac56c3621df0ef186ce5790da; derivation seal

sha256:6c4f143e466e9cafc4b5205774ab8c543b3a201d57cc9b87c1aa38770a0797b1; engine receipt

sha256:4ac2379caade53b802030b6fe2ee43633bb794d2a50b9f7cf4f3fd2982bf5cf1 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-CALIBRATION-001-4ac2379caade53b8.json; empirical hash

sha256:4b5eea2cc1d8a15f1e589ab0511927364ab680f5f8d7ed9fa3fee62332031bb7; measurement receipt

sha256:3ca1469a4ebf546a35e8913fb787a87358e5be94e75d96e21717da05b0f72d72.

78. Analytical selectivity and interference boundary

Claim identity: SFT-CHEM-ANALYTICAL-SELECTIVITY-001

Question and theorem. Analytical selectivity is exact distinguishability of the analyte response from every declared interferent response within a retained method-and-condition boundary; merged responses are recorded as interference.

analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001

- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analyte/interferent response support under one registered method and condition class.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained__method-and-condition-bounded__complete-response-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
target	analyte-response-held	response-without-analyte	Selectivity has no target.
alternatives	declared-interferent-support	other-components-erased	Competing support is hidden.
comparison	all-analyte-interferent-responses-compared	single-response-only	No discrimination is tested.
decision	distinguishable-analyte-response	name-based-assignment	A name cannot prove attribution.
interference	interference-boundary-retained	failed-alternatives-deleted	Unfavorable rows disappear.
conditions	method-and-condition-bounded	universal-selectivity	Response depends on method and conditions.
record	complete-response-comparison-trace	selectivity-scalar-only	The tested alternatives cannot be audited.
extra	no-extra-rule	free-selectivity-threshold	An arbitrary threshold can force success.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:3b6ea28b86431cb84b37ffd90b03311ea422cc2dc805d76224971189594641e8.

- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a7d04dc9d2f7bc12b43d5e5e8df3b7a4bcf26911119584e555d5d74f01160b64.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:18a0bf0c39f89ee23bcf5b4c7a0092ac625f88fea4c3d15c8e860f04698bc715.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:98ad52c643c40c247122849ca5db493c0f3ab8e4cb17c141826be16d1d16b702.

Independent reconstruction. Implementation hash

sha256:62b133e342f7dd1974b887e976c0df6a472e824750afb8ec0872087d4d5427bf; independent certificate sha256:fe120fe14ad5e2b636c45d8014360b4858a8c3d87b8d79139fc67719289f4487; external-validation hash sha256:208b713a88a9bfa39042681fadab45152cc4ddfcacac7f90ea134e10ec17a3d43. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Comparison records:

- analytical-selectivity-iupac-composite: predicted analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; source-derived analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks analyte discrimination, interference, non-universality or method/condition boundary, or if a tampered row is accepted.

Measurement receipt: sha256:da6f121d768e5e3cd227d4fa02a27a1ae951bed717f206f0509500450a09524c.

Isolation certificate: sha256:e400a8788587c5690c8d366b7d8e8b335e69773736267c741044f3d1df90f51f.

Custody certificate: sha256:c5b528577459443b0a49f2c21284d46b9012f08eb4f2c09ec59529ea2582d97f.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05564> (sha256:24cfb5654b5ba14eb99cc4dfe0c01efa7a251be91937a33000f362ba8a9730f6)

Registered target identities:

- analytical-selectivity-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analyte/interferent response support under one registered method and condition class.

Immutable identities. Source manifest

sha256:ff8fa9f556403b99cb8df4a5b8f602389cf76e2eec62851e961b0e9df189525f; derivation seal
 sha256:e3044df347a42d228e3b0e52ba0dfe680bc14e2a71dbe97beb680c3d7560040d; engine receipt
 sha256:b00b15ae23003f159f654f5f6bad65e0aae8268fe5769873ad0b3ab0e1859e8b at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SELECTIVITY-001-b00b15ae23003f15.json; empirical hash
 sha256:59a8da91f9ca9b1d0ef6fcd6aafb566445163284aa455228f415f976afa39a22; measurement receipt
 sha256:da6f121d768e5e3cd227d4fa02a27a1ae951bed717f206f0509500450a09524c.

79. Mass-spectral composition correspondence

Claim identity: SFT-CHEM-SPEC-MASS-001

Question and theorem. A mass-spectral record is the complete map from generated ionized carriers to retained mass/charge response classes and counted abundances, including precursor-fragment relations and observation conditions.

generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held__ionization-and-instrument-boundary__complete-carrier-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	generated-ionized-carriers	neutral-sample-label-only	No detected carrier class is generated.
identity	composition-and-isotope-support-held	carrier-composition-erased	Composition correspondence is impossible.
partition	mass-charge-classes-held	undifferentiated-response	Distinct carrier classes merge.
response	class-position-and-abundance-held	peak-position-only	Multiplicity support is lost.
fragments	precursor-fragment-map-held	fragment-origin-erased	Products cannot be related to precursors.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
conditions	ionization-and-instrument-boundary	universal-spectrum	Carrier production is method-bound.
record	complete-carrier-response-trace	spectrum-image-only	The composition map cannot be checked.
extra	no-extra-rule	free-peak-assignment	A library guess can select composition.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b30aae3c4fac29492c023e1ddecc2f8fcb59015da79eeadf2fb7ea23baaca22.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7023aee7d974957bfd7bd46f6c83b997fa078f56c68a69d46fb051980153b357.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5fbc7a983050ed25c72382178f46875fdf3fbb11d6ec7c91dc77044761a41e49.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e39d6c8b12dfe47a2149f642070ebdeba1fad62702440d164fd805569d2c2f98.

Independent reconstruction. Implementation hash

sha256:63f8223999cb6f43867dfa34e17c98dc20e6a5a1c49bc7286e9867776ab6c544; independent certificate sha256:68951ba69b1e384489a5e141bf0855dc916c1454d8548dcaab442b39f328392d; external-validation hash sha256:63571f9167c8b27006e3ed0cb71992f6cd7d04030ec1c7337a208e19ffea1aba. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- mass-spectroscopy-iupac-composite: predicted ions-separated-by-mass-to-charge__mass-spectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; source-derived ions-separated-by-mass-to-charge__mass-spectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks ion mass/charge separation, abundance, composition/fragment correspondence or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:77f7def06c3bc419f995ccf48f17f8d46ed68d2f835561979150017924e59336.

Isolation certificate: sha256:2b81b6c4e5626f4d9e1e9ce7b5152e70c1ef70b8967af710a3ce5f0ea2a7b1e4.

Custody certificate: sha256:2a1d5bc32a4354ccae5b3e58f09b16852cb112ec7ad240c15aee2376e5b7b20d.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/M03749>
(sha256:bb7b4631fab65d61fd3c0f8b6257ae91bc87da91ceea23a961f4c824f20b4e56)

Registered target identities:

- mass-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

Immutable identities. Source manifest

sha256:4b19515587686f660712aa91b2eccaf9918c1714c30bd5f711de9a2dedbc933; derivation seal
 sha256:5800598040da59b271b7dffad5de0528b95c6782508fc03815e14bc1cbae40f9; engine receipt
 sha256:e155dbf2c8cd4be28f11d225025424f0d4005a458783853b4588977745edc60e at
 receipts/engine/model_admitted/SFT-CHEM-SPEC-MASS-001-e155dbf2c8cd4be2.json; empirical hash
 sha256:21d06af9ba02bdae4c8382a1f7f65aeeacaa23b77e410ca9cc5401a32e644740; measurement receipt
 sha256:77f7def06c3bc419f995ccf48f17f8d46ed68d2f835561979150017924e59336.

80. Infrared recurrence and molecular distinction

Claim identity: SFT-CHEM-SPEC-INFRA-001

Question and theorem. Infrared organization is the complete condition-bound pattern of incident recurrence classes coupled to retained molecular vibrational state transitions, with whole-pattern rather than isolated-line identity.

incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPILING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-PHOTOCHEM-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern__whole-pattern-molecular-distinction__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
input	incident-infrared-support-held	unregistered-incident-recurrence	No excitation comparison is defined.
molecule	complete-molecular-carrier-held	molecular-identity-erased	Response cannot distinguish carriers.
transition	vibrational-state-transition-held	static-geometry-only	A spectrum requires a state change.
coupling	dipole-response-change-required	all-transitions-assumed-visible	Observation requires a changed interaction trace.
spectrum	complete-frequency-response-pattern	one-line-answer	Alternative recurrences are erased.
identity	whole-pattern-molecular-distinction	universal-one-line-identifier	Different carriers may share a line.
conditions	state-method-condition-boundary	condition-free-spectrum	State and method affect observation.
extra	no-extra-rule	free-infrared-line-rule	A lookup can force assignment.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b2ffe399a008258b480ff5f4c6f502fe06f1554a8544521f5d1ece040955890a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a603e622aeaed9f12c66289143a2bd1b22cec3d5a1b7d996404b77fa9d144334.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1d6b9d317916812bb704757723d4399c03e03f6d00a9e5e985f0bbc82060bbad.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3d6cea107788fcc04f575ed6126f2ad493ed4dc3187751c7aa2e56e002b9db24.

Independent reconstruction. Implementation hash

sha256:9623a7a8ad4f940f91f869bb1b7224df95a3399ea5ee013763d34c92a621724b; independent certificate sha256:81480db63e0f925d2062ec153fa02b0bb1df1d38b15c2471b5794949c032bd86; external-validation hash sha256:9529ae6b8296de786c171d9400d2fcf33d01478c255fb759e569522a707360a1. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- infrared-spectroscopy-iupac-composite: predicted infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; source-derived infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks infrared/vibrational correspondence, a retained frequency pattern, whole-spectrum molecular distinction or state/method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:66439cdd064a13c442bd3feb7372e4f0a016e24e4eae0ee368dcbacc91b7f8f9.

Isolation certificate: sha256:e4c85f682b1d10051b9f4667e9da502cd896e4b47d0219201be45dddb5396e19.

Custody certificate: sha256:f775a666c4853e98b0560c25dcbe0e626f85f0f743e1a5945b5ebc6f19336ff2.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08618> (sha256:0be3cc15940bf345e61af54ef378022de8f2c32b7e566b7d4ed25f69c553ab)

Registered target identities:

- infrared-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

Immutable identities. Source manifest

sha256:c07ca280f1bf8c27df6ee5f4c840dd086848066a03f5e4bb4b10cd3f5cca37df; derivation seal sha256:c1815e5aaa2678f960ef5ea2fab48ad01421689c29e088440df015d9a874a0d6; engine receipt sha256:b1d149a0252e24a9e4ae697f92c986f6ee3fd0bc34698d43d39b19f5cab0721b at receipts/engine/model_admitted/SFT-CHEM-SPEC-INFRARED-001-b1d149a0252e24a9.json; empirical hash sha256:30acd4a106af6be00731ab9c87d59bbb211d389aefda98155f76a354a97b0c43; measurement receipt sha256:66439cdd064a13c442bd3feb7372e4f0a016e24e4eae0ee368dcbacc91b7f8f9.

81. Electronic absorption and molecular-state distinction

Claim identity: SFT-CHEM-SPEC-UVVIS-001

Question and theorem. Ultraviolet-visible organization is the complete condition-bound response pattern of molecular or ionic carriers under retained transitions between ordered electronic states.

ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-PHOTOCHEM-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held__carrier-state-response-map__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
input	ultraviolet-visible-support-held	unregistered-radiation	No excitation support is identified.
carrier	molecular-or-ionic-carrier-held	chemical-carrier-erased	Absorption cannot be assigned.
state	paired-electronic-states	single-state-only	No transition occurs.
transition	higher-state-transition-held	energy-order-erased	Absorption direction is ambiguous.
response	wavelength-response-pattern-held	absorption-number-only	Spectral support is lost.
composition	carrier-state-response-map	response-detached-from-carrier	Molecular distinction cannot be tested.
conditions	state-method-condition-boundary	universal-response	Environment and method affect observation.
extra	no-extra-rule	free-electronic-band-rule	A fitted band can select the answer.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8f109438a2184ff5bd94ed5c3baa2779939b48fa5908c03a9494ead18ff4c43a`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bdaad010665344c3bc6d4315e843eea1f36ae20f9d6016475c3f06bac89b5953`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:256566afdc578e393902758a27a005c03759954a45509d5445c7b8427a425f5a`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4cf9ae33fa257656b34c7fe81af112869f07b904db81a4937acccd502270d63a`.

Independent reconstruction. Implementation hash

`sha256:09021bd00f42035e6774890cefc66fd971761de04775ef636494b024c7a45c8`; independent certificate `sha256:cdb28a031fb92bd003449669ff7410e83e34c5809bdcd5cc065f98c9471e5c7c`; external-validation hash `sha256:3b6f0f0ce8d093a1c6ece6539a8f88d62e09cea6999635a2c02be6f99743d299`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- uv-visible-spectroscopy-iupac-composite: predicted ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; source-derived ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks UV-visible absorption, electronic transition, retained carrier/state response or method boundary, or if a tampered row is accepted.

Measurement receipt: sha256:ab68ae086496094b34adbd6f2eb057ece27c88e0bee1c99e8dc2b6579499f9e.

Isolation certificate: sha256:bf41ea75746962dc9f01f22df3c1a0a081ba9342674ae8ce36bdf302a91d1895.

Custody certificate: sha256:69eab65a16c11a8b416bb56c52622248ca35f957a0c69f2335f2bf94b2fd5803.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/S05696> (sha256:bcf4022ed1743c6445d1c6a2dc88d616adad7f833c299734f1b3fc3b45652a7d)

Registered target identities:

- uv-visible-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

Immutable identities. Source manifest

sha256:3365544b98ddc2a8728c20b4b5ebaae3a1c83aadee3e8ee956b61be2574d6788; derivation seal

sha256:e01d957ce5faf81059e38d012818e21411f565c169add8875dc9a60c50836ff6; engine receipt

sha256:6286103d4fa407f19df94df7529d42c98998bce4f1873bfa1f71bad69af17e37 at

receipts/engine/model_admitted/SFT-CHEM-SPEC-UVVIS-001-6286103d4fa407f1.json; empirical hash

sha256:c226e877cef95636b84f27d10a5efee1a48cb1ae5dad9de2cf5b81b80b33dfe9; measurement receipt

sha256:ab68ae086496094b34adbd6f2eb057ece27c88e0bee1c99e8dc2b6579499f9e.

82. Rotational-vibrational molecular spectra

Claim identity: SFT-CHEM-SPEC-ROT-VIB-001

Question and theorem. A rotational-vibrational spectrum is the complete condition-bound transition organization of retained molecular orientation-recurrence and internal relative-displacement recurrence classes.

complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001

- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-SPEC-INFRARED-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace__complete-transition-frequency-pattern__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-molecular-geometry-held	point-label-only	Internal molecular motion is absent.
rotation	rotational-state-classes-held	orientation-history-erased	Rotational recurrence cannot be distinguished.
vibration	vibrational-state-classes-held	relative-displacement-erased	Vibrational recurrence cannot be distinguished.
composition	rotational-vibrational-composition	motions-collapsed-together	Joint spectral structure is lost.
transition	allowed-state-transition-trace	states-without-transition-map	No spectral event is generated.
spectrum	complete-transition-frequency-pattern	one-frequency-only	Resolved organization is erased.
conditions	state-method-condition-boundary	universal-spectrum	Population and resolution are bounded.
extra	no-extra-rule	free-rotor-oscillator-equation	An imported model can select the spectrum.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4323d8285eafe3506586879204c1788f573d36d9a713b75d9b52ab8745e41bfa`.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ca67c32795fdd77c1b6ba5a17ba0a5984f0c5433217db50572ebcc84a840f588`.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bef4f75c40ebc0d0e5933659a7ea33f77a3bad65f0ed79a5877261fbbf03c980`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:179ca61d9ff6baa1547972cfa7013cebb029156f005a5193a807e86ff46661d3.

Independent reconstruction. Implementation hash

sha256:833c5b45b93d721aa1ab38cc1261be7e6e312fd3be9721a6ba087236b26d7eb1; independent certificate sha256:416fbcd13691a5a9887624f22611dcd6b60a31f2f6e113167ace8efeb96a0183; external-validation hash sha256:d184133bed088a2864cb839289022239ee561c40235a83dd2e4d4d026f95c7e5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Comparison records:

- rotation-vibration-iupac-composite: predicted molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; source-derived molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks rotational/vibrational transitions, state composition, transition pattern or population/resolution boundary, or if a tampered row is accepted.

Measurement receipt: sha256:9e6705743272cc7a980dbff088a552cc9fa032666221bce0201399ba9c83dc25.

Isolation certificate: sha256:d6e4b8971fd420cca4caae0d29e24ca2a51f85f504f5ac52e284dea925f10e45.

Custody certificate: sha256:d6097c94596e1d228f13e4b8cc8fd50852a2dc8dcab8791cb81630c7fa40dd89.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: <https://goldbook.iupac.org/terms/view/08676> (sha256:fa1bdd34ab2e93cce09dd22311bba6d9c1e3ee4221a285510eca2b7ce8e3e647)

Registered target identities:

- rotation-vibration-iupac-composite from IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

Immutable identities. Source manifest

sha256:89d52a84711f33006402cfa91a8cd63bd29135b9e350468641221d220f5cd859; derivation seal

sha256:214611a96dc7ce7ed35126a179a111b50b66db259a3ac7be747818b07e920aef; engine receipt

sha256:c3ab121f56b37e99216f98fd7435428eee357b453bfda395ce375032e8de808f at

receipts/engine/model_admitted/SFT-CHEM-SPEC-ROT-VIB-001-c3ab121f56b37e99.json; empirical

hash sha256:6e5b5994425799fb11bed90270db138170775d0e10412ae47cd19fd829b2ae; measurement

receipt sha256:9e6705743272cc7a980dbff088a552cc9fa032666221bce0201399ba9c83dc25.

83. Complete analytical result and falsification record

Claim identity: SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001

Question and theorem. A complete analytical result retains the measurand, sample/analyte/matrix, method, conditions, reference chain, every raw and derived observation, uncertainty, failed and tampered controls, and an explicit falsification condition.

measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001
- SFT-CHEM-ANALYTICAL-CALIBRATION-001
- SFT-CHEM-ANALYTICAL-SELECTIVITY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
question	measurand-and-purpose-held	unstated-measurand	The result has no declared target.
sample	sample-analyte-matrix-record	sample-identity-erased	The result cannot be located.
method	method-and-condition-record	method-erased	The transformation cannot be reproduced.
reference	calibration-and-reference-chain	reference-erased	Traceability is lost.
observation	all-raw-and-derived-rows-held	accepted-values-only	Unfavorable evidence disappears.
uncertainty	uncertainty-and-resolution-held	single-value-certainty	Admissible alternatives are hidden.
falsification	failed-tampered-and-falsification-record	success-only-report	The claim cannot fail publicly.
extra	no-extra-rule	free-reporting-omission	Discretion can hide a contradiction.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0cab94f55d8f3ec6788fc21b068becad07822404dccecdfaa4aaef265212cad1.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1ee4fcf2da1f0174c3b1e3c980363e6bca1731c16e17bea678247da58b52565d.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:93a5466a246fc746f2db197b298846a920a9544ce5df6ee0b6ebd4bbb02b6726.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:739798d50cb8adcec6ce6d95e38de2516a30dd5354ebca6d8c46646e40ec048e.

Independent reconstruction. Implementation hash

sha256:2ca15f1a1c1c4a464dbbe10d8860ab9a34743a8ac02dfd0ab04a032a5c999881; independent certificate sha256:a82ede737db1b63ea05e93848b8c61587572baf23629fcc9fd3127be01fc3249; external-validation hash sha256:7809c436f7c32024339ec7030d9ecbb6b3618c0c77f20180bbb081a5fe7c65e9. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Comparison records:

- complete-analytical-record-authority-composite: predicted analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; source-derived analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks result identity, traceability/uncertainty, retained controls or explicit falsification boundary, or if a tampered row is accepted.

Measurement receipt: sha256:c7dfa8ab47c6d7b6328e0ce79bb0f55bb8c85329ff82744e552936a6ac22449a.

Isolation certificate: sha256:3c73e9d2545d47c2d1839231134fcbffa1ae7fc9808941257c095e078217d549.

Custody certificate: sha256:4423955a12bb76535a4949f952ddcbeae75aa34aeb77da18350c29ab5ecd4f69.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and National Institute of Standards and Technology:
<https://goldbook.iupac.org/terms/view/M03796>
(sha256:f27104514cc2af6ad31064e74244121a076fc3c795f3f636e4f7f8face60e9a1)

Registered target identities:

- complete-analytical-record-authority-composite from
IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

Immutable identities. Source manifest

sha256:9fd499aac36241625375469b74d0a4a57b88545b50a126df9bf9d2b1cf53c0f4; derivation seal
 sha256:19bd6f6461040e3eea0a099949c134066d4d2121ce5bc762ecd595c32e89e0b0; engine receipt
 sha256:e851e1cf0fa3a72278c1782e7858a862bbd9621d130034b3fdfeb33bf149ce50 at receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001-e851e1cf0fa3a722.json; empirical
 hash sha256:b6ace663ae95961dfe00c270b9ba6162f5bd4f03abe159ca77baf69c5ba5d0d6; measurement
 receipt sha256:c7dfa8ab47c6d7b6328e0ce79bb0f55bb8c85329ff82744e552936a6ac22449a.

18. Gblock Smithium

The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction.

84. Generated g-block structural prediction

Claim identity: SFT-CHEM-PRED-G-BLOCK-001

Question and theorem. The admitted Fold orbit capacities ordered by increasing joint principal/orbit cover, with principal rank breaking a cover tie, generate noble closures 2, 10, 18, 36, 54, 86 and 118; fill 8s at 119-120; and open the 5g block at 121. The known prefix is externally checked and the g-block placement remains a sealed unobserved prediction.

The exact walk has capacities 2,6,10,14,18; noble closures 2,10,18,36,54,86,118; 8s occupations at 119-120; and first 5g occupation at 121.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All finite subshell walks built from the admitted positive-rank capacities, complete valid principal/orbit cells, increasing joint cover and source-rank tie order before the atomic endpoint.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__increasing-joint-cover-then-principal-rank__first-5g-occupation-at-121__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.
order	increasing-joint-cover-then-principal-rank	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	first-5g-occupation-at-121	selected-future-coordinate	A selected coordinate is not forced.
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7029631ba2389d152c39777bc1b5fdc51d436ec265ac0641b9f3cef50e51fc70`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:250a87885493ce97efb4e3db74f1eb38976bc61c1cb2b6271fce279423030e19`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8e07b41044d8c88aea1e8e8bc250609590e03205172694eac518f05cc0012c3a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f0ea0abcc06647ebf2729218c6ad037a15810eced930bfe1b51e9914c4447ab`.

Independent reconstruction. Implementation hash

`sha256:e323670a7c5c5d5a5c69b3138b71f1339902d495c29f0e7a9569905a7ad707d3`; independent certificate `sha256:eec03859c29d9208b967fb97d8f354e997dcea1b0f8026466fb738a677ff1bfff`; external-validation hash `sha256:5e283a27959797e090c59a2f7f65d56cb204acc528b24f5f34d39b516cd13880`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-G-BLOCK-KNOWN-PREFIX: predicted `known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction`; source-derived `known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction`; exact match `True`

- deliberately tampered unfavorable control rejected

Falsification boundary: The generated known closure prefix differs from the official source, the 5g opening is not 121, a target enters before sealing, a future prediction is called observed, or a tampered row is accepted.

Measurement receipt: sha256:f90a1aabda3ab41d3955dcf4b1ed2e8f96540de855f7ca982d0eaa623a49e97e.

Isolation certificate: sha256:d11aee024e8a048521aa12fc28e2a03eca797223f52d918cd36ae5f7721eb501.

Custody certificate: sha256:f38e70328919f4e9f6ac89c2ce639f974e1f8bd41c9d40d67c849e49c9de9b91.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
 (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-G-BLOCK-KNOWN-PREFIX from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no imported aufbau list or observed shell width
- no V1/V2 answer as a derivational input
- no semantic numerical zero, negative or nonexact proof value
- no claim that element 121 has been observed

Immutable identities. Source manifest

sha256:b40e7b8ec262ceec1fe073069a7392a7f4bd2d01a0d6c5a89e5335954d33cd18; derivation seal sha256:289d7818e0571477f6a481a2145684d578166225407f8e84db7d7f6306042113; engine receipt sha256:3d646a5267b619daa6cd184814fbcbb36a85718702f9c5a338c275d1ad488bb6 at receipts/engine/model_admitted/SFT-CHEM-PRED-G-BLOCK-001-3d646a5267b619da.json; empirical hash sha256:72c25069eea465471b7de6aebaa729e84513f44faf0563dc492261ae794a39cc; measurement receipt sha256:f90a1aabda3ab41d3955dcf4b1ed2e8f96540de855f7ca982d0eaa623a49e97e.

85. Smithium element-126 standing prediction

Claim identity: SFT-CHEM-PRED-SMITHIUM-001

Question and theorem. The independently sealed nuclear recurrence places the next proton/neutron double closure at 126/184. The generated electronic walk gives element 126 the active configuration 8s2 5g6, mass coordinate 310, eight valence carriers and structurally admissible positive oxidation counts +2 through +8.

Smithium is the standing prediction $Z=126$, $N=184$, $A=310$ with active configuration 8s2 5g6, valence count 8 and predicted positive oxidation counts +2 through +8.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001
- SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001
- SFT-CHEM-PRED-G-BLOCK-001

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, nuclear/electronic coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All double-closure/electronic-fill consequences at the next generated proton closure after 82 using the admitted nuclear recurrence and complete subshell walk.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-joint-cover-fill-walk__next-double-closure-126-184-with-8s2-5g6__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.
order	sealed-joint-cover-fill-walk	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	next-double-closure-126-184-with-8s2-5g6	selected-future-coordinate	A selected coordinate is not forced.
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2b9281d1a762cc7bdb1579504ba24551a5b444ea492bab0951a4287e69276fbc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fef001517e2f8109bea8ec2718b66087f655d84ce9032d5dcf4bc6f32e033f52`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:3f0509e56d6e645f0f27d9d64972dc7be8ce260b8fc027edfd1ccecf7e3b2e6d.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e3c97ef11cec09172621574fe62e1ceeea7995cc9ff6eaac97785d3456443933.

Independent reconstruction. Implementation hash

sha256:dbf2dee9cba1c6c7ccb65a677bb3214e5f4aeefb986c74c37f57d5305bf0a956; independent certificate
sha256:c46787ad2d1ee4ec941ade01601086040d8389a934fb9f2de55f817cf3467695; external-validation
hash sha256:5e3d12781d9854b133abb666e2e465d4e7e5a92db4c46b727a63c168c3208db1. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: **True**. All registered rows were preserved: **True**. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY: predicted
official-table-through-118__smithium-126-standing-unobserved-prediction; source-derived
official-table-through-118__smithium-126-standing-unobserved-prediction; exact match **True**
- deliberately tampered unfavorable control rejected

Falsification boundary: A future verified element 126 lacks the sealed nuclear/electronic coordinates, the official known boundary differs, target data enter before seal, or a tampered row is accepted.

Measurement receipt: sha256:1f0a538e836b85d88fc7001c9d0c7c42a932f8913d40f946b9f518fe5e4833c4.

Isolation certificate: sha256:9cea68a9339c7df811842de45a9ed772999b45cd89c7a81d512aaa8152cd5d6c.

Custody certificate: sha256:dcbdf5cb5cfe46f2ad04414d23477b0cc211f2b88b5d7281aa3a6b5466b77a3a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed Smithium claim
- no imported island-of-stability coordinate
- no fitted nuclear coupling or chemical exception
- no semantic numerical zero, negative oxidation magnitude or nonexact proof value

Immutable identities. Source manifest

sha256:749eca244a42d91c3d13f9ba0c36d281c17744006c6493988dda57a677e160a3; derivation seal

sha256:4139210c4b7c81bc194e47786feef97691d365eb343dbc3a1d76a28e78038c6a; engine receipt

sha256:2f05e2fe825d74621cbdd063b397f6f5b9132a37ebc38cf2772c55942111c37c7 at

receipts/engine/model_admitted/SFT-CHEM-PRED-SMITHIUM-001-2f05e2fe825d7462.json; empirical

hash sha256:f3b32f3a2fd94d31e17ba472e47f7a769d3c79245b30d6394042692ec6247136; measurement

receipt sha256:1f0a538e836b85d88fc7001c9d0c7c42a932f8913d40f946b9f518fe5e4833c4.

86. Generated periodic endpoint standing prediction

Claim identity: SFT-CHEM-PRED-PERIODIC-ENDPOINT-001

Question and theorem. The independently sealed exact inverse fine-structure ratio and One binding ceiling admit positive whole atomic coordinate 137 and reject its successor 138. Chemistry therefore records 137 as the model's structural endpoint while the official observed table remains source-bounded through 118.

The model's exact structural periodic endpoint is positive atomic number 137; the IUPAC 2022 observed boundary remains 118, so 119-137 including the endpoint are unobserved standing predictions.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001
- SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001
- SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001
- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-CHEM-PRED-G-BLOCK-001
- SFT-CHEM-PRED-SMITHIUM-001

Generated grammar. Generate the complete eight-axis product of positive atomic carrier, exact binding boundary, greatest-whole construction, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All Chemistry endpoint consequences of the admitted exact atomic-existence boundary and positive atomic-number order, separated from the source-dated observed table boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-exact-binding-boundary__structural-endpoint-137-successor-138-excluded__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	positive-principal-and-orbit-ranks	borrowed-orbital-notation	A borrowed notation has no Fold support trace.
capacity	admitted-Fold-orbit-capacity	imported-subshell-widths	Imported widths would supply the answer.
order	sealed-exact-binding-boundary	memorized-aufbau-list	A memorized order is a target-bearing table.
coordinate	structural-endpoint-137-successor-138-excluded	selected-future-coordinate	A selected coordinate is not forced.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
provenance	complete-fill-and-dependency-trace	answer-only-prediction	An answer-only prediction cannot be reproduced.
target	official-table-opened-after-seal	official-table-readable-before-seal	The known table could select the walk.
status	known-prefix-validated-future-coordinate-unobserved	future-prediction-called-measured	That conflates unobserved consequence with empirical confirmation.
extension	no-extra-rule	free-exception-or-fit	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 9d93b8a29fb07ab3a7f8f5d4efe373c45864448cb4bc6b54ac7656b4f2ae0a7a.
- **tampered_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 314897440a8fb21e21bbe6d1be41620f1cc2b1435261325748c8d401dbc6ab28.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 4c7ebcab86120fc5c11fb4089e51b6de3225b9f60c844de96937fe2e3d2370a8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: a9260d97a3d1af2d03cce0d6cf6037fcfb8039e34a7f91e66000ce1a4e28ad78.

Independent reconstruction. Implementation hash

sha256: 419962bc01b6a1a8c4399dddc62b15cb7b621808d204ef66530146b89b8d85a6; independent certificate sha256: ee90ca1e5a00142feb28aacce9965f6f8aab8f8d9cdcec7a1ec2673ebc898507; external-validation hash sha256: 69a4057c8a7c7554c76a3aa71de7b535033a2fab3f9a498f9be0418d08b7908b. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY: predicted
official-table-through-118__endpoint-137-standing-unobserved-prediction; source-derived
official-table-through-118__endpoint-137-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: A verified neutral element beyond 137 satisfies the registered One-ceiling boundary, the exact prerequisite fails replay, the observed/source boundary is conflated with the structural law, or a tampered row is accepted.

Measurement receipt: sha256: 2b27c0299574bb1deea7d18b7dc0cb4d1562a24275c206e6b9664ee650c73e36.

Isolation certificate: sha256: b4318304b667dfd2262d19cebdb73044d925778b93851de42eec87fd5b5c398c.

Custody certificate: sha256: 51eea197c861b56a2f7d339a3e2e19d0df5c3d5a24d69df7f6248a5e0613358a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry:
https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf
(sha256: ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed absence promoted to impossibility
- no bounded endpoint scan
- no imported critical-charge value
- no claim that element 137 has been observed

Immutable identities. Source manifest

sha256:79311bd68d7c9ee7bc096c1c47696c737df00176d6cabc10b49b88f050d9ea66; derivation seal sha256:cce4325ca809e604d40b0f29f51767e1607e0a4568b84ec8809f8c32579ff3e5; engine receipt sha256:ab9af61c4eccd228e24a01a2fe8b5967116818bb285f7ddfc168e37e4db4b267 at receipts/engine/model_admitted/SFT-CHEM-PRED-PERIODIC-ENDPOINT-001-ab9af61c4eccd228.json; empirical hash sha256:9ab75a13a17bf7d8b1f1e7b29c6b21840fea3a048a8c73247f3f135d72a4c8c4; measurement receipt sha256:2b27c0299574bb1deea7d18b7dc0cb4d1562a24275c206e6b9664ee650c73e36.

19. Exact upstream prerequisite and validation appendix

These seven claims are not hidden assumptions. They are separately engine-admitted V3 laws with their own complete evidence.

87. Terminal exact inverse fine-structure Fold ratio

Claim identity: SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001

Question and theorem. The binary predecessor and generator-three recurrence force cover depths five and seven. Their typed tower, boundary, complete three-direction cover, returning One and finite one-direction-at-a-time promotion ladder force the exact terminal ratio 503846395469/3676744786 before measurement.

The terminal inverse fine-structure Fold ratio is exactly 503846395469/3676744786; its leading rung is exactly 34259/250.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-FOUNDATION-COUNT-001
- SFT-FOUNDATION-PART-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar. Generate the complete typed product of cover, depth, tower, boundary, return, join, promotion, termination, target custody and extension forms.

Boundary: All type-correct assemblies that use each forced binary/generator cover block in its generated role, admit only the One as recurrence correction, and promote each of the three interchangeable cover directions at most once.

The engine regenerated 1024 candidates and 1024 one-for-one decisions. Exactly one survived: least-complete-binary-covers__cover-generator-cross-lock__binary-at-up-depth__generator-at-boundary-rank__on

e-return-over-complete-cover__tower-plus-dilated-boundary__one-remaining-direction-per-rung__all-three-directions-promoted__measurement-inaccessible-until-seal__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
cover	least-complete-binary-covers	asserted-depths	Asserted depths have no support census.
depth	cover-generator-cross-lock	single-route-depths	One route cannot expose a changed carrier.
tower	binary-at-up-depth	free-leading-whole	A free leading whole is a parameter.
boundary	generator-at-boundary-rank	free-boundary-factor	A free coefficient is not structural.
return	one-return-over-complete-cover	fitted-small-correction	A fitted correction reads the target.
join	tower-plus-dilated-boundary	untyped-arithmetic-permutation	Permuting typed carriers erases their roles.
promotion	one-remaining-direction-per-rung	selected-refinement-series	A selected series adds unfixed terms.
termination	all-three-directions-promoted	infinite-or-truncated-by-fit	A fit-selected truncation is not closure.
target	measurement-inaccessible-until-seal	measured-alpha-visible	That makes measurement an answer-bearing premise.
extension	no-extra-rule	extra-scale-rule	An extra scale is a free parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:ca5b4e49ab3bedece92f0ec11667e4305baafe8624283410439868e1b79e151c.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:0d516338a2c561300eae04d43e460a74cae268ab52d41ce1ace635214b048ab5.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:937d252d67fbb9ad1f3a8bcb5000226d4d7bde3b511429cbe0ee917e4986843.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:debfd802d48d0d9c9b5e1df0a6bdbf950f19bd012c0bdf52cfeec2505494619b.

Independent reconstruction. Implementation hash

sha256:97d0cdb7077804e63fb1bab56244287f15fbc1ee6fdeece167b59a309604117f; independent certificate sha256:ec1a7c48b4269d74564b029d286350d600f1fec6a9383bf655ad7415714f71f4; external-validation hash sha256:d2a29ed3e7a828cd1adc21abffff803d43cce478e12935dc83818e0a79af26a67. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest

sha256:624239a4a1a47a06acec655b1a55db66675f5d32fe9f67c6a4305d52eb56b99a; derivation seal sha256:9dfbee307fc2a9b96180abd41ec62bdf2a131a6e7aa702951e7acbf0ad0459b1; engine receipt sha256:4999a85b5862afebcf24604c2191c0ce8b96aa57f988b3fab83b469c3506a7c9 at receipts/engine/model_admitted/SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001-4999a85b5862afeb.json; empirical hash None; measurement receipt None.

88. Atomic orbit-cell capacity from three-space boundary orientation

Claim identity: SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001

Question and theorem. At positive orbit rank r , three-space supplies one central orientation and the rank-two boundary supplies one held pair for every predecessor step. Two exclusion-distinct Fold labels occupy each orientation, forcing capacity $2(1+2(r-1))$ and the sequence 2, 6, 10, 14, 18 without observation.

For every positive orbit rank r , capacity is exactly $2(1+2(r-1))$; ranks one through five have capacities 2, 6, 10, 14 and 18.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-COMBINATORICS-001

Generated grammar. Generate the complete product of carrier, central class, successor extension, boundary multiplicity, Fold label, exclusion, capacity, generality, target custody and extension forms.

Boundary: All orbit-cell capacity laws generated from one central orientation, positive orbit succession, the complete rank-two boundary, two held Fold labels and exclusion.

The engine regenerated 1024 candidates and 1024 one-for-one decisions. Exactly one survived: held-orientation-cell__one-central-orientation__one-boundary-orientation-pair__forced-rank-two-pair__both-Fold-labels__one-of-each-label-per-cell__labels-times-complete-orientations__constant-boundary-pair-successor__widths-inaccessible-until-seal__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	held-orientation-cell	unlabelled-cell-count	An unlabelled count loses orientation identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
central	one-central-orientation	empty-numerical-origin	Numerical zero is not an SFT carrier.
successor	one-boundary-orientation-pair	free-rank-increment	A free increment is a parameter.
boundary	forced-rank-two-pair	single-or-three-direction-addition	One omits a boundary side and three re-adds the held normal.
label	both-Fold-labels	label-erased-or-selected	Erasure or selection violates complete Fold support.
exclusion	one-of-each-label-per-cell	duplicate-cell-label	Duplication destroys distinguishability.
capacity	labels-times-complete-orientations	linear-or-doubling-without-boundary	Those former survivors ignore the newly admitted boundary discriminator.
generality	constant-boundary-pair-successor	finite-width-list	A list has no next-rank proof.
target	widths-inaccessible-until-seal	width-list-visible	That reverses derivation and test.
extension	no-extra-rule	extra-degeneracy-rule	An added degeneracy is a parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:1ed87436e21f59d043967ca432026eabd71e5e1f8be1907692eab24d946c6d5.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:625315de8974967651ba198f65bcbdb903e36bfac7a0cc349d449aed6298445b7.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4bf7aa1fcc6d37bda661b6b898af0d9b3a48ba75d0f12afa04f8db6b363f4d61.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:1c45ec73b2de2f2c722af6fcddecc09f844cab496f9bb77bdbdda1f0c85f80ab.

Independent reconstruction. Implementation hash

sha256:86c0b909d7fb32a35993571e1c3646d129604a21264f1da55aabad2b205c428e; independent certificate
sha256:dd12a5fc73eacfd47d2eeaf16cc2925d756b0b88926041d958ecc2867474ec0; external-validation
hash sha256:4ac2066f98eb5a6803d8f83eb315180dfaccda40a4fdf6575c2208668792f67a. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form

- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest

sha256:7b624a62e7d39d55c7a235163b8d0f93f6ffeebf172a7b94bf8fd4790be796f8; derivation seal sha256:6cd7fec86b24f0f4624a70894b507f836d4358eb2229515b61498407d958f993; engine receipt sha256:2b8928b0ddf5b87a7bf56813a5f430ba4a99f715b1cfc1dd1db98106216e3d70 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001-2b8928b0ddf5b87a.json; empirical hash None; measurement receipt None.

89. Exact colour-sector Fold coupling

Claim identity: SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001

Question and theorem. The complete two-label Fold distinction transported over the generator-three carrier forces the exact positive coupling ratio two-thirds, with no measured or fitted interaction strength.

The exact generator-three colour-sector Fold coupling is 2/3.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-ASSEMBLY-001
- SFT-PHYS-FIELD-INTERACTION-CLASSES-001
- SFT-PHYS-MATTER-COMPOSITE-HADRONS-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar. Generate every product of carrier, numerator, denominator, orientation, normalization, target custody and extension forms.

Boundary: All exact normalized ratios formed by transporting the complete Fold-label support across one generator-three interaction carrier.

The engine regenerated 128 candidates and 128 one-for-one decisions. Exactly one survived: `generator-three-carrier__both-Fold-labels__complete-generator-support__labels-held-through-carrier__two-over-three__closures-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	generator-three-carrier	free-colour-count	A free carrier count is a parameter.
numerator	both-Fold-labels	selected-single-label	One label omits half the Fold distinction.
denominator	complete-generator-support	external-normalization	External normalization is fitted.
orientation	labels-held-through-carrier	labels-collapsed	Collapse loses the interaction distinction.
normalization	two-over-three	reciprocal-or-unscaled-count	Those forms reverse roles or omit normalization.
target	closures-inaccessible-until-seal	magic-list-visible	A closure list would fit the coupling.
extension	no-extra-rule	extra-coupling-rule	An added strength is a free parameter.

Adverse controls. All mandatory controls passed:

- **false_premise:** passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256:e4e7687a9a1d832e556174d015e0aa64f11f64f92e06734a6051d3b987b676db.
- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256:e2edad0e6d45799bbf3bb96bb8c4e89f8be5a715fc0d921e462d57b1d7d2e8f0.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:12e6496b82b65e70e5458cdc2c1f49daecc9625414c9a58da9cda056fea5e5c6.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256:5947e11fa098dddaf9eb6ce47ae72cd6b0492d99a75eeb6452299129b69e0996.

Independent reconstruction. Implementation hash

sha256:babedcbfa7852074ccccf0885740154b359515c4b90ca34e584e8544bc59f31b; independent certificate sha256:8a8b86ffa63d4555d21aaab69918f4d60af798ad0b31d7e9c2014eb38779646e; external-validation hash sha256:c9a0a0d4ee3a15a784312da4b2c7bff456b6a17ddc1b6073c1dc58244065e846. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest

sha256:55e3c26bda6cce369b8f47cf4b459b445cabed93342517e8e9ee93424b0ca57f; derivation seal sha256:4893061724f8e857a5efe97692521d1df83c013f9723fe61d4804c3bae4baa77; engine receipt sha256:d38dc32f0926ae5f06e432f6bd7546d8680a8ea648c211dfbe965fc44e8179e5 at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001-d38dc32f0926ae5f.json; empirical hash None; measurement receipt None.

90. Complete Fold nuclear-closure sequence

Claim identity: SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001

Question and theorem. Complete weak compositions across three spatial directions with two exclusion labels force the base filled-shell counts. The independently sealed two-thirds colour coupling reaches its first whole spin-orbit gap at angular count three and forces the reordered closure sequence 2, 8, 20, 28, 50, 82, 126, 184 with a depth-independent successor law.

The depth-independent nuclear closure law gives the exact prefix 2, 8, 20, 28, 50, 82, 126, 184; its next proton closure after 82 is 126 and the next closure after 126 is 184.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001
- SFT-PHYS-NUCLEAR-COLOUR-COUPPLING-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-QUANTUM-SPIN-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-NUCLEAR-BINDING-001
- SFT-PHYS-NUCLEAR-LEVELS-001
- SFT-PHYS-NUCLEAR-REACTIONS-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar. Generate the complete product of spatial composition, label multiplicity, base closure, coupling source, shift, threshold, reordering, recurrence, target custody and extension forms.

Boundary: All filled-shell closure schedules built from complete weak compositions over forced three-space, both exclusion labels and the first whole-gap action of the independently forced colour coupling.

The engine regenerated 1024 candidates and 1024 one-for-one decisions. Exactly one survived: `complete-three-direction-weak-compositions__two-exclusion-distinct-labels__positive-rank-composition-form__sealed-two-thirds-colour-coupling__top-angular-half-label-pair__least-positive-whole-gap__complete-top-orbit-capacity__piecewise-positive-successor-law__sequence-inaccessible-until-seal__no-extra-rule`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	complete-three-direction-weak-compositions	selected-shell-degeneracy	A selected degeneracy imports a model.
labels	two-exclusion-distinct-labels	single-or-duplicated-label	Omission or duplication violates Fold support and exclusion.
base	positive-rank-composition-form	closure-list	A list has no generative proof.
coupling	sealed-two-thirds-colour-coupling	fitted-spin-orbit-strength	A fitted strength reads the closures.
shift	top-angular-half-label-pair	free-shell-shift	A free shift is a parameter.
threshold	least-positive-whole-gap	selected-rank-threshold	A selected threshold fits the sequence.
reorder	complete-top-orbit-capacity	arbitrary-intruder-count	An arbitrary intruder count is fitted.
recurrence	piecewise-positive-successor-law	finite-eight-value-table	A table does not close later ranks.
target	sequence-inaccessible-until-seal	magic-numbers-visible	That is target leakage.
extension	no-extra-rule	extra-nuclear-rule	An added selector is a parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt

sha256:5c771cf5592a53cc672843d7cdadcfbe7e340db992858decb1ab49eb0d695e8b.

- **tampered_source:** passed; expected reject a changed source identity; observed reject a changed source identity; receipt sha256:6e2dc9948609f6a9bcffed8b9e06971fa9e84b851c60221af6c0726a7a4bf601.
- **tampered_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:8069e2016e09a7983c9c51f36313fbec85fe529516e27d9f31877bb68d776e78.
- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt sha256:9df3facdacb8b6591dceb69efd2b75b8b300056734782cf444b7e08cdb38fc6.

Independent reconstruction. Implementation hash

sha256:0c39e11b369cf6d388ac93d2942d49a037aa20f3cb2550601df04805eee0dcaa; independent certificate sha256:988e6fd342d2c09cabea9a075e383ac3d673f1e74279b820e4d79c2bbb3a37ea; external-validation hash sha256:4a0b588d1c3b8b4ea798c27237d953cdee1dcb0918ef3b436678c0aa798fccfa. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest

sha256:239be83e36611d89d86421f4dc8e517f20288187660251a1313146246bfe5133; derivation seal sha256:f063dc762cb725c66340c946953cd008664875456944811ad902315d6b79b69d; engine receipt sha256:35f7fe50e217d2a61383f4169c314f1d4d5b4671a1def5baa011d26a1d89af9f at receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001-35f7fe50e217d2a6.json; empirical hash None; measurement receipt None.

91. Exact atomic existence boundary

Claim identity: SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001

Question and theorem. A neutral atomic coordinate remains structurally admissible only while its positive whole charge does not exceed the exact inverse fine-structure Fold ratio. The unique greatest whole below that sealed ratio is 137, with 137 below and its successor 138 above the boundary.

The unique greatest positive whole atomic coordinate under the exact Fold binding ceiling is 137; $137 \leq 503846395469/3676744786 < 138$.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar. Generate the complete product of charge carrier, coupling, ceiling, endpoint construction, lower witness, upper witness, generality, target custody and extension forms.

Boundary: All greatest-positive-whole atomic-coordinate conclusions under the One binding ceiling and the independently sealed exact inverse fine-structure ratio.

The engine regenerated 512 candidates and 512 one-for-one decisions. Exactly one survived: `positive-held-charge-count__charge-over-sealed-inverse-ratio__complete-One-ceiling__greatest-whole-division-certificate__exact-lower-inequality__successor-above-plus-order__depth-independent-greatest-whole-law__observations-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
charge	positive-held-charge-count	unbound-number	An unbound number has no element identity.
coupling	charge-over-sealed-inverse-ratio	fitted-binding-equation	A fitted equation imports the endpoint.
ceiling	complete-One-ceiling	external-critical-charge	An external critical charge is target input.
endpoint	greatest-whole-division-certificate	bounded-neighborhood-walk	A chosen search interval does not close all successors.
lower	exact-lower-inequality	endpoint-label-only	A label has no inequality witness.
upper	successor-above-plus-order	single-next-value-check-only	One check without order closure does not exclude later values.
generality	depth-independent-greatest-whole-law	finite-coordinate-census	A finite census leaves later coordinates open.
target	observations-inaccessible-until-seal	observed-elements-visible	That would turn an observation boundary into a law.
extension	no-extra-rule	extra-size-parameter	A size correction is a different, parameterized model.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt
sha256: 4160fd712b17226fbf3a8cc047e5446f5375e7bd3fa14fd1c64641c517a5e38e.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt
sha256: eba05c35e0e176c852c05a3f09315b28f7bd0cbe56b2511e8b5f44780bd2472a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: fe957e7c888b14fab4f760796244349be17a05c08bb8a440bb09417b28d8a928.

- **boundary:** passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt
sha256 : a37c7574414b2ddd6b071742183623ac6758ef060b905ba5bbf38a6617730b62.

Independent reconstruction. Implementation hash

sha256 : 55d5c09b7d7696f1d73843966a64fb278a666537289fd9472b7c3e557a83e55c; independent certificate
sha256 : d4b358d1dc307a24df8800cec04b9db1f575e9e52f920356fabfbf7739b7e2d2; external-validation
hash sha256 : eb370cb694ac8bb024e4020fbed03b0cfbec1a28e9e0af2d2abf7141d17c999c. The independent
program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest

sha256 : 97c4ce7770ec307c488fa3e908f713ef94eb5057771ebba5848675592d99d879; derivation seal
sha256 : 6baa991e8391efbd50b0d7235617136cbf6b14f2df03464d9e05d075d2bd3476; engine receipt
sha256 : 5cedeb1cb99ec83a668db0c80839b248c93e83bdeefbd067fa379fb8e45eb17 at receipts/engine/model_admitted/SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001-5cedeb1cb99ec83a.json; empirical
hash None; measurement receipt None.

92. Post-seal CODATA check of the terminal inverse fine-structure ratio

Claim identity: SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001

Question and theorem. After V3 independently seals the exact terminal inverse fine-structure ratio, the complete NIST CODATA 2022 central value and stated standard uncertainty are converted to exact rational interval endpoints. The sealed ratio is checked for interval membership without fitting or rewriting it.

The exact terminal ratio 503846395469/3676744786 lies inside the complete 2022 CODATA interval 137.035999177 +/- 0.000000021.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001
- SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001
- SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001
- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-ORDER-LATTICE-001

Generated grammar. Generate the complete eight-axis post-seal relation, provenance, custody, row, exact-interval and no-extra-rule product.

Boundary: All exact post-seal comparisons between the immutable terminal ratio and the one complete registered CODATA inverse-alpha row including its uncertainty.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-fold-carrier__sealed-terminal-ratio-versus-complete-codata-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-terminal-ratio-versus-complete-codata-interval	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 2460ed9e9112ca2db7f5fec2ba9676993debc0b85cb8d1040342b8a51dd34e68.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 20a184a1749db08e208fa09d5609987741171219db6cfdb7e3c1db52cbf51806.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 3921cc46ead2ad22ee5ae1bb3ad7f8b9184ac08981590c2f580ab371eda892a3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 9d42267b960bf9dfdd82343e855943f7fd5e75c3f418041c4cee03165d0ad02f.

Independent reconstruction. Implementation hash

sha256: 31aaffa05397abd35a390640284e0b7e2c2bd7c1793d599a60aad3916ff1695; independent certificate sha256: 4b54876e16a0163c485b0fe3c7b79c5c12fd3c53c49dd90bdd7fa11ed7248116; external-validation hash sha256: bfb80b39991e704b6d7b7fc4a0a419ef35d89db78ec3db6f95bef05e54606135. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Comparison records:

- NIST-CODATA-2022-INVERSE-ALPHA: predicted sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; observed sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; exact match `True`
- deliberately tampered unfavorable control rejected

Falsification boundary: The sealed ratio lies outside the complete reported interval, the source row/hash changes, the uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt: `sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7`.

Isolation certificate: `sha256:b3e8f58f7e96427308176390df6c109d246ad97efe3734bcb6250ad59ec0cdb3`.

Custody certificate: `sha256:b5389bb0d662254fceddf1684eadabe1f77b1f3354e7f690b23622eaa5d61d54`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants:
<https://physics.nist.gov/cuu/Constants/Table/allascii.txt>
(`sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67`)

Registered target identities:

- NIST-CODATA-2022-INVERSE-ALPHA from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no CODATA value accessible to the formal derivation
- no fitted digit or best-match selection
- no floating-point comparison
- no omitted uncertainty or unfavorable control

Immutable identities. Source manifest

`sha256:fb0058c462a90eec0073158ae31c88e7bad16311f630152e83ed5ae277d5282a`; derivation seal

`sha256:5f16c61d3619ddeb76f43b0199161bf2920453b1ecbfa49e2e24a858fcf70d95`; engine receipt

`sha256:cfe0c0b068864707942c03909ef4dd16660476c808fe5e9567970c60d8640bbd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001-cfe0c0b068864707.json`;

empirical hash `sha256:5eed8e7391a59d9ffdbf11d158d63dc889357300a19a27203f3719606f5e6877`;

measurement receipt `sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7`.

93. Post-seal IAEA check of the nuclear-closure sequence

Claim identity: `SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001`

Question and theorem. After V3 independently seals the positive-rank nuclear recurrence, the complete first eight outputs are compared in order with the registered IAEA Nuclear Data Section magic-number record.

The sealed V3 closure prefix (2, 8, 20, 28, 50, 82, 126, 184) exactly matches the complete registered IAEA sequence.

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`

- SFT-PHYS-MEAS-TARGET-CUSTODY-001
- SFT-PHYS-MEAS-HOSTILE-PACKAGE-001
- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-MATH-EXACT-ARITHMETIC-001

Generated grammar. Generate the complete eight-axis post-seal ordered-sequence, provenance, custody, row-retention and no-extra-rule product.

Boundary: All exact order-preserving post-seal comparisons between the immutable first-eight closure output and the complete registered IAEA positive magic-number sequence.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-fold-carrier__sealed-closure-prefix-versus-complete-iaea-sequence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-fold-carrier	answer-only-scalar	An answer-only scalar erases the generated physical carrier.
relation	sealed-closure-prefix-versus-complete-iaea-sequence	imported-or-fitted-relation	An imported or fitted relation lets consensus or target data select the law.
provenance	source-bound-proof-trace	unbound-provenance	Unbound provenance cannot demonstrate which premises produced the result.
prediction	capability-closed-prediction	target-readable-prediction	A target-readable predictor can copy or optimize against the measurement.
record	separate-measurement-record	proof-measurement-conflation	Treating an external decimal record as a proof scalar violates the measured-value boundary.
rows	complete-registered-rows	selected-favourable-rows	Selecting favorable rows cannot falsify the relation.
generality	one-successor-closure	finite-answer-lookup	An answer lookup has no successor certificate.
extension	no-extra-rule	free-extra-rule	A free constant, scale or exception can select the measured answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6f99243f846465531b8e8d04eafd231fa57515365c2ce894180820a930c4f6be`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:856dc18ebff37c0d3ec1fe6a114c55e16f6bd261778ca0abd9f5830bf7c82bb8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:77a9bb7213b8ff916417351bf0dada8cf20622b8fef94e29c8a977e5c2c2eb9e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:745f64af58a7835963b0d86ea1dd07931f1d68b4035bc3d091798fd4b90ac198`.

Independent reconstruction. Implementation hash

`sha256:305b02c334ed9f2efcdc389e5a3601348dabb2f28d1b4b9e9a0c91f36ba7716b`; independent certificate `sha256:0c4548de901d5c3e1ab8a7be0eb1ce51e3801104365a796d733b907dd9ca9642`; external-validation

hash sha256:7098173bec13b8fbbba7b194c44cf71c486b50c1411683a505d48c61764a57da5. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Comparison records:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE: predicted sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; observed sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: Any sealed/source coordinate differs, either ordered sequence is incomplete, the source identity/hash changes, or a tampered comparison is accepted.

Measurement receipt: sha256: aae2b19de5952d1488ac8ec859b09819c1378406999367008709aa7b423a7f08.

Isolation certificate: sha256: 264c93c4ab7661fdc40750e550143a79427eeca64dc3cf4508db4d52fba2e913.

Custody certificate: sha256: 5ca4ac8aaff5471643b0ccb92f484218242a81626c97d4d9590494215809e94d.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: <https://www-nds.iaea.org/publications/indc/indc-nds-0452-part1.pdf> (sha256:ce642d62698b97fa509b8101a5661e578d6524f8705c19e38f7426efe7a0f6a6)

Registered target identities:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE from IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no magic-number value accessible to the formal derivation
- no selected favorable subsequence
- no fitted coupling or threshold
- no omitted source or tampered control

Immutable identities. Source manifest

sha256:1d815786712147e935c6d17ddb736acd9863124d56a42adab83e860b22b0d131; derivation seal

sha256:56679e7ef35dd6a054b48cb85d9a365e04483d1876703a9d2e8ef6e52858f811; engine receipt

sha256:e55e9e67e3e4b389dd46c7a5cd6260674d11a9c832d55835d92100936f365bbd at receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001-e55e9e67e3e4b389.json; empirical

hash sha256:bfb9984c3d44e47579ac8b6b710cc22f04875b3c99fbfc07a179ccb8905a036a; measurement

receipt sha256: aae2b19de5952d1488ac8ec859b09819c1378406999367008709aa7b423a7f08.

20. Integrated result and scientific meaning

The branch unifies chemical identity, exact composition, bonding, reactions, thermochemistry, interfaces, stereochemistry, polymers, analysis and periodic organization under one provenance-preserving state-transition framework. Chemical laws do not begin with a periodic-table diagram or a fitted Hamiltonian. They begin with generated distinguishability, held labels, complete joint support, exclusion, recurrence, conservation and observation classes. Conventional names appear only at the correspondence boundary.

The terminal sequence has two independent locks. First, three-space and its rank-two boundary force orbit orientation growth; the two Fold labels and exclusion force capacities 2, 6, 10, 14 and 18. Second, complete three-direction compositions and the separately sealed 2/3 colour coupling force the nuclear closures. The joint-cover fill walk then reproduces the known noble closures and extends to the 5g prediction. The terminal inverse-alpha object independently supplies the atomic existence ceiling. Agreement of the upstream known support does not constitute observation of the future coordinates.

21. Reproducibility route

The one-command repository validator rechecks schemas, source identities, engine unit/E2E coverage and every admitted receipt in order. The Chemistry publication verifier additionally requires 86/86 claim coverage, all evidence files, one survivor per claim, every control, every empirical custody record, paper inclusion of every receipt, exact evidence-map identities and a visually rendered PDF. This paper build is a read-only projection and never changes a derivation.

22. Limitations and falsifiers

- IUPAC correspondence through 118 does not confirm element 121, Smithium or element 137.
- The structural endpoint is the registered point-carrier One-ceiling result; other physical models cannot be silently substituted into its claim boundary.
- The predicted Smithium oxidation counts are structural model consequences awaiting direct chemical observation.
- Any official target mismatch, missing row, source-hash change, replay mismatch or accepted tampered control fails the corresponding claim.
- Open V1/V2 lineage groups outside Chemistry remain mandatory V3 work; this paper does not paper them over.

23. Conclusion

At its frozen boundary, Chemistry is fully admitted in V3: 86 required claims, every candidate census, one survivor each, complete controls, independent reconstruction and post-seal evidence where empirical. The paper preserves the difference between exact derivation, known-domain correspondence and unobserved prediction. Its authority is the open evidence chain from the premise-free root theorem to immutable receipts, not credential, consensus or opaque prediction.

References and authoritative records

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- Smith, Maria, *There Is No Nothing*, Ernos Labs methods and foundation paper.
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